Rural Spillovers of Urban Growth*

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Abstract

This paper estimates how urban employment growth affects population and non-farm employment in surrounding villages, and how these effects vary with distance from the city. Using census data for 256,000 Indian villages between 1998 and 2013, we identify these effects with a shift-share design that uses plausibly exogenous variation in nearby urban growth driven by national industry trends. A 10-percentage-point increase in nearby urban employment growth lowers village population growth by about 1 percentage point while raising non-farm employment growth by about 3.6 percentage points. Urban growth raises services employment and has no detectable effect on manufacturing. Near the city, the employment gains are largest in trade and services. Farther away, they are concentrated in resource extraction. A model in which growing cities draw workers from villages while creating demand for village services and resource products can account for these patterns and their variation with distance.

Keywords: Urban growth, Rural spillovers, Urban-rural linkages, Structural transformation.

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1 Introduction

Urbanization will continue to reshape the developing world for decades, as the world's cities are projected to add nearly one billion residents between 2025 and 2050, most of them in developing countries (United Nations 2025). Yet four of every five people below the international poverty line live in rural areas (World Bank 2020), and the world's rural population is not projected to peak until the 2040s (United Nations 2025). Whether these households benefit from urban growth depends in part on how growing cities affect the rural areas around them. Urban growth can benefit nearby villages by creating demand for what they produce, but it can also pull workers and capital toward the city, reducing the resources that remain in the countryside. The distinction between these "spread" and "backwash" effects goes back to Myrdal (1957) and Hirschman (1958). Which force dominates in practice, and how far from the city each one operates, has not been established causally for villages around growing cities in the developing world.

This paper estimates how employment growth in a city affects population and non-farm employment in surrounding villages, and how these effects change with distance from the city. We study India between 1998 and 2013, a period of rapid urban growth when about two-thirds of the rural population lived within 20 kilometers of a city. Our analysis covers roughly 256,000 villages and 1,500 cities.

Our empirical design allows each village to be affected by employment growth in every city within 100 km, with the contribution of each city increasing in its baseline size and declining with distance. Because urban growth may be endogenous to conditions in the surrounding countryside, we instrument this exposure by replacing actual city employment growth with growth predicted with a standard shift-share design. Specifically, we interact each city's baseline industry composition with national industry employment growth, and then construct village-level predicted exposure using the same size and distance weights as in the original exposure measure.

We show that, although the village-level instrument aggregates predicted growth across many cities, each with its own industry composition, it can be written as a standard shift-share instrument. For each industry, the share component is a weighted average of that industry's employment share in the cities surrounding the village, with weights that depend on city size and distance. This structure allows us to draw on recent work on shift-share designs (Goldsmith-Pinkham et al. 2020; Borusyak et al. 2022). Specifically, we adopt the shocks-based framework of Borusyak et al. (2022), in which identification comes from the exogeneity of national industry growth rates rather than of the exposure shares. We perform a series of validity tests that support the identifying assumptions of this framework in our setting.

We find that nearby urban growth slows village population growth while expanding village employment. A 10-percentage-point increase in nearby urban employment growth reduces village population growth by about 1 percentage point over the decade, relative to a mean of 13.4 percent. The estimated response varies with the distance at which urban growth takes place. It is smaller for growth close to the village, larger for growth 40–60 kilometers away, and close to zero for growth 80–100 kilometers away. The effect on village employment goes in the opposite direction. The same increase in urban growth raises employment growth by about 3.6 percentage points, relative to a mean of 20.2 percent. Urban growth raises services employment and has no detectable effect on manufacturing. The employment estimates also follow a

different spatial pattern from population. They are largest for urban growth within 40 kilometers, smaller between 40 and 80 kilometers, and larger again in the outermost band.

To account for these patterns, we develop a simple model of how a growing city affects the villages in its hinterland. The model separates two forces that push village population and employment in opposite directions. First, a migration pull that does not fade with distance draws workers toward the city and slows village population growth. Second, urban growth raises demand for what villages produce through two channels: services and resource-based products. Demand for village services declines with distance because urban consumers must travel to the village to consume them.[1] Demand for resource-based products also weakens with distance, and reaches villages that extract them from a fixed natural endowment. Where these demand channels reach the village, they create jobs, allowing employment to rise even as population falls. These mechanisms have distinct implications for which sectors respond to urban growth at different distances, which we use to interpret the sectoral results.

The sectoral responses are consistent with the model's implications. The extraction estimates are larger for the growth of more distant cities, and at these distances extraction grows more in villages with above-median forest cover, our proxy for the resource endowment.

To shed further light on the mechanisms behind these effects, we examine additional outcomes and how the responses vary across villages. Estimating the employment effect sector by sector, we find that the gains come in large part from activities that serve local demand, with the largest estimates in trade and services. We then split the activities that sell to final consumers by who buys from them. The estimates for retail and wholesale trade, which can sell to urban consumers reaching the village, are positive for urban growth within 40 kilometers. The estimates for repair, restaurants, and personal services, which sell mostly to the village's own residents, are negative for urban growth beyond 40 kilometers, at distances where the population response is also negative. Both patterns are consistent with the model, in which urban demand for village services fades with distance, while residents' demand moves with village population.

We next turn the focus from village establishments to village residents. Using the Population Census, we find that both population and the resident workforce grow more slowly with greater exposure to urban growth. The male population responds almost exactly as the total population does, consistent with workers leaving together with their households.

Finally, we investigate how the effects vary across villages. In less densely populated villages, where land is relatively more abundant, the employment estimates are about twice as large, and the difference comes mostly from resource extraction. The share of children in the population barely changes, which suggests that the population response is not driven by lower fertility.

Contribution to the Literature

This paper contributes to three literatures. The first studies the spatial incidence of urban growth and asks whether a growing city lifts the places around it or drains them. The second studies how local labor

[1] Consumer mobility makes these services effectively available to urban residents despite being non-tradable. It mirrors the mechanism in Blanchard et al. (2025), where consumer mobility allows city-produced non-tradables to serve demand from outside the city.

markets absorb a shock to labor demand and how far the effects of the shock travel. The third studies structural transformation and asks what moves rural labor out of agriculture.

First, we contribute to the literature on how urban growth affects the surrounding countryside. Since Myrdal (1957) and Hirschman (1958), the literature has distinguished between spread effects, through greater demand and market access, and backwash effects that pull labor and capital toward growing cities. Empirical work in regional science finds evidence of both in high-income countries. Nearby urban growth is associated with faster rural population growth in the southern United States and Canada, with the magnitude of the effect depending on the size of the urban center and declining with distance (Henry et al. 1997; Partridge et al. 2007). This work, however, treats urban growth as given. Meanwhile, work in economics has documented effects on the surrounding countryside using variation in proximity to large urban centers (Cuberes et al. 2021) or quasi-experimental variation in transport connections (Faber 2014), rather than variation in the growth of the urban center itself. Existing studies also typically measure aggregate changes in places without distinguishing between the responses of firms and households, making it difficult to distinguish a village where local firms expand while households move away from one where neither firms nor households respond.

In developing countries, the closest causal evidence comes from India, where studies have examined the rural effects of urbanization and migration. These designs do not, however, isolate how growth in particular cities affects the rural areas around them. Calì and Menon (2013) use district-level instrumental variables to show that urbanization reduced rural poverty between 1983 and 1999, primarily through greater demand for local agricultural products. Madhok et al. (2025) use income shocks in potential migrant destinations to instrument the emigration of Indian farm households, finding that migration reduces cultivation among migrant-sending households and shifts food production toward more remote areas. To the best of our knowledge, this is the first study to use plausibly exogenous variation in the growth of specific cities to estimate how both the population and the non-farm employment of the surrounding villages respond.[2]

Second, we contribute to the literature on how local labor markets adjust to shocks in labor demand. This literature has historically focused on the place that receives the shock. Blanchard and Katz (1992) find that in the postwar United States a fall in a state's labor demand was absorbed mostly through out-migration, and that employment did not return to its previous path. Later work finds that population adjusted too slowly to clear a local shock within a generation, and that differences in joblessness across local labor markets persisted for decades as a result (Amior and Manning 2018). Related work by Feyrer et al. (2017) asks how far the effects of a shock travel beyond the place that received it. They find that wage income from new oil and gas production in the United States accrued within 100 miles of the producing county at about three times the amount that accrued within the county itself. Studies of this kind measure economic activity across distance, such as income or employment (e.g., Hornbeck and Moretti 2024; Mansfield 2026), but few estimate how the population of the surrounding places responds. One study

[2]The closest designs are Partridge and Rickman (2008), who relate poverty in United States counties to their distance from metropolitan areas and to the industry-mix growth of the nearest one, Henry et al. (1997), who relate the growth of nearby urban cores to rural population and employment without instrumenting it, and Bartik (2024), who measures the effect of a shift-share demand shock to one United States county on employment in others.

looks at the reverse direction estimating the spillovers of productivity in rural areas on urban population and employment growth, finding that cities expand their manufacturing and service sector employment with modest population growth in response to higher productivity in rural hinterland (Grigoryeva 2026). These two responses need not move together: a shock can increase economic activity in surrounding places while drawing some of their residents toward the place that received it. The one study that reports both population and employment responses on the same distance bands examines a different shock, the establishment of India's special economic zones, and finds that both grew faster in municipalities within about ten kilometers of a zone (Gallé et al. 2024).

In addition, recent evidence from rural India suggests that population and employment may respond differently to the growth of a nearby city. Breza et al. (2021) document substantial rationing of paid work in rural labor markets, which means that village establishments can expand employment by hiring workers who already live in the village. At the same time, millions of villagers travel daily to nearby towns for work while continuing to live in their villages (Sharma and Chandrasekhar 2014), making employment and migration separate margins of adjustment. We contribute by measuring population and employment in the same villages, in response to the same measure of urban growth and across the same distance bands.[3] The two move in opposite directions: urban employment growth slows village population growth while raising non-farm employment growth, mostly in trade and services.

Third, we contribute to the literature on structural transformation and the exit from agriculture. Since Lewis (1954) and Harris and Todaro (1970), models of structural transformation have emphasized urban labor demand as a force drawing workers out of agriculture. More recent models introduce a spatial dimension, with changes in agricultural regions driven by national forces rather than by the growth of particular cities nearby (Michaels et al. 2012; Eckert and Peters 2025). In India, Fan et al. (2023) estimate that productivity growth in consumer services such as retail and restaurants accounts for about half of the 18-percentage-point fall in agriculture's share of employment between 1987 and 2011. Empirical work, meanwhile, has focused mostly on structural transformation following changes that directly affect rural areas. Emerick (2018) finds that a wetter monsoon in India increased village non-farm employment without increasing migration. Asher and Novosad (2020) find that new paved roads reduced agricultural employment among village workers but generated little additional employment at village firms. In Brazil, Bustos et al. (2016) find that labor-saving soy technology reduced agricultural employment and increased local manufacturing employment, with about a third of the adjustment occurring through migration. Together, these studies provide evidence on how rural economies adjust to changes in agricultural productivity and market access. The role of growth in nearby cities on rural structural transformation has received much less empirical attention.

A related literature on the rural non-farm economy has long argued that demand from cities can support local non-farm activity (Haggblade et al. 2010; Lanjouw and Lanjouw 2001). We provide evidence consistent with this channel by linking shocks to employment growth in particular cities to sectoral changes

[3]Population growth is measured between the 2001 and 2011 Population Censuses and employment growth between the 1998 and 2013 Economic Censuses, while the urban employment growth to which both respond is measured over 1998–2013. The two responses therefore cover different periods and are not directly comparable in magnitude.

in surrounding villages. Employment at village establishments grows in trade and services near cities and in resource extraction, primarily forestry, farther away.

The remainder of the paper proceeds as follows. Section 2 describes the data and the definitions of cities, villages, and village-city distances, and documents descriptive facts about the urban system and its rural hinterland. Section 3 presents the empirical design, the validity checks, and the main results. Section 4 develops the model that can account for them and derives its implications for each sector. Section 5 examines the mechanisms behind the main results in light of the model, and Section 6 concludes.

2 Data and Context

2.1 Data

Our analysis uses data from the Socioeconomic High-resolution Rural-Urban Geographic Dataset on India, SHRUG (Asher and Novosad 2019). The SHRUG is a large harmonized data source describing socioeconomic development in India at the village and town level for the period 1990 to the present. It combines data from multiple different sources, of which our analysis draws on two: demographic and location amenities from the Population Census (1991, 2001, 2011), and firms and employment from the Economic Census (1990, 1998, 2013). Importantly, it provides these data using time-consistent geographies. Villages and towns are combined as needed to have unchanging borders since 1990, using an identifier called the SHRUG ID or *shrid*.

The Economic Census (EC) is a comprehensive enumeration of all non-farm economic establishments in India: private and public, formal and informal, manufacturing and services, urban and rural. To categorize and classify industries throughout the panel, the SHRUG uses new industrial codes, called *shrics*, which are consistent from the first to last round of the economic census. For consistency across rounds, the SHRUG drops all establishments in agriculture, government administration and defense before generating employment by shric (i.e. industry). We use these data to measure both urban and rural employment growth in our analysis.

The Population Census (PC) is a decennial enumeration of India's population. Among many other variables, it contains village-level data on both demographics (population by age and social group, etc.) and village amenities (road access, schools, health centers, electrification, etc.). We use these variables to construct both outcomes and controls, as described in Appendix Table A2.

The multiple rounds of the Population and Economic Censuses allow us to construct long-difference measures of employment and population, which constitute our main outcomes of interest. Due to data availability, the start and end years differ across sources. Specifically, for Economic Census variables, long differences are measured between 1998 and 2013, while for Population Census variables they are measured between 2001 and 2011. We use the 1990 Economic Census and the 1991 Population Census to

measure growth and industry composition before the study period, in the validity tests of Section 3.1 and in robustness checks.[4]

Urban Areas

Our definition of urban areas (i.e., "cities") is based on the global urban boundaries constructed by the Development Data Lab (DDL) (Development Data Lab 2026). This is a country-calibrated, multi-source construction designed to improve on the Global Human Settlement Layer (GHSL)'s Urban Centre Database, whose fixed global thresholds perform poorly in low-income countries with dense rural populations, making DDL's boundaries particularly well suited to our India-focused analysis. The choice matters, because different delineation methods can produce substantially different sets of metropolitan areas, as Bosker et al. (2021) show for Indonesia. DDL delineates each agglomeration's contiguous built-up core from three satellite-derived measures: population density (Tatem 2017), built-up surface (Pesaresi and Politis 2023), and night-time lights (Elvidge et al. 2021). A grid cell is classified as urban wherever at least two of the three agree. The population-density cutoff follows the GHSL standard of 1,500 residents per square kilometer, while the built-up and night-lights thresholds are calibrated country by country, so the boundaries adapt to settings such as the densely populated Indo-Gangetic plain, where a single global rule tends to misclassify dense rural land as urban. The construction is deliberately not fully automated: the resulting boundaries were hand-inspected for the largest cities and countries and manually corrected where they were clearly off, and the DDL team continues to refine them. DDL's boundaries, delineated for 2025, include only agglomerations whose population estimate, built from the GHS-POP grid (Schiavina et al. 2023), exceeds 50,000 residents.

For our application, we overlay DDL urban-boundary polygons with SHRUG geographies and identify all local administrative units (*shrids*) whose polygons intersect a DDL boundary. Any such shrid is classified as urban. We then aggregate all urban shrids overlapping the same boundary into a single analysis unit corresponding to the urban agglomeration, which we refer to as a "city" throughout the paper. This procedure yields 1,507 cities that can be matched to SHRUG localities and measured with census data, and these are our urban units of analysis. The city descriptives in Figure 2 cover the 1,455 of them with both a population change over 2001–2011 and an employment change over 1998–2013. City population and employment are the sums of the corresponding shrid-level counts within each agglomeration. Growth rates are computed from these city-level totals.

These cities span a wide range. The typical city is modest in size, with a median of about 70,000 residents in 2001, though the distribution is highly skewed, ranging from small cities to multi-million-resident metropolises. In the typical city, services account for roughly three-quarters of non-farm employment, with manufacturing accounting for most of the remainder (Appendix Table B1). Over the study period these cities grew rapidly, with non-farm employment, in services and in manufacturing alike, growing far faster than population (Appendix Table B1).

[4]The SHRUG maps the 1990 round to the same harmonized industry codes and time-consistent boundaries as the later rounds, although it locates a smaller share of 1990 establishments in villages and towns, because no location directory is available for that round (Asher and Novosad 2019; Asher et al. 2021).

Rural Areas

We treat as villages all SHRUG administrative units (shrids) that fall outside every DDL urban boundary and that we observe in both the Population and the Economic Censuses. Because shrids keep their borders across census rounds, a village that was reclassified as a census town by 2011 continues to be treated as a village. This avoids a potential mechanical source of slower rural population growth near cities, since new census towns account for almost 30 percent of India's urban growth over the decade (Pradhan 2013). In addition, in order to reduce the noise associated with growth rates computed from very small baseline counts, we restrict the sample to villages that have at least 100 residents and at least 10 non-farm jobs at baseline. Because our exposure measure, defined in Section 3, aggregates the cities within 100 km of a village, the estimation sample further keeps only villages with at least one city within that radius. Roughly 256,000 villages meet these criteria and make up our core estimation sample.

These villages are small and predominantly agricultural: the median village has about 1,250 residents and 36 non-farm jobs, and lies some 16 km from the nearest city (Table 1; extended statistics in Appendix Table A1). Roughly three-quarters of main workers are cultivators or agricultural laborers. Basic infrastructure is uneven: about 87 percent of villages have electricity, but only about two-thirds have a paved approach road, and adult literacy averages just under 60 percent.

Village-City Distances

We measure how closely each village is connected to surrounding cities using geodesic (great-circle) distances, the shortest "as-the-crow-flies" path over the Earth's surface, from the village centroid to the centroid of the *closest urban neighborhood* (urban shrid) within each city.

Let r index a village and let u index a city. Each city u consists of a set of urban shrids $\mathcal{S}_u$. Denote by c_r the centroid of village r and by c_s the centroid of urban shrid $s \in \mathcal{S}_u$. The distance from village r to city u is defined as

$$d_{ru} \equiv \min_{s \in \mathcal{S}_u} \; d_{\text{gc}}(c_r, c_s),$$

where $d_{\text{gc}}(\cdot, \cdot)$ is the geodesic (great-circle) distance between two points on the WGS84 ellipsoid.

In particular, it more accurately reflects proximity to urban activity for large cities, where villages located on the urban periphery may be close to economic activity despite being far from the overall agglomeration centroid. Figure 1 illustrates the measure for a sample region around Udaipur. Panel (a) links each surrounding village to its nearest city and colors it by that distance, measured to the centroid of the closest of the city's constituent neighborhoods rather than to the city's center. Proximity varies sharply across the region, and even a village's nearest city can lie tens of kilometers away. Because a village usually has several cities within reach, it can also be linked to more distant, and often larger, cities, as panel (b) shows for one village.

Table 1: Village Summary Statistics

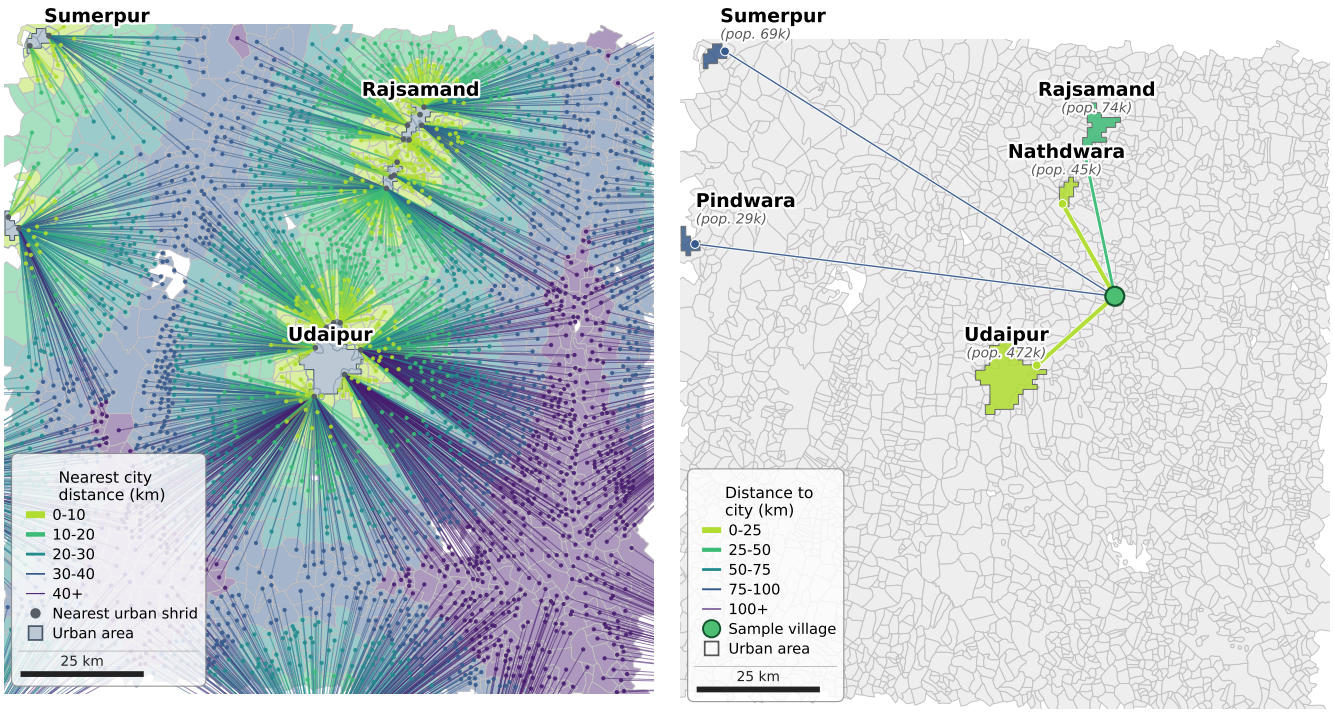
	N	Mean	Std Dev	p50
Panel A: Outcomes				
Population growth (log change, 2001–2011)	256,167	0.134	0.156	0.131
Total employment growth (log change, 1998–2013)	256,167	0.202	1.035	0.257
Services employment (log change)	252,670	0.359	0.962	0.392
Manufacturing employment (log change)	190,445	0.069	1.295	0.000
Cultivators (log change)	250,473	-0.165	0.830	-0.081
Male population (log change)	256,167	0.132	0.163	0.129
Female share of main workers (share diff)	252,197	0.010	0.156	0.011
Panel B: Village Characteristics				
Log population 2001	256,167	7.138	0.873	7.132
Log male population 2001	256,167	6.469	0.870	6.465
Log total employment 1998	256,167	3.792	1.059	3.584
Log population density 2001	256,161	5.890	0.964	5.862
Latitude	256,167	22.688	4.953	23.221
Longitude	256,167	80.626	5.420	79.496
Distance to nearest city (km)	256,167	19.488	13.356	16.000
Urban employment exposure	256,167	0.323	0.365	0.338
Bartik instrument (retail excl)	256,167	0.334	0.061	0.345
Panel C: Baseline Shares				
Services employment share 1998	256,167	0.371	0.245	0.348
Tradable employment share 1998	256,167	0.293	0.259	0.217
Resource extraction employment share 1998	256,167	0.025	0.117	0.000
Cultivator share 2001	256,167	0.486	0.247	0.477
Ag laborer share 2001	256,167	0.242	0.204	0.202
Female share of main workers 2001	256,167	0.239	0.163	0.222
Literacy rate 2001	256,167	0.594	0.159	0.608
Has power supply 2001	252,851	0.873	0.333	1.000
Has paved road 2001	252,019	0.675	0.469	1.000
SC share 2001	256,167	0.183	0.178	0.142
ST share 2001	256,167	0.135	0.262	0.001

Notes: The sample consists of the approximately 256,000 rural villages within 100 km of a city that enter the main analysis, with non-missing baseline population and employment; outcomes, baseline covariates, and the distance to the nearest city are winsorized at the 1st and 99th percentiles, while the urban employment exposure and the instrument are not. Outcomes (Panel A) are log differences over the study period, from the Population Census (2001–2011) for population variables and the Economic Census (1998–2013) for employment variables, except the female share of main workers, which is the 2001–2011 change in the share. Resource extraction employment (Panel C) is the Economic Census agriculture-and-mining category, which contains no farm employment and is primarily forestry (see Section 5).

Figure 1: Linking Villages and Cities: A Sample Region around Udaipur

(a) Each village linked to its nearest city

(b) One village linked to all nearby cities



Notes: The figure shows how we link villages to cities in a sample region around Udaipur. Gray polygons are cities, delineated by the DDL urban boundaries. All other points are villages. A village's distance to a city is measured to the centroid of the nearest of the city's constituent neighborhoods, not to the city's center. Panel (a) links each village to its nearest city and shades it by that distance, from green near a city to purple in the remote hinterland. Even a village's nearest city can lie tens of kilometers away. Panel (b) draws one village's distance to every city within reach, showing that a village is typically linked to several cities of different sizes and distances. Scale bars in kilometers.

We use d_{ru} in two ways. First, when assigning a village to a single reference city for descriptive profiles, the village is matched to the city u with the smallest d_{ru}. Second, when constructing distance bands and village-level exposure to urban shocks in the empirical analysis, we include all cities within a fixed radius and either group them into distance rings or form continuous weights based on d_{ru} and baseline city size.

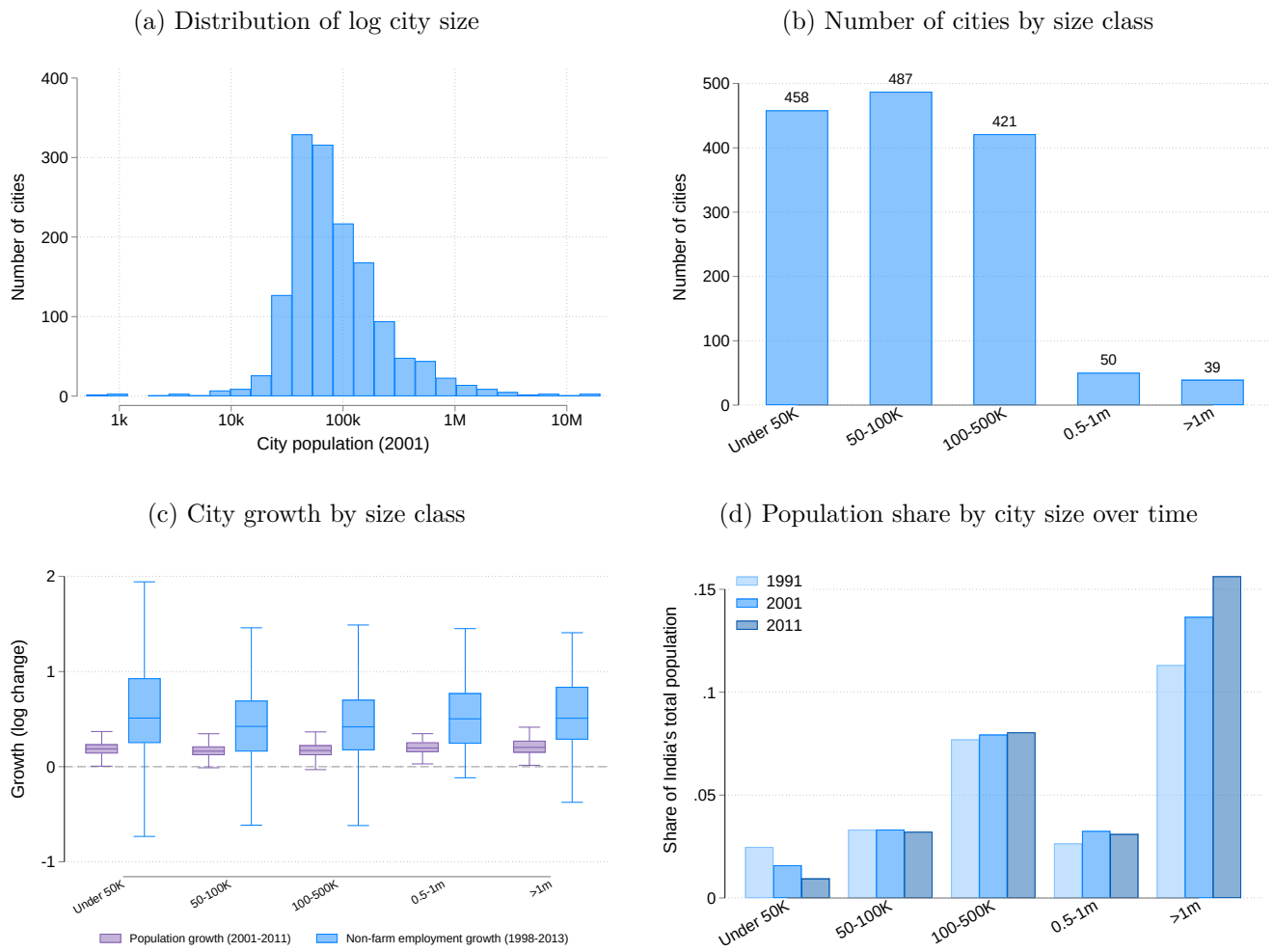
2.2 Descriptive Facts

India's rapid growth in recent decades has been accompanied by a persistent spatial divide, as in most of the developing world (Lagakos 2020): economic activity is disproportionately concentrated in urban centers, while a large share of the population continues to reside in rural areas. The divide is not particular to India. Across 65 developing countries, Young (2013) finds that the urban-rural gap accounts for about 40 percent of average within-country inequality. Chand et al. (2017) estimate that cities accounted for just 29.1 percent of the labor force in 2011–12 but generated 53.1 percent of output. Concerns that villages have been left behind have been central to policy debates and have motivated major programs aimed at raising rural living standards and accelerating rural growth. The evidence on these efforts is mixed: the flagship rural workfare program (NREGA) raised rural wages and generated large welfare gains for poor households

(Imbert and Papp 2015), whereas the Prime Minister's Rural Road Program did not have detectable effects on agricultural income, firm growth, or poverty, its main effect being to shift labor out of agriculture into wage work (Asher and Novosad 2020). While agriculture remains the largest single source of rural labor demand, non-farm work already provides close to two-thirds of rural income (Chand et al. 2017).

The urban system around these villages grew over this period, and its population concentrated in the largest cities. Nearly two-thirds of its cities have fewer than 100,000 residents in 2001, and 458 fall below 50,000. Thirty-nine of them exceed one million (Figure 2, panels a and b). Across every size class, non-farm employment grew far faster than population, and population itself grew in cities of all sizes (panel c), so the urban expansion of this period showed up more in jobs than in population. The largest cities gained ground over these two decades: their share of the national population rose from roughly 11 to 16 percent between 1991 and 2011, while the smallest cities' share fell (panel d). The same holds for economic activity. Desmet et al. (2015) show that services in India increasingly concentrated in high-density clusters over these years, unlike in the United States, where they grew fastest in medium-density places. Within cities, this employment growth appeared in both manufacturing and services, at comparable rates across the two sectors (Appendix Figure B1).

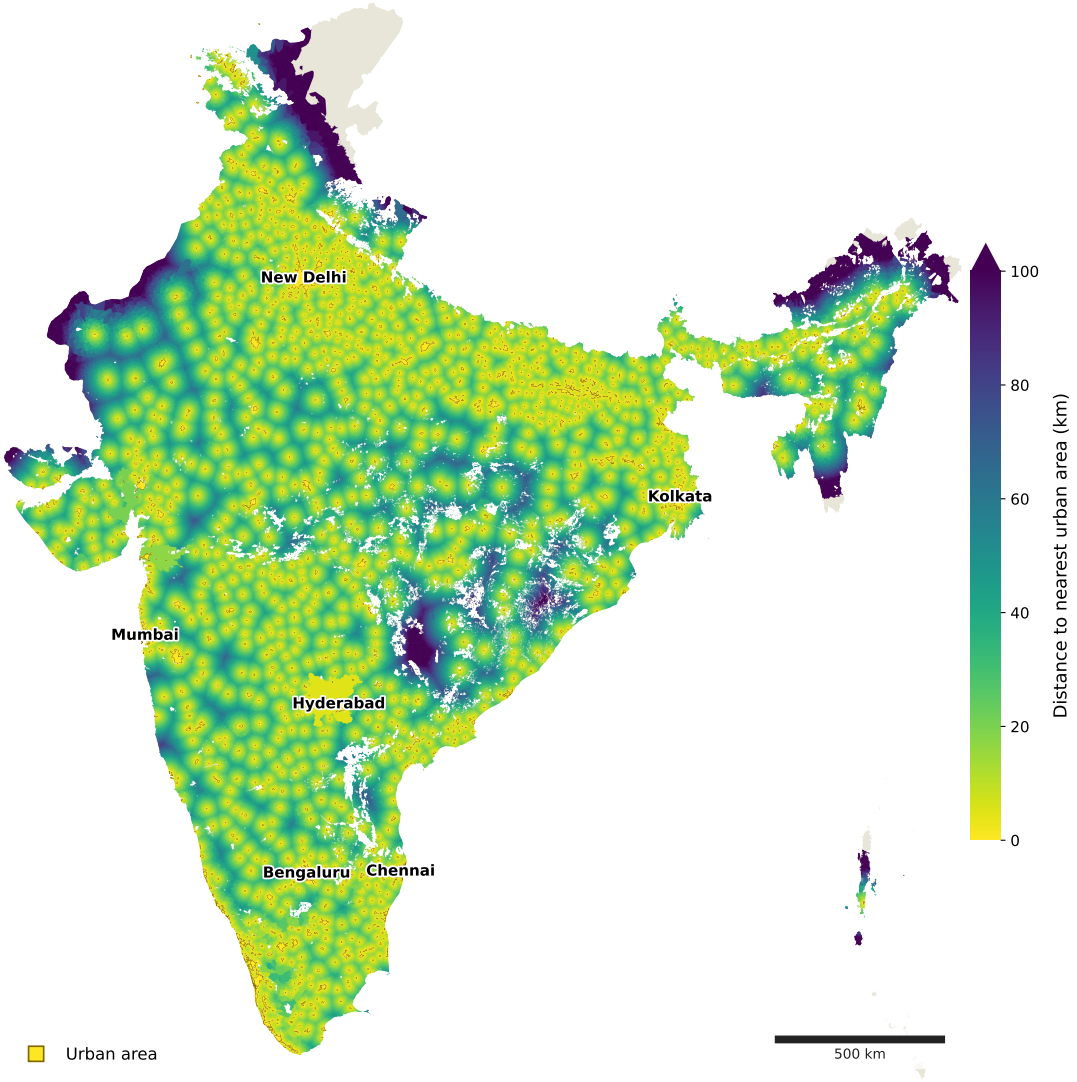
Figure 2: The Urban Hierarchy: Cities



Notes: The Indian urban system on the DDL city definition (DDL urban boundaries), across the paper's five baseline (2001) size classes (Under 50K, 50–100K, 100–500K, 0.5–1m, >1m). Panel (a): distribution of log city population in 2001. Panel (b): number of cities in each size class. Panel (c): boxplots of city population growth (long difference in log population, 2001–2011) and non-farm employment growth (long difference in log employment, 1998–2013, using $\ln(x+1)$) by size class; boxes span the interquartile range with the median, whiskers the adjacent values, outside values omitted, dashed line at zero. Panel (d): each size class's share of India's total population (summed over all localities) across the three Population Census years, with each city assigned to a class by its population in that year. Size classes use baseline Census population, so cities below 50,000 in 2001 appear even though the DDL product applies a 50,000 floor on its current population estimate. Panels (a) to (c) cover the same 1,455 cities. Panel (d) covers the 1,440 of them with a 1991 population, and all 1,455 in 2001 and 2011.

Villages lie close to this urban system almost everywhere. Figure 3 shades each rural locality by its distance to the nearest city. Outside a few sparsely settled or topographically constrained regions, most of the country falls within a few tens of kilometers of a city, and the remote hinterland forms only a small share of the map. Proximity to a city is the rule rather than the exception, so urban growth has the potential to reach rural areas across much of the country.

Figure 3: Geographic distribution of urban areas and rural proximity

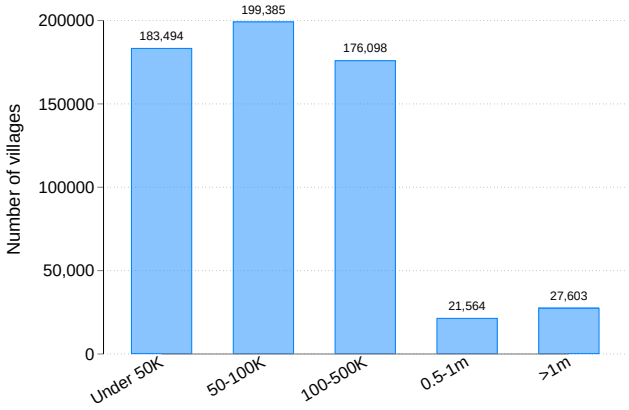


Notes: Each rural locality (SHRID) is colored by its geodesic distance (km) to the nearest urban area, delineated by the DDL urban boundaries: the contiguous built-up core of each agglomeration. Lighter (yellow-green) shading marks villages closest to an urban area; darker (blue-purple) shading marks the more remote hinterland; urban footprints are filled yellow (distance 0, the near end of the color scale). Distance is measured to the nearest constituent urban SHRID of the city, not to its centroid (see Section 2.1). Outside a small set of sparsely settled or topographically constrained regions, most of the country lies within a few tens of kilometers of a city.

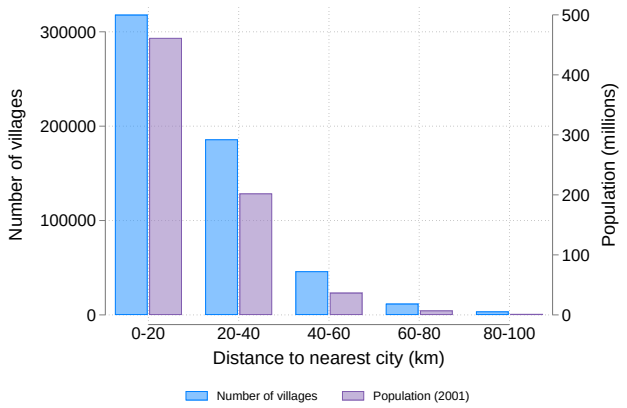
Most villages are nearest a small or mid-sized city. Figure 4, panel (a), classifies each rural locality by the size of its nearest city: villages are numerous around cities from under 50,000 up to 500,000 residents, while relatively few (fewer than one in ten) are nearest a city above half a million. Panel (b) shows how close they lie: just over half of villages, and about two-thirds of the rural population, fall within 20 kilometers of a city, and both counts drop off sharply beyond 40 kilometers. Distance to a city is short for the great majority of rural India, and shorter still for the places where most rural people live.

Figure 4: The Urban Hierarchy: Villages

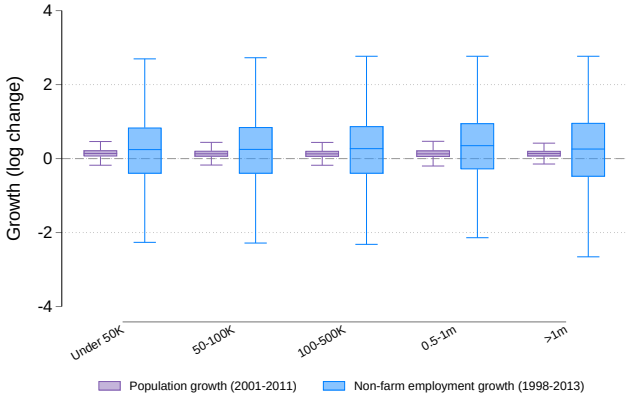
(a) Number of villages by nearest-city size



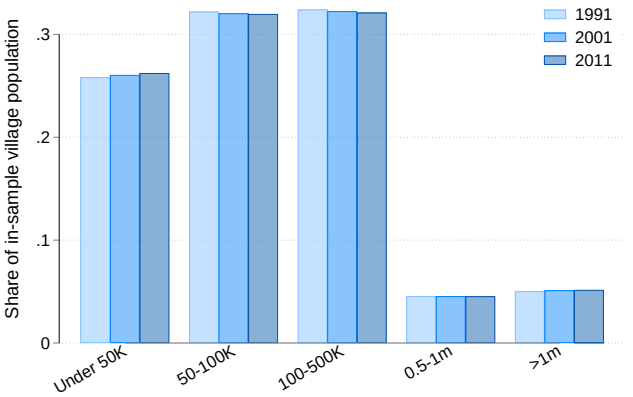
(b) Villages and population by distance to nearest city



(c) Village growth by nearest-city size



(d) Village population share by nearest-city size over time



Notes: Rural localities classified by the size of their nearest DDL city, across the five baseline (2001) size classes. Panels (a),(b) cover all rural localities (∼610K), defined as localities that do not overlap a DDL city boundary; panels (c),(d) the ∼256K in-sample analysis villages. Panel (a): number of villages by nearest-city size class. Panel (b): villages by geodesic distance to the nearest city, in 20-km bins from 0–20 to 80–100 km; the left axis counts villages and the right axis sums their 2001 population. Panel (c): boxplots of village population growth (2001–2011) and non-farm employment growth (1998–2013, using $\ln(x+1)$) by nearest-city size class; boxes span the interquartile range with the median, whiskers the adjacent values, outside values omitted, dashed line at zero. Panel (d): each nearest-city size class's share of total in-sample village population across the three Population Census years, with each village assigned in every year to the class of its nearest city's 2001 population.

Over the study period, the village non-farm economy grew quickly, and the growth was led by services. Village population grew by 13 percent between 2001 and 2011 and non-farm employment by 20 percent between 1998 and 2013, which is about the same rate per year. Both grew around cities of every size (panel c), but many more villages lost jobs than lost people: employment fell in 39 percent of villages, population in 14 percent. When villages are grouped by the 2001 size of their nearest city, each group's share of village population barely changed across the three censuses (panel d), which implies that villages grew at similar rates whatever the size of their nearest city. Within employment, the gains were concentrated in services: village services employment expanded around cities of every size, while manufacturing growth was zero at the median and highly dispersed across villages (Appendix Figure B2). Nationally, services rose from

about 35 percent of GDP in 1984 to more than half in 2010, while their share of employment rose only modestly (Amirapu and Subramanian 2015).[5]

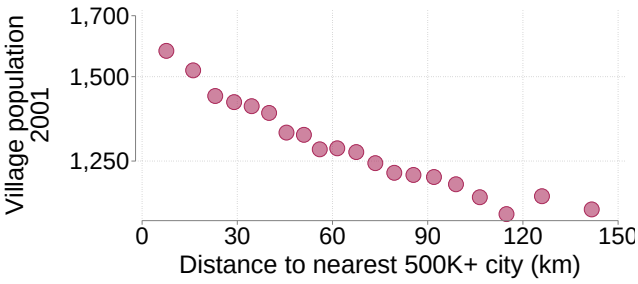
About 45 percent of village residents are recorded as working at baseline, counting both main workers, who worked at least six months of the year, and marginal workers, who worked less. Women make up 24 percent of main workers. Over the decade their numbers grew faster than men's. Resource extraction, mostly forestry, is absent from 89 percent of villages, but where it is present it accounts for 22 percent of non-farm employment on average. It is also more common far from cities: 10 percent of villages within 20 kilometers of their nearest city have any extraction employment, compared with 23 percent of those more than 60 kilometers away.

Villages closer to a large city are larger and more economically active, and they also grew somewhat faster, though the link to growth is loose. Figure 5 plots village outcomes against distance to the nearest city of at least half a million residents. In levels, proximity and scale go together: baseline population, employment, and services employment are all lower the farther a village lies from a large city, so villages nearby are bigger and more active than distant ones (left column), the same proximity gradient in rural non-farm activity that Fafchamps and Shilpi (2005) document for Nepal and Deichmann et al. (2009) for Bangladesh. The growth gradients are weaker. Employment, and within it services and manufacturing, grew markedly faster near large cities, with the premium concentrated in the first 40 kilometers, while population growth shows no systematic decline with distance (right column). These are raw, unconditional patterns, however: near-city and far-city villages differ in many ways beyond their exposure to urban growth, so distance alone cannot isolate the effect of a growing city. Separating that effect is the task of the empirical strategy we develop next.

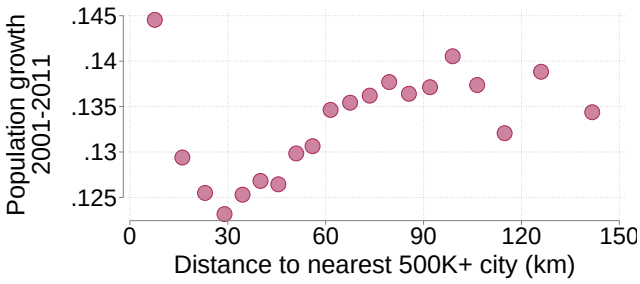
[5]The services-led expansion is sharper in output than in aggregate employment, where construction and informal activity absorb much of the employment growth (Kotwal et al. 2011). Within the non-farm enterprise sector we measure, captured by the Economic Census and excluding agriculture and public administration, services nonetheless lead employment growth, consistent with our village-level estimates.

Figure 5: Village Outcomes by Distance to the Nearest 500K+ City: Log Levels and Growth

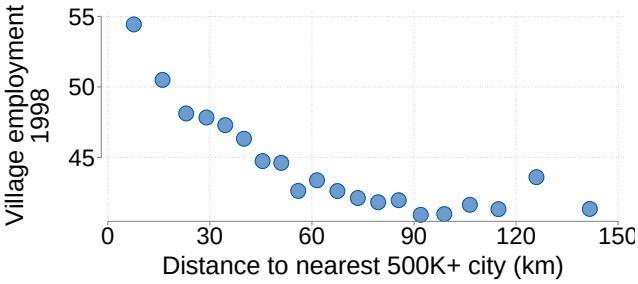
(a) Log population 2001



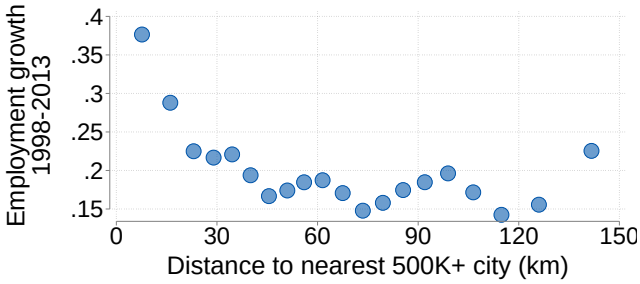
(b) Population growth 2001–2011



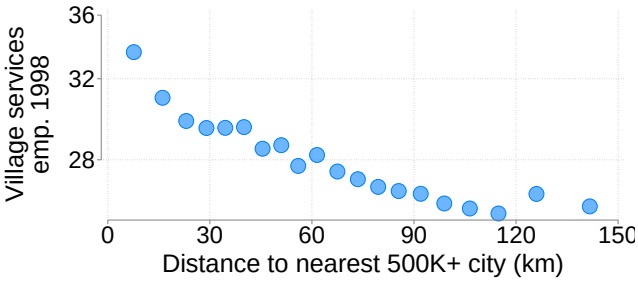
(c) Log employment 1998



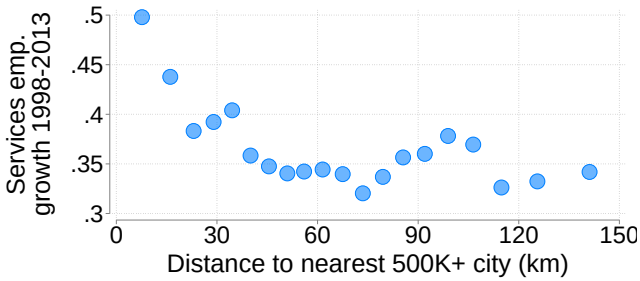
(d) Employment growth 1998–2013



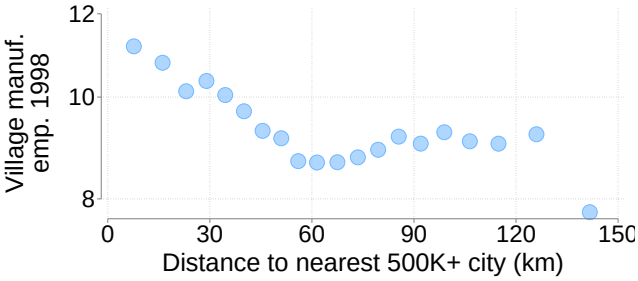
(e) Log services emp. 1998



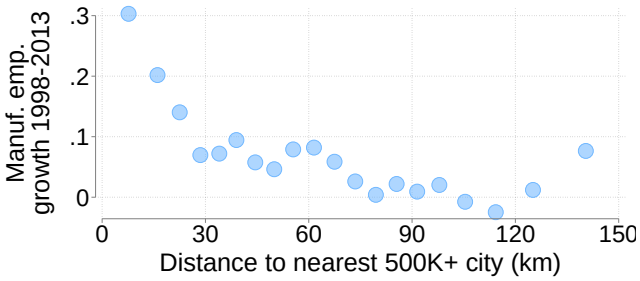
(f) Services emp. growth 1998–2013



(g) Log manuf. emp. 1998



(h) Manuf. emp. growth 1998–2013



Notes: Each panel is a binscatter (20 quantile bins) of a village-level outcome against distance to the nearest *large* city, one with a baseline (2001) population of at least 500,000. The left column is the baseline log level and the right column its long-difference growth (population 2001–2011; employment, services, and manufacturing 1998–2013; employment variables enter as $\ln(x+1)$), with rows for total population, total non-farm employment, services employment, and manufacturing employment. Distance is the shortest distance to any such city within 150 km; because large cities are sparse it ranges widely (median ∼64 km), so the panels extend to 150 km. Sample: the ∼232,000 analysis villages with at least one city of 500,000 or more within 150 km; the ∼24,000 villages with none are omitted. Each panel's y-axis is scaled to the outcome's range.

3 Empirical Design and Main Results

3.1 Empirical Design

Our objective is to estimate how urban growth affects the economic outcomes of surrounding rural areas. We index rural areas ("villages") by subscript r and urban areas ("cities") by subscript u. In the simplest setting, with a single city, we consider the following relationship:

$$\Delta Y_r = \alpha + \beta\, \Delta E_u + \boldsymbol{X_r'}\boldsymbol{\gamma} + \Delta\epsilon_r, \tag{1}$$

where ΔY_r denotes the long difference in the rural outcome of interest between the baseline and endline years available for that outcome. In our main outcomes, population changes are measured using the Population Census long difference (2001–2011), while employment changes are measured using the Economic Census long difference (1998–2013). ΔE_u denotes city employment growth over 1998–2013, $\boldsymbol{X_r}$ is a vector of predetermined village controls, and $\Delta\epsilon_r$ is the error term.

Village Exposure to Multiple Cities

In practice, a village may be exposed to shocks originating in multiple nearby cities. We therefore allow rural outcomes to depend on employment growth in all cities located within a given distance band around the village. Let $b \equiv [l, h]$ denote a distance band, where l and h are, respectively, the lower and upper bounds of distance from village r to city u.

Our main specification allows the effect of urban employment growth on village outcomes to depend on both the size of the city and its distance from the village, such that growth in larger and closer cities is expected to have a stronger impact than growth in smaller or more distant cities. This is in line with evidence from Brazil that the labor-market linkages between a city and the rural areas that supply its migrants are stronger for larger cities and for rural areas closer to them (Busso et al. 2021). Specifically, we aggregate employment growth across all cities within band b around village r, weighting each city by its baseline employment and its distance to the village:

$$\Delta Y_r = \alpha + \beta_b\, \Delta AE_{rb} + \boldsymbol{X_r'}\boldsymbol{\gamma} + \Delta\epsilon_r, \tag{2}$$

where β_b captures the effect on a village of urban employment growth in the cities within band b of that village, and the aggregate urban employment growth in band b around village r is defined as

$$\Delta AE_{rb} = \sum_u \omega^b_{ru}\, \Delta E_u. \tag{3}$$

The weights ω^b_{ru} assign relative importance to each city u within band b of village r according to the city's baseline employment and its distance to the village:

$$\omega^b_{ru} = \frac{\mathbf{1}^b_{ru}\, E_{u0}\, d^{-\theta}_{ru}}{\sum_{u'} \mathbf{1}^b_{ru'}\, E_{u'0}\, d^{-\theta}_{ru'}}, \tag{4}$$

where $\mathbf{1}^b_{ru}$ is an indicator equal to one if the distance between village r and city u lies within band b, and zero otherwise, E_{u0} denotes city employment in the baseline year (1998), and d_{ru} is the distance between village r and city u. The term $d_{ru}^{-\theta}$ captures the decay of a city's influence with distance. In our main specification, we set $\theta = 1$ and assess robustness to alternative values of θ. Ganning et al. (2013) compare weighting approaches for rural communities in the United States and find that inverse-distance weights over multiple cities perform slightly better than attributing all urban influence to the nearest city.

In our preferred specifications, we condition on a standard set of predetermined village characteristics. First, we control for the baseline level of the dependent variable (e.g., $\ln Y_{r0}$), which flexibly accounts for differences in initial scale and potential mean reversion and improves precision. Second, we control for the log of baseline village population (Population Census 2001), a summary measure of initial village size. Third, we include baseline geographic coordinates (latitude and longitude) to absorb broad spatial gradients in outcomes and access to urban markets. Finally, we include state fixed effects, so that the estimates compare villages within the same state, absorbing state-level policies, institutions, and growth trends. Baseline controls are measured in the baseline year corresponding to each outcome source (2001 for Census-based outcomes; 1998 for Economic Census outcomes). We winsorize the main outcomes and baseline covariates at the 1st and 99th percentiles to limit the influence of extreme values. The urban employment exposure measure and the shift-share instrument are left unwinsorized, so that the instrument keeps its exact form as exposure shares multiplied by national industry shocks. In all specifications we report robust standard errors clustered at the subdistrict level (5,473 clusters), and we show in Section 3.3 that our results are robust to alternative approaches to inference.

Composite Shift-Share Instrument

Ordinary least squares estimates of equation (2) may be biased if urban employment growth is correlated with unobserved shocks affecting nearby rural economies. For example, increases in agricultural prices or the diffusion of new technologies may raise demand for goods and services produced in cities, causing urban employment to expand in response to improvements in surrounding rural areas. Henderson et al. (2017) show that, in Sub-Saharan Africa, service-oriented cities can be reliant on rural demand, and therefore subject to the same economic shocks as their rural hinterlands. Rural shocks can also reach cities through capital, as in Brazil, where the savings from an agricultural productivity boom financed industry and services in urban regions rather than in the countryside (Bustos et al. 2020). In India, this is a salient concern given the fast growth of the service sector nationwide in recent decades. More generally, cities may grow because they are embedded in regions experiencing favorable demographic or infrastructural trends, which could independently influence village outcomes. For instance, the Golden Quadrilateral highway upgrade, which was built during our study period, raised manufacturing activity within about ten kilometers of its route (Ghani et al. 2016). To address these concerns, we instrument for aggregate urban employment growth using a shift-share design.

Our instrument is constructed directly at the village level. The starting point is the observation that villages are differentially exposed to national industry trends through their proximity to cities with different initial industry compositions. Villages located closer to cities that were initially specialized in industries

that subsequently expanded at the national level are more exposed to positive labor demand shocks than villages surrounded by cities specialized in declining industries.

Formally, we first construct, for each city u, a standard shift-share predictor of city employment growth,

$$Z_u = \sum_i s_{ui}\, g_i, \tag{5}$$

where s_{ui} denotes the share of industry i in city u's employment in the base year, and g_i is national employment growth in industry i over the period 1998–2013, measured as the long difference in log employment. We then aggregate these city-level predicted shocks to the village level using the same distance- and size-based weights employed in the construction of ΔAE_{rb}. Specifically, the village-level shift-share instrument for aggregate urban employment growth in band $b \equiv [l, h]$ around village r is defined as

$$AZ_{rb} = \sum_u \omega^b_{ru}\, Z_u, \tag{6}$$

where the weights ω^b_{ru} are given by equation (4). Intuitively, the instrument is a weighted average of the growth predicted for a village's nearby cities, with more weight on the larger and closer ones. It is therefore higher for villages whose most heavily weighted cities were initially specialized in industries that grew faster nationally. We use AZ_{rb} as an instrument for the aggregate urban employment growth measure ΔAE_{rb} in equation (2). Appendix Table B2 reports the first stage under alternative standard-error and control specifications. The instrument is a strong and significant predictor of ΔAE_{rb}, with a first-stage F-statistic above 150 in our preferred specification.

Identification

Recent work has clarified the conditions under which shift-share instruments identify causal effects and how inference should be conducted in such settings. Two broad identification frameworks have been proposed. One relies on the exogeneity of the exposure shares, while allowing for potentially endogenous shocks (Goldsmith-Pinkham et al. 2020). The other allows exposure shares to be endogenous but requires the shocks component to be exogenous, leveraging quasi-random variation in national industry trends (Borusyak et al. 2022).

Note that, although our instrument aggregates city-level shift-share shocks using distance- and size-based weights, it retains a standard shift-share structure. To see this, substitute equation (5) into equation (6), to obtain:

$$AZ_{rb} = \sum_i \tilde{s}_{rib}\, g_i, \tag{7}$$

where the village-level exposure shares are given by

$$\tilde{s}_{rib} = \sum_u \omega^b_{ru}\, s_{ui}. \tag{8}$$

Thus, our village-level instrument can be written as a conventional shift-share instrument, with exposure shares $\tilde{s}_{rib}$ that are linear combinations of city-level industry shares. This representation allows us to directly apply the insights of the recent shift-share literature to our setting.

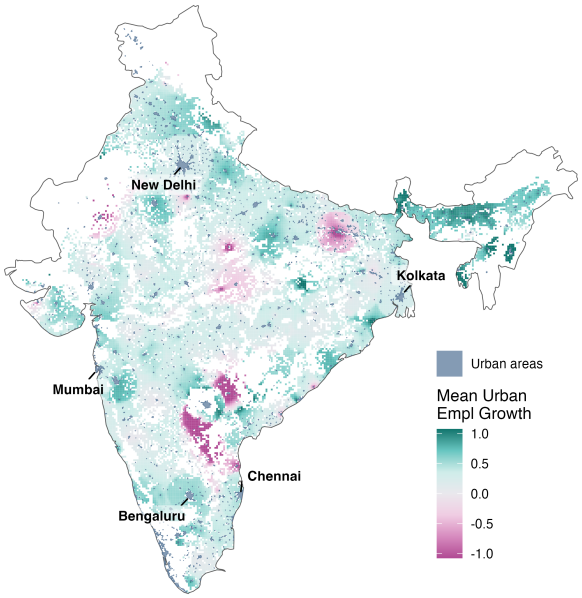
In our context, identification based on the exogeneity of the village-level exposure shares $\tilde{s}_{rib}$ is *a priori* unlikely to be valid. These shares contain information about the size and spatial distribution of nearby cities, which are themselves plausibly correlated with long-run rural development patterns. In India, for example, villages located closer to historically large cities tend to differ systematically in infrastructure access, market integration, and public investment exposure, all of which may affect rural outcomes independently of contemporary urban employment growth. In the terms of Borusyak et al. (2025), shares that reflect the overall industrial composition of nearby cities are generic rather than tailored: they capture a village's exposure to many kinds of shocks at once, not only to the labor-demand shocks the design intends to use.

We therefore adopt the shocks-based identification framework. Specifically, we assume that national industry employment growth g_i is as good as randomly assigned across industries, conditional on controls. Following Borusyak et al. (2022), this requires the growth of each industry to be uncorrelated with the unobserved determinants of outcomes in the villages exposed to it, averaged across villages with the exposure shares $\tilde{s}_{rib}$ as weights. This assumption is more plausible in our setting, as national industry trends reflect aggregate technological change, trade exposure, and policy shifts that are unlikely to be driven by conditions in specific rural localities. To further strengthen the credibility of this assumption, we exclude retail trade from the industries used to construct the shift-share instrument, since urban retail employment growth may at least partially reflect local demand (including that of people living in surrounding rural areas but shopping in the city) rather than exogenous national trends. Because the instrument is built from a restricted set of industries, these exposure shares do not sum to one across villages. Following Borusyak et al. (2022), we control for the incomplete-shares correction $1 - \sum_i \tilde{s}_{rib}$.

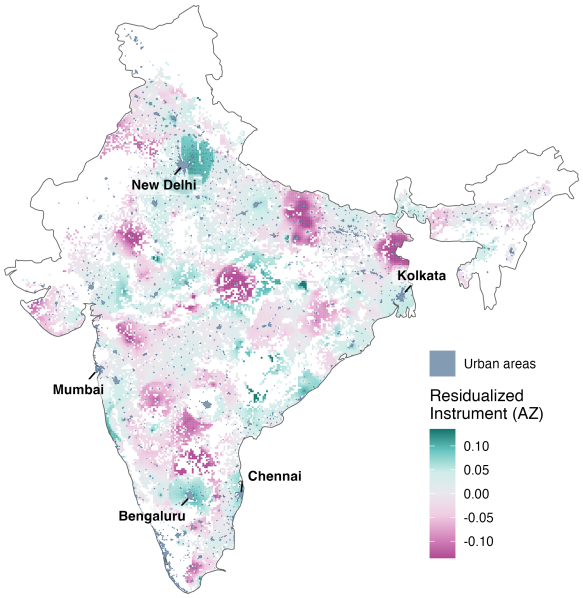
Figure 6 shows how this identifying variation is distributed across the Indian geography. Panel (a) maps our endogenous regressor: the urban employment growth villages are exposed to. This variable is strongest near large, fast-growing cities. Panel (b) maps the variation the population estimates use: the instrument, which reflects the baseline industry mix of nearby cities combined with national industry growth, residualized on the controls of the population specification. About three quarters of the instrument's variance is preserved after this residualization. The employment specifications add the baseline level of employment to these controls. The remaining variation is local and spread across the country's geography, takes both signs within every state, and does not simply follow proximity to the largest cities.

Figure 6: Identifying Variation in the Shift-Share Design

(a) Exposure



(b) Instrument (AZ), residualized



Notes: The two panels map the identifying variation in the shift-share design across rural India. Panel (a) plots the mean urban employment-growth exposure that each village faces, the endogenous urban-labor-demand object our design instruments. Panel (b) plots the variation the estimates use: the village-level shift-share instrument AZ of equation (6), which aggregates the city-level shift-share predictors Z_u over each village's nearby cities using the baseline distance- and size-based weights, and combines national industry employment growth over 1998–2013 (excluding retail) with the baseline (1998) industry composition of employment in those cities, residualized on the controls of the population specification (state fixed effects, latitude, longitude, log baseline village population, and the incomplete-shares correction) over the estimation sample. The color scale is symmetric around zero. Village exposure to nearby cities is measured over geodesic (great-circle) village–city distances. Urban areas (cities, delineated following the DDL urban boundaries) are shown in gray; rural villages are colored by the indicated quantity.

Validity of the Design

For the shocks-based design to identify a causal effect, two conditions need to hold. First, the national industry growth rates g_i must be as good as randomly assigned, conditional on our controls: the baseline urban employment of the industries that grew faster over 1998–2013 cannot be concentrated near villages displaying systematically different trajectories. Second, the instrument must combine many industry shocks, with none of them dominating. With only a few shocks, random correlations between industry growth and village unobservables would not average out, even if each shock were as good as random.

Both conditions refer to industries rather than villages. Borusyak et al. (2022) show that our village-level IV regression is numerically equivalent to a weighted IV regression at the level of the industry shocks, in which each industry's observation is the exposure-weighted average of the villages it reaches. The identifying variation thus comes from the 87 industry shocks, and the validity tests are conducted at this level.[6]

[6]In practice, each test residualizes the village-level variable on the controls, averages it to the industry level using the exposure shares, and runs the resulting industry-level regression weighted by each industry's average exposure. We implement this with the `ssaggregate` package of Borusyak et al. (2022).

Table 2 examines whether the industry shocks satisfy the two conditions. Panel A characterizes the shocks, following the practice recommended by Borusyak et al. (2025). Industries do not enter the instrument symmetrically. Each industry's weight is its exposure-weighted average share across villages, so industries that are prominent in the cities near many villages account for more of the identifying variation than industries concentrated in a few places. To measure how concentrated these weights are, we compute the effective number of shocks as the inverse of their Herfindahl index. It gives the number of equally weighted industries that would supply the same independent variation, and for our instrument it is 22.7 out of 87. The largest single industry accounts for about a tenth of the total weight, so no small group of industries dominates the design.

Table 2: Validity of the Shift-Share Design

Panel A: The identifying shocks (importance-weighted)

	Value
Number of industry shocks	87
Effective number of independent shocks (1/HHI)	22.7
Largest importance weight	0.103
Mean of the shocks	0.466
Standard deviation of the shocks	0.465

Panel B: Instrument relevance

	Estimate	SE
First-stage coefficient, village level	1.483	(0.118)
First-stage F, village level (Kleibergen–Paap)	158.9	
First-stage F, shock level (exposure-robust)	11.1	

Panel C: Shock-level balance (baseline characteristics on the shock)

	Coefficient	Exp.-robust SE	p	Conditional p
Log population 2001	-0.019	(0.022)	0.396	–
Log employment 1998	-0.000	(0.021)	0.996	0.550
Cultivator share 2001	0.004	(0.005)	0.389	0.373
Literacy rate 2001	0.010	(0.004)	0.011	0.113
Services emp. share 1998	0.007	(0.006)	0.231	0.798
Latitude	-0.181	(0.252)	0.474	–
Longitude	-0.288	(0.234)	0.222	–

Panel D: Shock-level pre-trends (pre-period outcome on the shock, 1990-vintage weights)

	Coefficient	Exp.-robust SE	p	RI p	Contemp. coef.	Ratio
Population growth, 1991–2001	-0.0017	(0.0009)	0.054	0.116	-0.0027	0.65
Employment growth, 1990–1998	0.0024	(0.0036)	0.506	0.608	0.0095	0.25
Services emp. growth, 1990–1998	0.0054	(0.0039)	0.173	0.256	0.0080	0.67
Manuf. emp. growth, 1990–1998	0.0035	(0.0043)	0.409	0.534	0.0006	5.56

Notes: Diagnostics for the shocks-based identification, computed on the population-growth estimation sample. In Panel A, an industry's importance weight is its exposure-weighted average share across villages, and the effective number of shocks is the inverse Herfindahl index of these weights. In Panel B, the village-level rows report the first-stage regression of exposure on the instrument under the main specification, and the shock-level F comes from the equivalent shock-level regression of Borusyak et al. (2022). Panels C and D report shock-level regressions of predetermined village characteristics (Panel C) and pre-period outcome growth (Panel D) on the national industry employment-growth shock, with exposure-robust standard errors. Balance regressions residualize the characteristic on the incomplete-shares correction; the conditional column instead residualizes on the full main-specification controls (state fixed effects, latitude, longitude, log baseline population, and the sum of shares) and is not defined for characteristics that are themselves controls. Pre-trend regressions use the main-specification controls and 1990-vintage exposure shares, which predate the pre-period growth windows; in the population row the baseline control is log 1991 population. The RI column reports randomization-inference p-values from 1,000 permutations of the shocks. The contemporaneous column repeats the regression for the outcome measured over the study period, and the ratio divides the pre-trend coefficient by it; the manufacturing ratio is uninformative because its contemporaneous coefficient is close to zero. All p-values are two-sided.

Panel B reports the strength of the first stage. In the village-level regressions the instrument predicts urban employment-growth exposure strongly, with a Kleibergen–Paap F statistic of 158.9. Given that the identifying variation comes from the industry shocks, however, the relevant statistic is the one computed at industry level. The equivalent shock-level regression yields an exposure-robust first-stage F of 11.1, compared with a conventional rule of thumb of ten. Because the statistic exceeds that threshold by a small margin, we verify in Section 3.3 that the main effects remain significant under inference methods that are robust to a weak first stage.

Panel C tests whether the shocks are as good as randomly assigned with respect to baseline village characteristics. A village receives a high instrument value when it is exposed to industries that grew fast nationally, so the concern is that those industries were concentrated near villages that already differed from the rest. Each row regresses the average characteristic of the villages exposed to an industry on that industry's national employment growth. Six of the seven characteristics are unrelated to the shocks. The exception is the literacy rate, whose correlation is significant at the 5 percent level but becomes indistinguishable from zero once we condition on the main-specification controls.

Panel D assesses the presence of pre-trends, asking whether the villages most exposed to subsequently growing industries were already on different paths before the study period. Each row regresses pre-period growth in an outcome on the shocks. The exposure shares are measured in 1990, so they pre-date the pre-period growth windows and do not capture that growth. We find no association between the shocks and pre-period growth in employment, services, or manufacturing. In the case of village population, the pre-trend coefficient is significant at the 10 percent level. This significance does not hold, however, under inference that does not rely on large-sample approximations.

In addition to validating the shocks-based design, we assess whether the alternative route, which relies on the exogeneity of the exposure shares, is credible in our setting. Goldsmith-Pinkham et al. (2020) show that a shift-share estimate can be written as a weighted average of the estimates that each exposure share would yield on its own as an instrument. They propose a measure of how much the estimate depends on each share, the Rotemberg weights. Appendix Table B3 reports these weights and tests whether the shares are exogenous. Even though the share-based instrument does not predict pre-period growth, the aggregate exposure share is correlated with eight of nine predetermined village characteristics after we condition on the main-specification controls. Appendix Table B4 also reports the population estimates that the five highest-Rotemberg-weight shares yield when each is used alone as an instrument. These estimates differ sharply. The difference could reflect invalid shares or effects that vary across industries, and the overidentification test cannot separate the two. The balance evidence therefore supports our decision to base identification on the shocks-based approach, following Borusyak et al. (2022).

One additional concern with our shift-share approach relates to inference. Specifically, villages exposed to cities with similar industry compositions are affected by the same shocks, so their residuals are correlated even when the villages are far apart. Clustering at the subdistrict level allows for correlation among neighboring villages but treats distant ones as independent, so it does not capture this dependence. To address this concern we show in Section 3.3 that our main results remain significant when using "exposure-robust" standard errors (Borusyak et al. 2022), which are computed at the level of the shocks.

Distance Rings Specification

To characterize how the influence of urban growth varies with distance, we also estimate specifications that allow spillover effects to differ flexibly across narrower distance rings.[7] Let $\{b_k\}_{k=1}^{K}$ denote a partition of the baseline 100 km neighborhood into K non-overlapping distance bands. In our implementation we use 20-km rings up to 100 km, so that $K = 5$ and $b_1 = [0, 20]$, $b_2 = (20, 40]$, $\ldots$, $b_5 = (80, 100]$. For each band b_k, we construct the corresponding band-specific exposure measure

$$\Delta AE_{rb_k} \;=\; \sum_{u} \omega_{ru}^{b_k} \, \Delta E_u, \tag{9}$$

where the weights $\omega_{ru}^{b_k}$ are defined exactly as in equation (4), but using the band indicator $\mathbf{1}_{ru}^{b_k}$. We then estimate the following specification, which allows the data to allocate identifying variation jointly across rings by considering all bands simultaneously:

$$\Delta Y_r \;=\; \alpha \;+\; \sum_{k=1}^{K} \beta_{b_k} \, \Delta AE_{rb_k} \;+\; \boldsymbol{X}_{\boldsymbol{r}}'\boldsymbol{\gamma} \;+\; \Delta\epsilon_r. \tag{10}$$

To adapt our identification strategy to this new specification we construct band-specific predictors. Specifically, for each distance band b_k, we aggregate the city-level shift-share predictor at the band level using the same $\omega_{ru}^{b_k}$ weights as in equation (4):

$$AZ_{rb_k} \;=\; \sum_{u} \omega_{ru}^{b_k} \, Z_u, \tag{11}$$

and we use these band-specific predictors to report reduced-form decompositions (replacing ΔAE_{rb_k} with AZ_{rb_k} in equation (10)). We report reduced-form estimates for this specification.[8]

We address the fact that some villages have no cities in a given band by setting ΔAE_{rb_k} and AZ_{rb_k} to zero when the set of cities in b_k is empty and including indicators for whether a village has no city in a given band, so that differences between "zero exposure" and "no nearby cities" are not conflated. All regressions include the same set of village-level controls as the main specification, including the corresponding band-specific sum of included exposure shares as a control (i.e., the analogue of $\sum_i \tilde{s}_{rib}$, constructed at the level of each ring). The band predictors introduce no new identifying assumption, since they aggregate the same

[7] The alternative to rings is to measure the effect within disks of increasing radius around the shock, as Feyrer et al. (2017) do for the income effects of new oil and gas production in the United States. Because each disk contains all the smaller ones, its estimate describes how the total effect accumulates with distance rather than where it arises. James and Smith (2020) re-estimate those effects with rings entered jointly, which attribute the effect to the distance at which it occurs, and Feyrer et al. (2020) accept in their reply that rings are the better choice when the question is where the effect arises and when a location is exposed to shocks at several distances at once, as in our paper.

[8] Estimating equation (10) by two-stage least squares would express the ring coefficients in terms of realized rather than predicted urban employment growth. Doing so requires solving the five first-stage equations jointly rather than one ring at a time, and that joint system is not estimated precisely enough for the rescaling to be informative. Feyrer et al. (2020) note the same difficulty, since splitting an instrument across bands weakens the first stage. The reduced-form coefficients do not depend on it: each is the response to the growth that the national shocks predict for the cities in its ring, identified under the same condition as the aggregate design.

city-level predictors as the composite instrument in equation (6), restricted to the cities in each ring, and therefore depend on the same requirement that the national industry growth rates be as good as randomly assigned. All band regressions additionally control for the baseline population of cities within 100 km, entered separately for four city-size categories (10,000 to 50,000, 50,000 to 100,000, 100,000 to 250,000, and above 250,000), to account for differences across villages in the amount of urban population around them.

The ring specification shows how the response of village outcomes changes with the distance at which predicted urban growth takes place. Although the ring predictors require no new identifying assumption, two features of the ring-by-ring design limit how the individual coefficients can be read. First, each ring's predictor also predicts part of the growth realized in other rings, and the predictors differ in how strongly they predict growth in their own ring, which implies that a ring's coefficient can also reflect responses to growth at other distances. Second, each predictor is a weighted average of the predicted growth rates of the ring's cities, and for the typical village the outer rings contain more cities and more urban employment. This implies that the same change in the predictor corresponds to more predicted new urban jobs in the outer rings. We therefore interpret the ring coefficients as describing how the response to predicted urban growth varies with distance, rather than as separate effects of growth at each distance.

3.2 Main Results

Table 3 reports our four main outcomes: growth in village population, in total employment, and in its services and manufacturing components. The three panels present, in turn, OLS estimates as a benchmark, the reduced form that regresses each outcome directly on the shift-share instrument, and 2SLS estimates, which instrument nearby urban employment growth with the same shift-share predictor. The first stage is strong, with F-statistics above 150 in every column. The signs are consistent throughout the table: nearby urban growth lowers village population growth while raising village employment and its services component. The manufacturing estimates, in contrast, are indistinguishable from zero in the reduced form and in 2SLS.

In column 1, the OLS estimate for village population is negative but small. Instrumenting moves the estimate to -0.100, an order of magnitude larger. The 2SLS coefficient implies that a 10-percentage-point increase in nearby urban employment growth lowers village population growth by about 1 percentage point over the period, a sizable effect relative to its mean of 13.4 percent. Expanding cities, in other words, slow population growth in the villages within their reach.

In contrast to population, employment rises with exposure to nearby urban growth. The 2SLS estimate for total employment (column 2) implies that a 10-percentage-point increase in nearby urban employment growth raises village employment growth by about 3.6 percentage points, a substantial effect relative to its mean of 20.2 percent. Urban growth raises services employment (column 3) and has no detectable effect on manufacturing (column 4). Employment at village establishments therefore grows even as population

Table 3: Effects of Urban Employment Growth on Village Outcomes, by Industry

			Employment by Industry	
	Population	Total Employment	Services	Manufacturing
	(1)	(2)	(3)	(4)
Panel A: OLS				
Urban emp growth	-0.009***	0.130***	0.107***	0.119***
	(0.003)	(0.019)	(0.015)	(0.021)
Panel B: Reduced Form				
Bartik instrument	-0.148***	0.532***	0.454***	0.036
	(0.019)	(0.129)	(0.094)	(0.157)
Panel C: 2SLS				
Urban emp growth	-0.100***	0.359***	0.308***	0.026
	(0.016)	(0.090)	(0.067)	(0.112)
State fixed effects	✓	✓	✓	✓
Geography (lat/lon)	✓	✓	✓	✓
Log baseline population	✓	✓	✓	✓
Baseline outcome	✓	✓	✓	✓
Incomplete-shares correction	✓	✓	✓	✓
N	256,167	256,167	252,670	190,445
First-stage F	158.92	159.25	153.06	207.69
Mean dep. var.	0.134	0.202	0.359	0.069
Mean baseline level	1,879	99	65	39

Notes: Outcomes are log differences (population, 2001–2011; employment, 1998–2013). The services and manufacturing columns are estimated on villages with positive employment in that sector at baseline (1998) and a valid endline record, which is why their N differs. Standard errors clustered at the subdistrict level in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Services in column (3) is the broad aggregate of trade, transport, and other services. Section 5.1 separates its components.

growth slows, consistent with some residents leaving while local labor demand strengthens.[9] A similar separation of the two outcomes has also been observed under migration restrictions. In Chinese counties connected by the national highway system, output fell while population did not respond, which Faber (2014) attributes to the household registration system. These patterns suggest that part of the spillover comes from the demand a growing city generates for local services. We further explore the composition of these employment gains in Section 5.

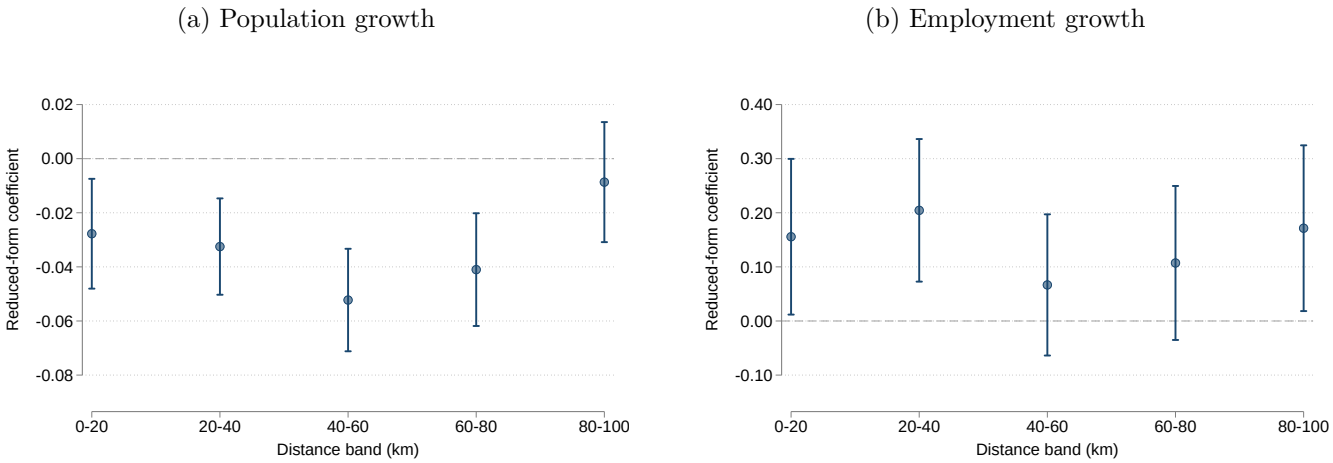
We next ask how these effects vary with the distance at which urban growth takes place. Figure 7 reports reduced-form coefficients from a single specification that enters five 20-kilometer distance bands jointly, letting the data allocate the response across rings. Because nearly every village has cities in several bands at once, the coefficients describe how the response varies with the distance at which the predicted urban growth takes place, not with how remote the village is. In panel (a), the population coefficients

[9] In the agricultural lean season, at least a quarter of rural men who want paid work at the going wage cannot find it: Breza et al. (2021) find that withdrawing 24 percent of a village's male workforce for a few weeks left wages and total employment unchanged. New demand for labor in the village can therefore be met by people who already live there.

are negative in each of the first four rings and close to zero in the outermost 80–100 km ring. They are smallest for urban growth in the nearest bands and largest for growth 40–60 kilometers away.

In panel (b), the employment coefficients are largest for urban growth within 40 kilometers, with the largest estimate at 20–40 kilometers, consistent with a demand channel that operates over short distances. They are smaller for growth 40–80 kilometers away and larger again in the outermost 80–100 km ring. Section 5 decomposes employment growth by sector and shows that the employment estimates for the most distant urban growth come largely from resource extraction, primarily forestry, rather than from trade or services, which implies that they have a different source from the estimates near the city. Appendix Figure B3 reports the same specification for cultivators, male population, and manufacturing employment.

Figure 7: Reduced-Form Effects by Distance Band



Notes: Panel (a) population growth 2001–2011; panel (b) total employment growth 1998–2013. Each point is the reduced-form coefficient on the band-specific Bartik instrument, which aggregates the predicted growth of the cities in the indicated 20-km distance band around the village, from a single regression that enters all five band-specific instruments simultaneously. The band instruments follow the same retail-exclusion construction as the main instrument. Controls: band-specific sums of included exposure shares, band-specific no-city indicators, total baseline city population within 100 km in four city-size categories, and the main-specification village controls (state fixed effects, latitude, longitude, log baseline population, the outcome baseline, and the overall incomplete-shares correction). Population and employment are estimated on the 256,151 villages for which the baseline city-population controls are available. Standard errors clustered at the subdistrict level; bars are 95% confidence intervals. The vertical scale differs between the panels. Appendix Table B12 reports exposure-robust standard errors, AKM0 confidence sets, and randomization-inference p-values for these coefficients.

3.3 Robustness

We assess the robustness of our results to alternative inference approaches. A first concern, specific to shift-share designs, is that conventional standard errors overstate precision: villages exposed to cities with similar industry compositions share the same national shocks, so their residuals are correlated even when the villages are far apart (Adão et al. 2019). Following Borusyak et al. (2022), we compute exposure-robust standard errors at the shock level. These are between roughly thirty percent and nearly four times larger than the conventional subdistrict-clustered ones. The population, employment, and services effects remain significant with these standard errors, while manufacturing remains indistinguishable from zero (Table 4). A second concern is that the identifying variation comes from a limited number of national shocks, so

the usual large-sample approximations may be unreliable. We address it with two procedures that are more reliable when the number of shocks is small. The first, following Adão et al. (2019), retains every candidate effect size the data cannot reject, testing each candidate as if it were the true effect. These are the AKM0 intervals in Table 4. The second is randomization inference: we reshuffle which industry received which national shock 1,000 times and ask how often the reshuffled shocks predict the outcome as strongly as the actual shocks do. A reshuffling across the whole economy assumes that a services industry could just as well have received a manufacturing industry's shock, so we also permute shocks only within broad sectors. The population, employment, and services effects remain significant throughout: their confidence intervals exclude zero, and their randomization p-values are below 0.05 under both permutation approaches. Appendix Table B5 reports the same tests with clustering at the district and state levels and for additional outcomes.

Table 4: Instrument Strength and Robust Inference

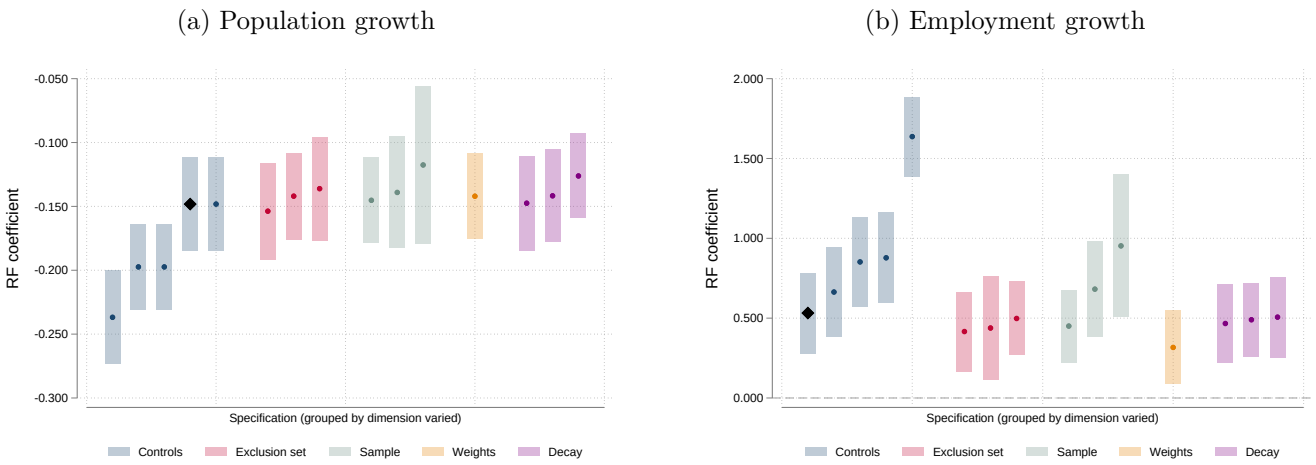
	2SLS est.	Subdist. SE	BHJ SE	AKM0 95% CI	RI p
Population growth	-0.100	(0.016)	(0.048)	[-0.335, -0.009]	0.046
Employment growth	0.359	(0.090)	(0.115)	[0.074, 0.790]	0.022
Services employment growth	0.308	(0.067)	(0.112)	[0.021, 0.713]	0.036
Manufacturing employment growth	0.026	(0.112)	(0.423)	[-1.837, 1.134]	0.975
First-stage F (Kleibergen–Paap, conventional): 158.9					
First-stage F (BHJ shock-level): 11.1					
Effective number of shocks (1/HHI): 22.7					

Notes: Each row reports the 2SLS estimate of the effect of nearby urban employment growth on the village outcome, under the main specification: state fixed effects, latitude and longitude, log baseline village population, the baseline level of the outcome, and the incomplete-shares correction. Four inference methods are reported: conventional subdistrict-clustered standard errors (in parentheses); exposure-robust standard errors computed at the industry-shock level ("BHJ SE"); a null-imposed 95% confidence set constructed by inverting the exposure-robust test at the shock level over a grid of candidate values ("AKM0 95% CI"), a procedure that remains valid when the first stage is weak; and a randomization-inference p-value from 1,000 permutations of the national industry shocks, based on the t-statistic of the shock-level reduced-form regression ("RI p"). The bottom panel reports the first-stage F statistic from the conventional (Kleibergen–Paap) computation and from the shock-level (BHJ) regression, together with the effective number of independent shocks (1/HHI of the shock importance weights). Sample and outcomes as in Table 3.

Figure 8 reports the reduced-form results under fifteen variants, grouped by the choice each one changes. How we build the instrument matters little. We leave retail out of the shift component because its growth partly reflects local demand rather than national trends, and the population, employment, and services estimates have the same sign and a similar magnitude when we remove that exclusion, when we extend it to repair, restaurants, and personal and social services, or to agriculture and wholesale trade, and when we drop railway transport and education, the two industries with the largest Rotemberg weights, whose employment is largely administered by the state (Appendix Table B6). The same holds for the rate at which a city's influence decays with distance. Our baseline weights each city by the inverse of its distance from the village, and a weaker decay, a steeper one, and no decay at all leave the three estimates within the confidence interval of the preferred one. Dropping the control for the baseline level of the outcome, which accounts for differences in initial scale and mean reversion, also leaves their sign unchanged, with the employment estimate about a quarter larger. What moves the magnitudes is the

geographic controls: without state fixed effects the employment and services responses are roughly three times larger, reflecting the cross-state variation that the preferred specification deliberately absorbs, though their sign is unchanged. Manufacturing is indistinguishable from zero in thirteen of the fifteen variants, and the two exceptions have opposite signs.

Figure 8: Reduced-Form Specification Curve: Population and Employment



Notes: Each bar reports the reduced-form estimate of the outcome on the shift-share instrument under one of fifteen specification variants, grouped by the dimension the variant changes relative to the main specification: the control set, the industries excluded from the instrument, the estimation sample, the weighting, and the distance-decay parameter θ in the exposure weights. The three sample variants restrict the villages included by each village's distance to its own nearest city; the 100 km radius over which exposure is measured is the same in every variant. Within a group, bars are ordered by the point estimate. The marker is the point estimate and the shaded bar spans its 95% confidence interval, with standard errors clustered at the subdistrict level in every variant. The black diamond marks the main specification's reduced form: state fixed effects, latitude, longitude, and log total village population in 2001, retail exclusion set, $\theta = 1$, unweighted, baseline-outcome control included. Panel (a): population growth 2001–2011. Panel (b): total employment growth 1998–2013. Appendix Figure B4 repeats the exercise for all main outcomes; Appendix Table B6 reports the estimates in a table.

To be sure the effect is not tied to a particular distance range, we restrict the sample by each village's distance to its own nearest city, keeping only villages whose nearest city is closer than 60 kilometers, then only those whose nearest city is at least 20 kilometers away, and then only those at least 10 kilometers away, since a village that close can be reshaped by the city directly. We also weight villages by population, so that larger places count for more. The population, employment, and services estimates have the same sign in every one of these variants. The employment estimate is larger in the samples that leave out the villages closest to a city, and smaller when villages are weighted by population. All three estimates also have the same sign when each of the thirty-two states and union territories in our sample is dropped in turn, and manufacturing stays indistinguishable from zero in every one of those drops (Appendix Figure B5). Across all fifteen variants the preferred specification is not an outlier. Appendix Figure B4 reports the same variants for services and manufacturing.

Our estimates remain close to those of the main specification under three additional checks (Appendix Table B7). One concern is that the national shocks could reach a village directly, through the industries located in the village itself, rather than through the growth of nearby cities. We add a control for the village's own exposure to the shocks. This control sums the national growth rates of the industries in the

village, each weighted by the industry's share of village employment in 1998. The employment estimate moves from 0.359 to 0.337 and the population estimate from −0.100 to −0.099. Another concern is that national growth in the extraction and wholesale industries could raise village employment in the same industries directly. When we remove these industries from the instrument, the employment estimate moves from 0.359 to 0.340 and the extraction estimate from 0.519 to 0.432. A third concern is that our control for village population comes from the 2001 Census, three years into the period over which we measure employment growth. It could therefore capture part of the effect we estimate. When we replace it with village population from the 1991 Census, the employment estimate moves from 0.359 to 0.310 and the services estimate from 0.308 to 0.268.

A last concern is that the DDL boundaries are delineated for 2025, after our study period. A shrid that became urban during or after the period is therefore classified as urban for the whole period. This affects both which shrids we include in our village sample and the baseline industry composition of the cities behind the instrument. To address this concern, we adjust the definition of the city and count a shrid inside the DDL boundaries as part of a city only if it contained a statutory town or a census town in the 1991 Census.[10] Unlike in our main specification, we therefore include in the village sample the shrids inside the boundaries that contained no 1991 town, and we exclude from it the shrids outside the boundaries that contained one. We drop the cities without a 1991 town, as well as those whose 1991 towns had fewer than 10,000 residents in total or are all missing from the 1998 Economic Census. We then recompute each city's baseline employment, industry composition, and distance to each village, and rebuild the exposure measure and the instrument. Under the adjusted definition there are 1,202 cities, whose 1,888 shrids account for 91.8 percent of the 1998 employment of the cities in our main specification. The village sample now includes 10,674 shrids that our main specification treats as part of a city, most of them within a few kilometers of a city. The estimates have the same sign as in the main specification and are larger in magnitude (Appendix Table B7). The employment estimate rises from 0.359 to 0.607 and the services estimate from 0.308 to 0.471. The population estimate moves from −0.100 to −0.142.

4 Theoretical Framework

This section presents a simple model of how a growing city shapes the economy of the villages around it. Urban growth reaches a representative village through three channels: it pulls workers toward the city, it raises demand for the village's services, and it raises demand for the resource-based products that some villages extract from a fixed natural endowment. The first channel draws workers away from the village, while the other two raise the demand for what it produces. The same two forces appear in models of economic geography, where a growing center draws workers away from the places around it while giving them better access to its market (Krugman 1993; Fujita et al. 1999). Together these channels determine

[10] The Census classifies as a statutory town any place with a municipality, corporation, cantonment board, or notified town area committee. A census town is any other place with at least 5,000 residents, a population density of at least 400 per square kilometer, and at least 75 percent of its male main workers outside agriculture.

how the village's population, wages, and employment in farming, extraction, and services respond at each distance. Formal derivations and proofs are left to Appendix C.

4.1 Setup

The representative village lies at distance d from the city and produces three goods.[11] The *crop* good (agriculture) sells in national markets at a price normalized to one. The *resource* good (timber, forest products, quarried materials) is produced from a fixed natural endowment and sells to buyers in the city and the region around it, at a price p_R that rises as the city grows. In rural India, about a fifth of the fuelwood that households collect is consumed in nearby towns and cities (Bošković et al. 2023). The *non-tradable* good (retail, services, construction) is produced and consumed within the village, at a price p_N^V set by local supply and demand. The village has $\bar{N}$ workers, a fixed endowment of land $\bar{T}$, and a resource endowment $\bar{F}$, which can be zero. Only villages with $\bar{F} > 0$ operate an extraction sector. Each worker chooses among farming, extraction, local services, or migrating to the city. Those who stay, L_T in farming, L_R in extraction, and L_N in services, form the village population, $N_V = L_T + L_R + L_N$.

Villagers consume two of the three goods, the crop good and the non-tradable good, since the resource good is sold outside the village. They have Cobb–Douglas preferences over the two,

$$U(c_T, c_N) = c_T^{1-s}\, c_N^{s}, \qquad s \in (0,1), \tag{12}$$

where s is the non-tradable expenditure share.[12] The implied indirect utility is $V(w, p_N) = \ln w - s \ln p_N$, up to an additive constant.

As commonly assumed in models of agricultural production (Restuccia et al. 2008), the crop sector combines labor with the village's fixed land in a Cobb–Douglas production function,

$$Y_T = A_T\, L_T^{\alpha}\, \bar{T}^{1-\alpha}, \qquad \alpha \in (0,1). \tag{13}$$

Land is fixed, so each additional worker adds less output than the one before. The extraction sector uses a similar production function, with the resource endowment as the fixed factor,

$$Y_R = A_R\, L_R^{\gamma}\, \bar{F}^{1-\gamma}, \qquad \gamma \in (0,1), \tag{14}$$

and sells its output at the price p_R. The non-tradable sector uses only labor and a constant returns technology. We further assume free entry, such that the price equals the marginal cost, $p_N^V = w_V / A_N$, where A_N is labor productivity in the non-tradable sector. Land and the resource endowment earn the

[11] Each sector in the model is defined by where its output is sold. Village manufacturing has no single market of this kind. The Economic Census records under it both goods sold within the village, such as food processing and construction materials, and goods sold in wider markets. We therefore leave manufacturing outside the model and make no prediction for its response.

[12] Cobb–Douglas preferences hold this share fixed. Using household expenditure data from India's National Sample Survey, Comin et al. (2021) and Fan et al. (2023) both find instead that the share of spending on services rises with income. A rising share would give urban visitors, who are richer than villagers, a larger services share than residents and a larger weight in village services demand.

revenue that remains after wages are paid. For simplicity, we assume that both are owned by absentee landlords.

The crop price is set in national markets and is thus exogenous to local economies (Moretti 2011), implying that a village's crop employment responds to urban growth only through the village wage.

We assume that the village's distance to the city matters in three ways. First, it raises the cost of migrating. Following Bryan and Morten (2019) and Morten and Oliveira (2024), a worker who moves to the city gives up $\mu(d)$ in utility, a cost that is partly pecuniary (e.g., transport, resettlement) and partly not pecuniary (e.g., distance from family and community),[13] and that rises with distance without bound. Second, it makes it costly for urban consumers to reach the village. Services are consumed where they are produced, so the village sells them to city residents only when those residents travel to it, as in Faber and Gaubert (2019). Only a fraction $\phi(d)$ of the city's demand for services reaches the village, and that fraction falls with distance.[14] Third, distance weakens the response of the resource price to the city's growth, $a(d)$. Extraction in the Indian countryside is driven by urban and industrial demand (Asher et al. 2020), but a village also sells what it extracts to other buyers, such as smaller towns and local traders, whose prices do not move with the growth of this city. Shipping to the city is costly, increasingly so with distance, which implies that the growing city accounts for a smaller share of the sales of villages farther away. The price these villages receive therefore responds less to the city's growth. How far each demand channel reaches depends on how fast ϕ and a fall.

We do not model the city's economy in detail. It affects the village only through three objects: the urban wage w_U, the urban population N_U, and the urban services price p_N^U, each a function of urban productivity A_U. We study how an exogenous increase in A_U, which raises both N_U and w_U, affects village outcomes.

4.2 Equilibrium

Workers who remain in the village move freely across its three sectors. This implies that wages equalize at a single village wage w_V.[15] In crops and in extraction, this wage equals the marginal value product of labor. In services, it is set by local demand.

Each consumer spends the share s of income on services. The village's services are bought by two groups. The first is the village's own residents.[16] The second is the $\phi(d)N_U$ urban consumers who travel

[13] Imbert and Papp (2020) find that this cost is large in India even for seasonal moves. Seasonal migrants in their data accept earnings 35 percent lower on local public works rather than migrate.

[14] In practice, this term can represent urban residents who travel to the village and spend their income on the goods and services sold there. It can also reflect the spending of village residents who live close enough to the city to commute to jobs there and who spend their earnings in the village. Because we do not observe commuting, we remain agnostic about which of the two this term mainly captures.

[15] We assume that the village labor market clears. In the agricultural lean season, paid work in rural India can instead be rationed (Breza et al. 2021), and new demand for labor could then be met without a rise in the wage. Neither census records village wages, which means that our data cannot distinguish these two forms of adjustment.

[16] Residents' spending is an important source of demand for village enterprises. Egger et al. (2022) find that randomized cash transfers to households in rural Kenya raised the revenue of local enterprises, and that consumption rose among nearby households that received no transfer.

to the village. Both groups spend in proportion to their incomes, namely the village wage w_V for residents and the urban wage w_U for visitors.

Together these conditions yield the labor demanded by each sector, given the village and urban wages,

$$L_T = \left[\frac{\alpha A_T \bar{T}^{1-\alpha}}{w_V}\right]^{\frac{1}{1-\alpha}}, \qquad L_R = \left[\frac{\gamma p_R A_R \bar{F}^{1-\gamma}}{w_V}\right]^{\frac{1}{1-\gamma}}, \qquad L_N = sN_V + \frac{s\, w_U \phi(d) N_U}{w_V}. \tag{15}$$

Employment in each sector decreases in the village wage. The three sectors draw on the same pool of village labor, and the wage that clears that market is the one at which their combined demand equals the village population,

$$L_T + L_R + L_N = N_V. \tag{16}$$

If the village lies far enough from the city that no urban demand reaches it, services employ the share s of residents, regardless of the village's endowments. The remaining workers sort into crops or extraction, with the relative shares depending on the village's land and resource endowments. Land per worker also affects the level of the village wage. Land and labor are complements in crop production, which means that villages with more land per worker have a higher marginal product of labor and, in consequence, higher wages.

Each worker decides whether to stay in the village or move to the city, comparing the utility of the two locations. Village utility is $V^V = \ln w_V - s \ln p_N^V$, and urban utility net of the migration cost is $V^U = \ln w_U - \mu(d) - s \ln p_N^U$. Workers also differ in idiosyncratic tastes for the two locations, which we model with type I extreme-value shocks with dispersion $\sigma > 0$, in the tradition of discrete location choice models (Kennan and Walker 2011).[17] The share who stay in the village therefore follows the logit form $N_V/\bar{N} = [1 + \exp((V^U - V^V)/\sigma)]^{-1}$.

Combining this condition with equation (16), we obtain the village population,

$$N_V = \frac{\bar{N}}{1 + \exp\big([V^U - \mathcal{V}(N_V)]/\sigma\big)}, \qquad \mathcal{V}(N_V) := (1-s) \ln w_V(N_V) + s \ln A_N, \tag{17}$$

where $w_V(N_V)$ solves equation (16). A larger village population depresses the village wage, and with it village utility, which makes staying less attractive at the margin. The two sides of this equation therefore cross exactly once, and the village equilibrium exists and is unique for any strictly positive parameters (see Lemma 1 in Appendix C). All other village outcomes, including sectoral employment, wages, and welfare, are determined by N_V once the village's distance and the city's size are known. Applying the implicit function theorem to this equation yields the response of each to urban growth.

4.3 The Population Response

Consider a marginal increase in urban productivity A_U, holding the village-side parameters and the distance decay parameters fixed. Workers compare the utility of the two locations, both of which are positively

[17] We model the move to the city as permanent, abstracting from the temporary, seasonal, and circular migration (Bryan et al. 2014; Morten 2019), and from the effects of commuting on the village labor supply and on the decision to stay.

affected by urban growth. The village gains population if village utility rises by more than city utility, and loses population if city utility rises by more.

Two forces, both operating through the village wage, raise village utility. The first is urban demand for the village's services. An additional urban consumer who reaches the village spends $s\,w_U$ there. The second is urban demand for the village's resource output. A larger city raises the price the extraction sector receives, and therefore the value of what an extraction worker produces. Together the two forces raise the village wage by

$$\left.\frac{\partial w_V}{\partial N_U}\right|_{N_V} = \Phi(d) + \Psi(d), \qquad \Phi(d) = \frac{s\,w_U\,\phi(d)}{\Lambda}, \qquad \Psi(d) = \frac{a(d)\,w_V\,L_R}{(1-\gamma)\,\Lambda}, \tag{18}$$

where

$$\Lambda = \frac{L_T}{1-\alpha} + \frac{L_R}{1-\gamma} + (L_N - sN_V) \tag{19}$$

measures how much village labor demand contracts when the wage rises. The larger Λ is, the smaller the wage increase in response to a given rise in demand. The urban demand behind both channels fades with distance, since ϕ and a both fall. In villages without a resource endowment there is no extraction sector, and Ψ is zero at every distance.

A third force creates incentives to migrate by making the city more attractive. That increase in attractiveness is the urban wage gain net of the urban cost-of-living increase, per unit of urban population growth, namely

$$\tilde{M} = \frac{1}{w_U}\frac{dw_U}{dN_U} - s\,\frac{1}{p_N^U}\frac{dp_N^U}{dN_U}. \tag{20}$$

We assume $\tilde{M} > 0$, consistent with real urban wages rising with city size. Because the migration cost is not affected by urban productivity, the migration pull does not fade with distance, unlike the urban demand behind the two village-side channels.

Combining the three forces, village population rises with urban growth when the gain in village utility exceeds the migration pull, and falls when the pull is larger,

$$\frac{dN_V^*}{dN_U} > 0 \quad \Longleftrightarrow \quad (1-s)\,\frac{\Phi(d)+\Psi(d)}{w_V} > \tilde{M}. \tag{21}$$

The migration pull on the right of this comparison stays flat, while the part of the left side that comes from urban demand for services falls at every distance. The part that comes from resource demand falls as well, provided resource demand decays at least as fast as the extraction sector's weight in village labor demand rises with distance (see Section C4). Under that condition the two sides cross at most once, and they may never cross. In villages without a resource endowment they cross at most once with no condition.

To compare the combined strength of the demand channels with the migration pull, we evaluate them where the demand channels are strongest, at the city's edge, and take their ratio,

$$R := \frac{(1-s)\,[\Phi(0)+\Psi(0)]}{w_V\,\tilde{M}}. \tag{22}$$

where the village wage w_V is also evaluated at the city's edge. Two regimes follow from this comparison under the same condition (see Proposition 1 in Appendix C). When $R \leq 1$, the demand effect does not exceed the migration pull even at the city's edge. The two sides never cross, and village population falls at every distance beyond the city's edge. When $R > 1$, the demand effect is the larger of the two close to the city, and the two sides cross at a distance d^*. Village population rises in villages closer to the city than d^*, and falls in those farther away. A stronger demand effect places that crossover farther from the city, and a stronger migration pull places it closer.

Equation (21) sets the sign of the population response. Its size, in turn, depends on how many workers are close to the margin between staying and moving. That number falls with distance as long as most workers stay in the village and the migration cost rises with distance faster than village utility declines as urban demand for its services fades. Thus, far from the city, the fall in population shrinks toward zero (see Section C2).

Because the two demand channels can fade at different rates, the population response depends on both of them together, and does not disentangle the speed at which each one fades with distance. In order to separate them, we turn to the response of employment in each sector.

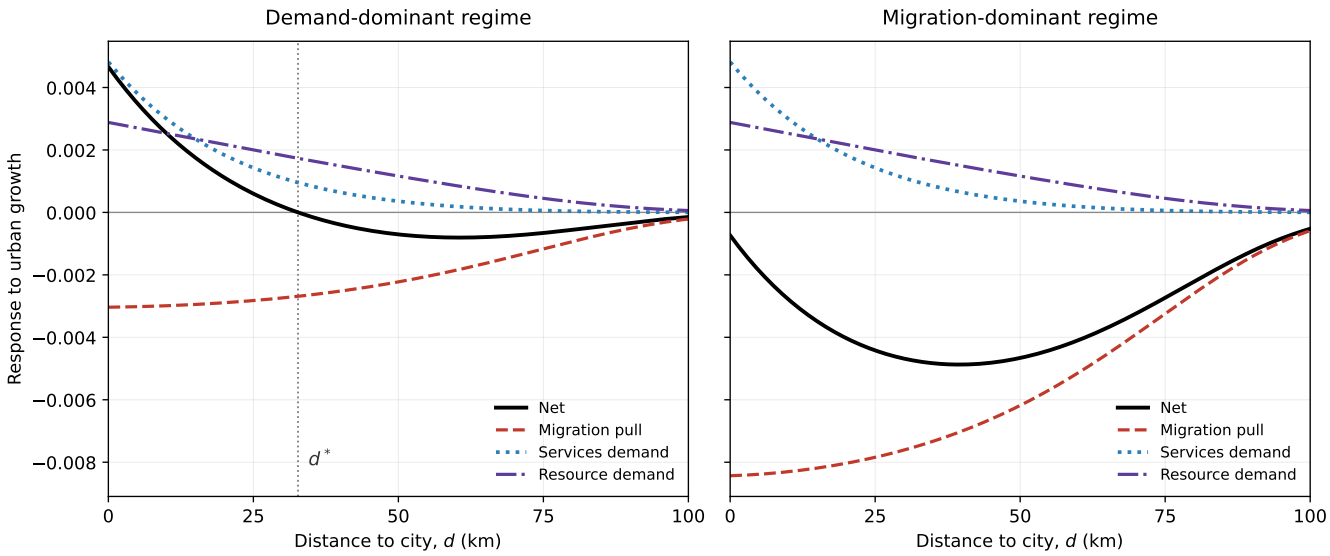
Figure 9 illustrates the two regimes of the population response. The two panels share the same demand channels and differ only in the strength of the migration pull. Each curve is the contribution of one of the model's three forces, with their sum in black. Even though the migration pull does not fade with distance, its contribution to the total effect shrinks far from the city, because rising migration costs leave few workers close to the margin between staying and moving. By contrast, the services and resource contributions shrink because urban demand for the village's services and for its resource output fades with distance. In the parameterization drawn, resource demand dies out more slowly than services demand. In the left panel the demand effect exceeds the migration pull near the city ($R > 1$). Population rises there and falls beyond the crossover d^*. In the right panel the migration pull exceeds the demand effect at every distance ($R < 1$). Population falls throughout. In this parameterization the decline is small near the city, where the demand channels offset most of the pull, largest at intermediate distances, and small again far out. The model can generate this pattern when the migration cost rises without bound and R is close to one. The distance at which the decline peaks depends on the parameters (see Section C2).

4.4 Wages and Sectoral Reallocation

The village wage rises with urban growth at every distance, even where population falls. This is driven by three different forces. The first is urban demand for the village's services, $\Phi(d)$. The second is urban demand for its resource output, $\Psi(d)$. Both raise the demand for village labor. The urban demand behind each of them fades with distance. The third force comes from migration. When the city draws a worker away, labor in the village becomes scarcer and the wage rises.[18] Imbert and Papp (2015) find that this is the mechanism through which public hiring under India's employment guarantee raised private rural wages. Urban growth makes the city more attractive by the same amount at every distance, but how much

[18] Hornbeck and Moretti (2024) document this effect across cities in the United States. When a city's productivity growth draws workers from other cities, real earnings rise for the workers who stay behind.

Figure 9: The two regimes of the population response to urban growth



Notes: The figure plots the three terms of the population response dN_V^*/dN_U in equation (31), evaluated at the model's equilibrium at each distance. The black line is their sum. Both panels use the same demand side, with services demand decaying at $\delta_\phi = 0.045$ and resource demand at $\delta_a = 0.012$ with $a_0 = 0.20$. The other parameters are $s = 0.25$, $\alpha = 2/3$, $\gamma = 0.6$, $\sigma = 0.25$, and $w_U = 2.5$; the migration cost is $\mu(d) = \mu_0 e^{\delta_\mu d}$ with $\mu_0 = 0.03$ and $\delta_\mu = 0.033$. The remaining parameters are set so that, in the baseline equilibrium, a village 16 km from the city, the median distance in the sample, retains 90 percent of its workers, urban visitors account for two percent of its employment, and the extraction sector employs five percent. The left panel assumes a migration pull that is weaker than the combined demand effect at the city's edge ($\tilde{M} = 0.009$, $R = 2.5$), while the right panel assumes a migration pull stronger than the demand effect at every distance ($\tilde{M} = 0.025$, $R = 0.9$).

labor it draws away depends on how many workers are close to the margin between staying and moving. That number falls with distance, because of rising migration costs. This third force therefore weakens far from the city as well. Appendix C derives the closed form and shows that $dw_V^*/dN_U > 0$ at every distance.

To assess how urban growth affects the welfare of the villagers who stay, we use the real wage, namely the village wage deflated by the prices of the two goods they buy, the crop good and services (see Section C3). Under this measure urban growth makes villagers better off at every distance, including where it draws people away.[19] Because the crop price is set in national markets, a higher wage buys more of the crop good. Meanwhile, even though the services price rises with the wage, services take only the share s of villagers' spending. Thus villagers keep the fraction $1 - s$ of any proportional rise in the village wage.

The rising wage reallocates labor across the village's three sectors. Crop employment contracts at every distance (see Proposition 2 in Appendix C). Because the village sells its crop output at a fixed national price, urban growth never shifts the demand for crop labor. With that demand unchanged, employment in the sector adjusts only to the wage,

$$\frac{dL_T^*}{dN_U} = -\frac{1}{1-\alpha}\,\frac{L_T^*}{w_V}\,\frac{dw_V^*}{dN_U} < 0. \tag{23}$$

[19] We do not observe village wages or prices. The closest evidence that a local demand shock raises them in a developing country is Aragón and Rud (2013), who find that a Peruvian gold mine's purchases of local inputs raised real income, rents, and local food prices in districts up to 100 kilometers away.

The wage rises for different reasons at different distances. Near the city, the increase comes both from urban demand for the village's services and for its resource output, which compete for village labor, and from the migration pull. Farther away, fewer workers remain close to the margin of migrating and the contribution of the migration margin dissipates. When the migration cost rises fast enough with distance, what is left is whichever of the two demand channels still reaches the village at that distance.

Extraction employment can rise or fall, depending on which of two effects dominates. Urban growth raises the price the sector receives for its output, which raises its demand for labor. Urban growth also raises the village wage, which reduces employment in the sector. Extraction employment rises when the first effect is larger, that is, when the response of the resource price is large enough, namely

$$\varepsilon_a(d) > T_R(d), \tag{24}$$

where $\varepsilon_a(d)$ is the elasticity of the resource price to city size, and $T_R(d)$ measures how much upward pressure the other two forces, urban demand for services and migration, put on the village wage at that distance (see Proposition 3 in Appendix C). Far from the city, where urban demand for services has faded and few workers remain close to the margin of migrating, $T_R(d)$ approaches zero. The response of the resource price fades with distance as well, which implies that extraction expands far from the city when that response fades more slowly than the other two forces, and contracts there when it fades faster than at least one of them.

Services employment can expand or contract, depending on the interaction of three forces. First, the change in population shifts residents' own demand for services. Second, the urban consumers who travel to the village add their demand to that of residents. This force is strongest near the city and fades with distance. Third, higher wages raise the cost of hiring the workers who serve urban visitors, since residents' own spending on services rises with the wage they earn. In a village that loses population and lies far enough from the city that urban demand for services no longer reaches it, none of these forces pushes services employment up. Out-migration to the city lowers residents' local demand. This contrasts with villages located near the city, where visitors' demand can outweigh the higher cost of hiring and migration-driven drops in demand, and services employment can rise even if population falls (see Proposition 4 in Appendix C).

4.5 Employment Composition at Longer Distances

Far enough from the city, urban demand for services no longer reaches the village and the fall in population becomes negligible. Urban growth then leaves population and services employment effectively unchanged, since the services sector at these distances only serves residents (and employs the share s of them).

Urban demand for the resource good can reach farther than the other two forces. Where that is the case, urban growth raises the resource price and the extraction sector expands. The village wage increases as a result, and the higher wage in turn leads crop employment to contract. Since urban growth has little effect on population and on services employment at larger distances, the additional extraction jobs come largely out of crop employment (see Proposition 5 in Appendix C). Urban growth at these distances therefore

changes which activities employ village workers rather than how many of them work. These responses are concentrated in villages with a resource endowment, since the other sectors' demand does not respond to city growth in villages this far out.

We make no assumption about how fast urban demand for the resource good fades relative to urban demand for services.

4.6 Implications

The model can account for two patterns found in Section 3.2. First, the negative effect of urban growth on village population can be largest at intermediate distances. Close to the city, urban demand for services offsets part of the migration pull. Far from the city, migration costs are high enough that few workers are on the margin between staying and moving. At intermediate distances, the village is close enough to the city that moving there is still relatively inexpensive, but too far to benefit directly from urban demand. Whether the overall effect on population is most negative at these distances depends on the relative strength of these forces (see Section C2 in Appendix C). Such a pattern is an agglomeration shadow at intermediate distances, of the kind that Hornbeck et al. (2024) identify around ancient ports and Tang et al. (2025) around China's large cities. Second, non-farm employment rises even where population falls. The services sector near the city sells to urban visitors as well as to residents, which means that villages can gain non-farm jobs while losing people.

The model also has implications for how employment in each sector responds to urban growth. First, crop employment contracts at every distance (Proposition 2 in Appendix C). Urban growth raises the village wage at every distance, through urban demand for the village's services and resource output and through the labor scarcity that the migration pull creates. Crop employment responds only to that higher wage. Since the crop price is set in national markets, urban growth does not raise the value of what crop workers produce. The crop sector therefore employs fewer workers at the higher wage, making labor available to the village's other sectors or to further migration to the city.

Second, extraction employment expands where the rise in the resource price outweighs the rise in the village wage (Proposition 3 in Appendix C). A higher resource price raises the value of what an extraction worker produces, while urban demand for services and the migration pull raise the wage the sector must pay. Close to the city, both the resource price and the village wage respond strongly to urban growth. Farther away, both responses weaken. Extraction can therefore respond more strongly to more distant urban growth, as long as the response of the village wage weakens with distance by more than the response of the resource price. These responses arise only in villages with a resource endowment, $\bar{F} > 0$.

5 Additional Results and Mechanisms

This section explores the mechanisms behind the main results in light of the model. We first decompose the employment effects, then explore how farming and extraction respond in light of the model, then turn to village residents, and close by asking how the results vary across villages.

5.1 The Composition of the Employment Effects

The positive employment effect comes in large part from the activities that serve local demand. Muralidharan et al. (2023) show that when a reform raised private wages and incomes in rural India, village non-farm enterprises grew despite the higher wages, which they attribute to local demand. Table 5 estimates the employment effect separately for seven sectors that together account for all Economic Census employment in our sample. The services category is defined as service activities other than trade and transport, including hotels and restaurants, personal services, finance, business services, education, and health. Among these activities, the largest responses are in trade and services, with 2SLS estimates close to 0.29 in both cases, followed by construction at about half that size.[20]

Table 5: Employment Effects by Sector

	Trade	Services	Construction	Resource extraction	Manufacturing	Transport	Facilities
Urban emp growth	0.285***	0.290***	0.145**	0.519***	-0.079	-0.069	-0.008
	(0.081)	(0.083)	(0.064)	(0.104)	(0.111)	(0.088)	(0.035)
AKM0 95% CI	[-0.125, 1.063]	[-0.028, 0.649]	[-0.415, 0.704]	[0.203, 1.757]	[-1.505, 0.847]	[-0.747, 0.317]	[-0.204, 0.229]
RI p-value	0.152	0.043	0.512	0.009	0.879	0.694	0.942
Mean dep. var.	0.248	0.464	0.172	0.140	-0.064	0.377	0.149
N	256,167	256,167	256,167	256,167	256,167	256,167	256,167

Notes: Each column reports the 2SLS estimate of the effect of nearby urban employment growth on the log difference of Economic Census employment in the indicated sector between 1998 and 2013, under the same specification, sample (N = 256,167), and controls as Table 3. The seven sectors partition total Economic Census employment. The resource extraction sector covers forestry and logging, fishing, oil and gas, and mining and quarrying establishments, with forestry and logging the dominant component; it contains no crop or livestock agriculture. Standard errors in parentheses are clustered at the subdistrict level. The two bottom rows report the null-imposed AKM0 95% confidence interval and the randomization-inference p-value, defined as in Table 4; Appendix Table B8 reports these together with the conventional and exposure-robust standard errors. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Manufacturing, transport, and facilities are indistinguishable from zero on average.[21] A shock that starts in the countryside can move labor the other way: in Brazil, labor-saving soy technology moved workers from agriculture into local manufacturing and left services unchanged (Bustos et al. 2016), and in India a wetter monsoon raises village employment in manufacturing as well as in construction (Emerick 2018). The one sector that does not fit the local-demand pattern is resource extraction, primarily forestry, which is the largest estimate in the table. Unlike trade and services, the extraction effects are larger for urban growth at greater distances (Appendix Table B9). To measure each sector's contribution to the far-ring effect, we break total employment growth into the sum of sector contributions and re-estimate the band specification on each sector separately (Appendix Table B10). Resource extraction accounts for about two-thirds of the 80–100 km response, while the trade and services contributions are small. The far-ring employment estimates in Figure 7 therefore come mostly from extraction, with a sizable contribution from manufacturing in the outermost ring, rather than from the services demand channel. The extraction

[20] The broad services aggregate used in Table 3 combines this category with trade and transport.

[21] As noted in Section 4.1, the model does not include manufacturing. The crop logic applies to the part of its output sold in wider markets and the services logic to the part sold locally. The Economic Census does not record how much of each a village produces.

estimate also remains significant under the alternative inference approaches of Section 3.3, reported in the bottom rows of the table.

If demand from nearby cities drives the employment effect, it should be concentrated in private establishments rather than in government ones. Table 6 tests this idea, estimating the employment effect separately by ownership, by whether workers are hired for a wage, and by establishment size. The point estimate for private employment is larger, though the two are not statistically distinguishable. To see whether the new jobs come from employers or from villagers working on their own account, we separate workers hired for a wage from the self-employed and unpaid family workers. The effect is positive in both cases. Employment also grows in establishments with 20 or more workers, which usually serve markets beyond the village, and these estimates are positive in every distance band (Appendix Table B9). The large-establishment effect also remains significant under the alternative inference approaches of Section 3.3.

Table 6: Employment Effects by Establishment Type

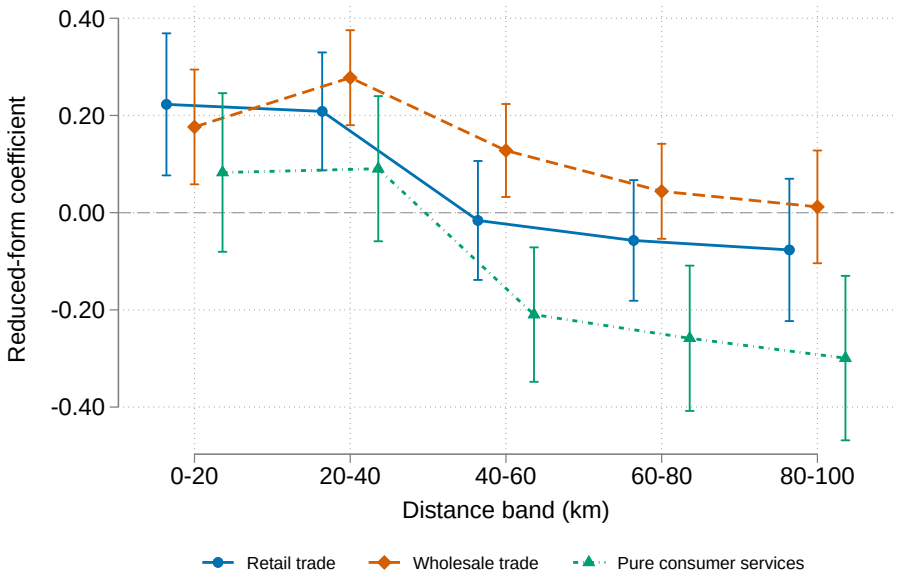
	Private	Government	Hired	Non-hired	20+ establishments
Urban emp growth	0.334***	0.208**	0.236***	0.324***	0.361***
	(0.104)	(0.099)	(0.083)	(0.115)	(0.090)
AKM0 95% CI	[-0.053, 0.980]	[-0.520, 0.614]	[-0.038, 0.568]	[-0.208, 1.016]	[0.075, 0.793]
RI p-value	0.053	0.357	0.057	0.129	0.016
Mean dep. var.	0.203	0.247	0.443	0.182	0.224
N	256,167	256,167	256,167	256,167	256,167

Notes: Each column reports the 2SLS estimate of the effect of nearby urban employment growth on the log difference of Economic Census employment in the indicated category between 1998 and 2013, under the same specification, sample (N = 256,167), and controls as Table 3. The categories split the same total employment as Table 5, by ownership, by hiring status, and by establishment size; non-hired workers are own-account and unpaid family workers, and the last column is employment in establishments with 20 or more workers. Standard errors in parentheses are clustered at the subdistrict level. The two bottom rows report the null-imposed AKM0 95% confidence interval and the randomization-inference p-value, defined as in Table 4; Appendix Table B8 reports these together with the conventional and exposure-robust standard errors. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Figure 10 estimates the employment effect by distance band separately for three groups of activities that sell to final consumers: retail trade, wholesale trade, and repair, restaurants, and personal and social services. The estimates for retail and wholesale trade are positive for urban growth in the nearest bands, and close to zero for growth beyond 40 kilometers in retail and beyond 60 kilometers in wholesale. This pattern is consistent with the model's services channel, given that urban demand for village services declines with distance. The fact that the response extends 40 to 60 kilometers is in line with recent evidence from smartphone location data in three African countries showing that consumers travel tens of kilometers to purchase services (Blanchard et al. 2025). Repair, restaurants, and personal and social services follow a different pattern. Their estimates are close to zero for urban growth within 40 kilometers and negative for growth farther away. Unlike retail and wholesale trade, which can also sell to urban visitors, these activities sell mostly to the village's own residents, whose number grows more slowly as nearby cities grow. In the

model, employment in services sold to residents moves with village population, which responds negatively to urban growth 40 to 80 kilometers away.

Figure 10: Consumer Services by Distance Band



Notes: Reduced-form coefficients on the Bartik instrument interacted with 20-km distance bands, from the same joint five-band specification as Figure 7, for the three mutually exclusive components that together make up consumer-facing trade and services: retail trade, wholesale trade, and pure consumer services (repair, restaurants, and personal and social services). Appendix Table B17 reports average effects for pure consumer services alone, for that category combined with retail trade, and for the three categories together. Retail trade is excluded from the instrument's shift component, so the retail series has no mechanical link to the instrument. Standard errors clustered at the subdistrict level; bars are 95% confidence intervals. Appendix Table B12 reports exposure-robust standard errors, AKM0 confidence sets, and randomization-inference p-values for these coefficients.

5.2 Farming and Extraction in Light of the Model

We next examine the two sectors whose output is sold outside the village, crops and extraction. In the model, crop employment contracts at every distance, because urban growth raises the village wage while leaving the crop price unchanged. We measure crop employment with the number of agricultural laborers, the farm workers hired for a wage, whose employment depends most directly on the village wage. The estimates in panel (a) of Figure 11 are negative for urban growth at every distance, consistent with the model, and largest for growth 40–60 km away.

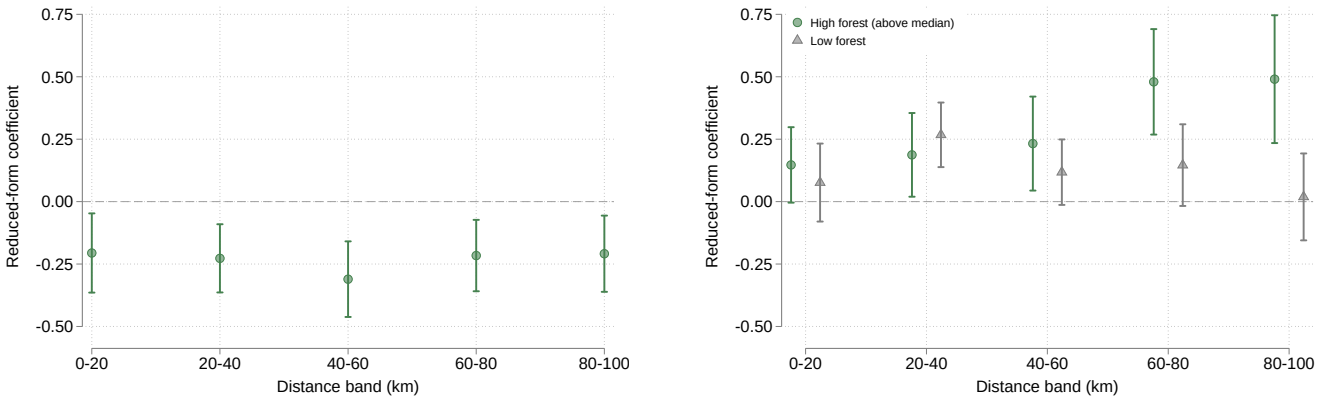
The model also implies that the extraction response requires a resource endowment. To examine this, we estimate the extraction response separately for villages above and below the median of baseline forest cover, our proxy for the endowment.[22] Within 60 km, the two groups respond similarly (panel (b)). Beyond that distance, extraction grows more in high-forest villages, with estimates of 0.48 and 0.49 in the two farthest bands, while the estimates for low-forest villages are much smaller.

[22] Baseline forest cover is the village's mean tree cover in 2001 from the MODIS Vegetation Continuous Fields data distributed with SHRUG, split at the sample median. Land abundance cannot stand in for the resource endowment, because it conflates that endowment with the crop land endowment, a fixed factor in a different sector.

Figure 11: Agricultural Laborers and Extraction by Forest Cover

(a) Agricultural laborers

(b) Extraction, by baseline forest cover



Notes: Each point is the reduced-form coefficient on the band-specific Bartik instrument from one joint 5-band regression, with the same specification and controls as Figure 7. Panel (a): growth in the number of agricultural laborers (Population Census main workers, 2001–2011), full sample. Panel (b): growth in extraction employment (the Economic Census agriculture-and-mining bin, primarily forestry, 1998–2013), estimated separately for villages above and below the sample median of baseline forest cover (mean 2001 tree cover from MODIS Vegetation Continuous Fields; median 5.6 percent). Vertical bars show 95% confidence intervals; standard errors clustered at the subdistrict level. Appendix Table B12 reports exposure-robust standard errors, AKM0 confidence sets, and randomization-inference p-values for these coefficients.

5.3 Effects on Village Residents

The estimates so far count jobs at establishments located in the village. Because workers can commute, a place's employment and its resident workforce need not move together (Monte et al. 2018). Table 7 turns to the people who live there, using the Population Census, which records how many residents each village has and how many of them work. Both grow more slowly where exposure to urban growth is higher, and the resident workforce grows more slowly than the population, so a smaller share of residents work. The male population responds almost exactly as the total population does, which is consistent with workers leaving together with their households.

Table 8 reports the effects on residents' work separately for women and men. Exposure lowers the female share of main workers. This is in line with what Gallé et al. (2024) find around India's special economic zones, where the decline in agricultural employment is concentrated among women and the increase in services employment among men.

Panel B turns to jobs at establishments located in the village, measured with data from the Economic Census. The employment estimates for women and men are positive and of similar size. The female share of main workers and men's establishment employment also remain significant under the alternative inference approaches of Section 3.3.

Table 7: Resident Population and Work

						Main workers by sector	
	Population	Male pop.	Workforce	Main workers	Marginal workers	Farm	Non-farm
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Reduced Form							
Bartik instrument	-0.148***	-0.150***	-0.230***	-0.037	-0.193	-0.152*	0.299***
	(0.019)	(0.019)	(0.032)	(0.058)	(0.128)	(0.087)	(0.078)
Panel B: 2SLS							
Urban emp growth	-0.100***	-0.101***	-0.155***	-0.025	-0.131	-0.102*	0.201***
	(0.016)	(0.016)	(0.026)	(0.039)	(0.088)	(0.061)	(0.058)
AKM0 95% CI	[-0.335, -0.009]	[-0.351, -0.008]	[-0.646, -0.037]	[-1.282, 0.355]	[-0.807, 1.549]	[-1.961, 0.544]	[-0.154, 0.944]
RI p-value	0.046	0.054	0.023	0.929	0.759	0.809	0.280
N	256,167	256,167	256,167	256,167	256,167	256,167	256,167
Mean dep. var.	0.134	0.132	0.139	0.059	0.159	0.005	0.115

Notes: Outcomes are log differences of Population Census resident counts between 2001 and 2011, estimated on the same sample and with the same specification and controls as Table 3. The workforce is the sum of main workers, who worked at least six months of the reference year, and marginal workers, who worked less than six months. Because each column is a regression on a separate log change, and the log change in the workforce weights its two components differently in each village, the workforce estimate need not lie between the estimates for main and marginal workers. Farm main workers are cultivators and agricultural laborers; non-farm main workers are those in household industry and all other work. Standard errors in parentheses are clustered at the subdistrict level. For each column, the table also reports the null-imposed AKM0 95% confidence interval and the randomization-inference p-value computed at the shock level, defined as in Table 4. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

5.4 Heterogeneity by Village Characteristics

Table 9 examines which villages drive these effects, splitting the sample at the median of two baseline characteristics and estimating the effects separately for each half. The cultivator share (i.e., the share of resident main workers who cultivate land, whether as owners, tenants, or sharecroppers) captures the potential role of access to land in shaping the outcomes, while population density captures how scarce village land is.

The results in Panel A show that the population and employment responses are similar in villages with high and low cultivator shares. Services employment, however, grows less where the cultivator share is high. A possible explanation is that, to take up a non-farm job, a cultivator would have to leave land idle or pay others to work it, while an agricultural laborer would only change employers. Consistent with this, Fernando (2022) finds that inheriting agricultural land reduces entry into non-farm work. Next, we explore whether slower population growth reflects fewer births rather than migration. We find that the share of children in the population barely responds to exposure, and children's numbers decline somewhat more than the total population (Appendix Table B18). This suggests that the population result reflects out-migration rather than lower fertility. The share of children aged 0 to 6, however, also responds to child survival and to the migration of families with children.

Table 8: The Gender Recomposition of Village Work

	Female	Male
	(1)	(2)
Panel A: Resident main workers (Population Census, 2001–2011)		
Main workers	-0.634***	0.121***
	(0.089)	(0.036)
AKM0 95% CI	[-2.874, 0.103]	[-0.846, 0.428]
RI p-value	0.097	0.580
Mean dep. var.	0.190	0.047
Farm main workers	-0.753***	0.012
	(0.118)	(0.053)
AKM0 95% CI	[-3.628, 0.326]	[-1.591, 0.579]
RI p-value	0.176	0.965
Mean dep. var.	0.118	0.002
Non-farm main workers	-0.123**	0.299***
	(0.061)	(0.061)
AKM0 95% CI	[-0.813, 0.404]	[-0.138, 1.287]
RI p-value	0.619	0.215
Mean dep. var.	0.380	0.056
Female share of main workers	-0.122***	
	(0.014)	
AKM0 95% CI	[-0.375, -0.014]	
RI p-value	0.032	
Mean dep. var.	0.010	
N	252,197	
N	256,167	
Panel B: Establishment employment (Economic Census, 1998–2013)		
Employment	0.305***	0.361***
	(0.106)	(0.086)
AKM0 95% CI	[-0.485, 1.214]	[0.154, 0.727]
RI p-value	0.362	0.010
Mean dep. var.	0.709	0.090
Female-owned establishments	-0.322***	
	(0.102)	
AKM0 95% CI	[-1.903, 0.798]	
RI p-value	0.626	
Mean dep. var.	0.866	
N	256,167	

Notes: Each entry is the 2SLS estimate of the effect of nearby urban employment growth on the indicated outcome for the column's gender, under the same specification, sample, and controls as Table 3. The two panels measure work in different frames and are not directly comparable: Panel A counts village residents by working status from the Population Census, while Panel B counts persons engaged at establishments located in the village from the Economic Census. Outcomes are log differences of counts, except the female share of main workers, which is the 2001–2011 change in the share itself and is estimated on the 252,197 villages where the share is defined in both years. Standard errors in parentheses are clustered at the subdistrict level. Each row also reports the null-imposed AKM0 95% confidence interval and the randomization-inference p-value computed at the shock level, defined as in Table 4. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 9: Heterogeneous Effects by Village Characteristics

	Pop. growth	Emp. growth	Services emp.
	(1)	(2)	(3)
Panel A: Cultivator share (2001)			
High	-0.130***	0.528***	0.240**
	(0.026)	(0.177)	(0.111)
Low	-0.164***	0.642***	0.679***
	(0.017)	(0.131)	(0.106)
High − low	0.035	-0.114	-0.439
	(0.060)	(0.303)	(0.122)
RI p-value, equal effects	0.678	0.758	0.040
N	256,167	256,167	252,670
Panel B: Population density (2001)			
High	-0.101***	0.308**	0.386***
	(0.019)	(0.135)	(0.124)
Low	-0.144***	0.640***	0.349***
	(0.029)	(0.197)	(0.112)
High − low	0.043	-0.332	0.037
	(0.106)	(0.220)	(0.236)
RI p-value, equal effects	0.790	0.210	0.883
N	256,161	256,161	252,664
Mean dep. var.	0.134	0.202	0.359

Notes: Same specification and sample as Table 3: reduced-form regression of each outcome on the retail-exclusion shift-share instrument, with main-specification controls and subdistrict-clustered standard errors. Each panel splits villages at the median of one baseline characteristic and reports the reduced-form coefficient in the high and low subsamples. The high-minus-low row reports the difference between the two coefficients, which comes from a pooled specification that interacts the instrument and every control with the group indicator. The number in parentheses below the difference is its exposure-robust standard error. The next row reports the randomization-inference p-value for the null that the effect is the same in both groups. The exposure-robust first-stage F statistic is about 12 in villages with a high cultivator share and 10 in those with a low share, and about 1 in high-density and 44 in low-density villages. The cultivator share is the share of cultivators among resident main workers in 2001; the density split drops the six villages with missing baseline density. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Panel B shows that the employment estimate is about twice as large in villages with low population density, where land is more abundant.[23] To see which sectors drive this difference, we estimate the employment effect by sector separately for high-density and low-density villages (Appendix Table B19). By

[23]Population density is an imperfect measure of land abundance because it also reflects village size. In our sample it captures primarily variation in land area: the correlation of log density with the log area of the locality polygon is −0.63 (versus 0.36 with log population), and the regressions control for baseline population. Land abundance is also distinct from the model's resource endowment. The crop land endowment and the resource endowment are separate fixed factors in separate sectors. Whether a village can expand extraction depends on the resource endowment, which Section 5.2 measures directly with forest cover.

far the largest difference is in resource extraction, which grows much more in land-abundant villages. Land abundance is therefore associated with higher employment in extraction, the same sector behind the employment gains from urban growth at the greatest distances in our rings specification.

6 Conclusion

This paper shows that growing cities in India both slow population growth in the villages around them and raise their non-farm employment. Using information on both outcomes for roughly 256,000 villages across India, we identify these effects from variation in nearby urban growth driven by national industry trends. A 10-percentage-point increase in nearby urban employment growth lowers village population growth by about 1 percentage point over the decade, relative to a mean of 13.4 percent, while increasing non-farm employment growth by about 3.6 percentage points, relative to a mean of 20.2 percent. Urban growth raises services employment and has no detectable effect on manufacturing.

The composition of the employment gains also varies with distance from the city. Nearby, the estimates are largest in trade and services, consistent with urban demand reaching the village. Farther away, the gains are concentrated in resource extraction, primarily forestry. A simple model can account for these patterns. Urban growth pulls workers toward the city at every distance, while the demand it creates for village services and resource products weakens with distance. Where this demand is strong enough, village employment can grow even as population falls. Depending on the relative strength of these forces, the negative effect on population can also be largest at intermediate distances, where villages are close enough for workers to move to the city but too far away to benefit directly from urban demand.

Overall, our results suggest that a growing city can pull workers from surrounding villages while also creating demand for what those villages sell. The first force slows village population growth, while the second creates non-farm jobs. Whether villagers themselves end up better off is a question our data cannot answer directly, because wage and income data are not available at the village level for the years we study. In villages more exposed to nearby urban growth, however, we do observe that village establishments add jobs even as population growth slows. This points to one way in which villages can benefit from nearby urban growth, even when some of their residents leave. A growing city is thus not only a destination for workers who leave surrounding villages but also a source of jobs for those who stay.

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A Data appendix

Table A1: Extended Summary Statistics

	N	Mean	Std Dev	p10	p50	p90
Panel A: Population & Demographics (2001–2011)						
Population growth (log change)	256,167	0.134	0.156	-0.028	0.131	0.300
Male population (log change)	256,167	0.132	0.163	-0.038	0.129	0.306
Children population (log change)	256,144	-0.066	0.296	-0.387	-0.066	0.254
Children share (share diff)	256,144	-0.030	0.029	-0.065	-0.029	0.004
Cultivators (log change)	250,473	-0.165	0.830	-1.099	-0.081	0.674
Female share of main workers (share diff)	252,197	0.010	0.156	-0.186	0.011	0.203
Panel B: Employment & Firms (1998–2013)						
Total employment (log change)	256,167	0.202	1.035	-1.143	0.257	1.459
Services employment (log change)	252,670	0.359	0.962	-0.869	0.392	1.537
Manufacturing employment (log change)	190,445	0.069	1.295	-1.558	0.000	1.705
Panel C: Structural Transformation						
Services share change	233,788	0.066	0.292	-0.294	0.057	0.440
Manufacturing share change	189,885	-0.075	0.254	-0.401	-0.049	0.192
Panel D: Controls & Baselines						
Log population 2001	256,167	7.138	0.873	6.043	7.132	8.235
Log male population 2001	256,167	6.469	0.870	5.366	6.465	7.573
Log total employment 1998	256,167	3.792	1.059	2.565	3.584	5.273
Log density 2001	256,161	5.890	0.964	4.725	5.862	7.109
Latitude	256,167	22.688	4.953	15.212	23.221	28.319
Longitude	256,167	80.626	5.420	74.380	79.496	88.067
Distance to nearest city (km)	256,167	19.488	13.356	6.000	16.000	38.000
Population of nearest city	256,163	2.71e+05	8.19e+05	33065.000	76709.000	4.39e+05
Panel E: Heterogeneity Variables						
Services share 1998	256,167	0.371	0.245	0.059	0.348	0.714
Resource extraction share 1998	256,167	0.025	0.117	0.000	0.000	0.010
Cultivator share 2001	256,167	0.486	0.247	0.157	0.477	0.833
Ag laborer share 2001	256,167	0.242	0.204	0.009	0.202	0.537
Female share of main workers 2001	256,167	0.239	0.163	0.035	0.222	0.468
Literacy rate 2001	256,167	0.594	0.159	0.378	0.608	0.790
SC share 2001	256,167	0.183	0.178	0.000	0.142	0.415
ST share 2001	256,167	0.135	0.262	0.000	0.001	0.559
Has power supply 2001	252,851	0.873	0.333	0.000	1.000	1.000
Has paved road 2001	252,019	0.675	0.469	0.000	1.000	1.000
Panel F: Instruments						
Urban employment exposure	256,167	0.323	0.365	0.054	0.338	0.660
Bartik instrument, retail exclusion	256,167	0.334	0.061	0.260	0.345	0.398
1 minus sum of shares, retail exclusion	256,167	0.282	0.060	0.206	0.283	0.361
Bartik instrument, agriculture/wholesale/retail exclusion	256,167	0.324	0.061	0.251	0.335	0.388

Notes: Sample restricted to the main estimation sample ($\approx$ 256,000 villages). All continuous outcomes are winsorized at the 1st and 99th percentiles. Log-change variables are $\ln(Y_{t+1}+1)-\ln(Y_t+1)$. Panels: (A) population and demographics (2001–2011 Census); (B) employment (1998–2013 Economic Census); (C) structural-transformation share changes; (D) baseline controls and geography; (E) baseline shares and heterogeneity variables (the splits in Table 9 use the cultivator share and log density; the others serve the balance tests of Table 2); (F) shift-share instrument components under two exclusion-set variants (retail, and agriculture/wholesale/retail).

Table A2: Data Sources and Core Variable Definitions

Construct	Source	Years	Definition and construction
Geographic units and urban-rural classification			
Time-consistent localities	SHRUG	1990–present	SHRUG harmonizes villages and towns into time-consistent polygons so outcomes and covariates are comparable over time at fixed geographic units.
Cities (urban agglomerations)	DDL urban boundaries + SHRUG overlay	v0.1 boundaries (2025 reference)	Any locality whose polygon intersects a DDL urban-boundary polygon is classified as urban and aggregated to the urban-boundary level.
Villages (rural localities)	DDL urban boundaries + SHRUG overlay	v0.1 boundaries (2025 reference)	All SHRUG localities outside the urban-boundary polygons, regardless of administrative designation.
Village-to-city distance	SHRUG geometry + DDL classification	Static	Geodesic (great-circle) distance from the village centroid to the centroid of the closest urban neighborhood within a city.
Population outcomes and baseline characteristics			
Population level	Population Census (via SHRUG)	2001, 2011	Total resident population in each locality.
Population growth	Population Census (via SHRUG)	2001–2011	Long difference in log population between 2001 and 2011.
Demographic composition	Population Census (via SHRUG)	Baseline (2001)	Population shares by sex, age, social groups, and literacy; used in the summary statistics, the balance tests, and the heterogeneity splits.
Paved road access	Population Census Village Directory (via SHRUG)	Baseline (2001)	Indicator for whether the village has a paved approach road; used in the summary statistics and the balance tests.
Geographic controls	SHRUG geometry	Static	Latitude and longitude of the locality centroid.
Employment outcomes and economic structure			
Non-farm employment level	Economic Census (via SHRUG)	1998, 2013	Total employment in non-farm establishments aggregated to each locality under SHRUG's harmonized industry system.
Employment growth	Economic Census (via SHRUG)	1998–2013	Long difference in log non-farm employment between 1998 and 2013.

Continued on next page.

Table A2 continued.

Construct	Source	Years	Definition and construction
Sectoral employment and growth	Economic Census (via SHRUG)	1998–2013	Employment in manufacturing and services and corresponding long differences; used in sectoral spillover analysis.
Establishment size	Economic Census (via SHRUG)	1998, 2013	Employment in establishments with 20 or more workers; used as an outcome in the employment decomposition.
Urban growth exposure and shift-share instrument			
Urban employment growth exposure	Economic Census + DDL boundaries + SHRUG	1998–2013	For each village, a distance-weighted average of employment growth across all cities within 100 km, with weights increasing in baseline city employment and decaying with distance.
Shift-share instrument	Economic Census industry composition and national shocks	1998–2013	Bartik-style predictor combining baseline city industry shares with national industry growth shocks, excluding selected industry groups to reduce the risk that measured urban growth responds to rural shocks; aggregated to villages using the same spatial weights as the exposure measure.

Notes: Outcomes are constructed as long differences in logs: population changes are measured over 2001–2011, while employment changes are measured over 1998–2013. Cities are defined using the DDL urban boundaries and villages are SHRUG localities outside those boundaries. Village-to-city distance is computed as the minimum geodesic (great-circle) distance from the village centroid to any urban neighborhood centroid within a city. In the main regressions, outcomes and baseline covariates are winsorized at the 1st and 99th percentiles, while the urban employment exposure measure and the shift-share instrument are left unwinsorized.

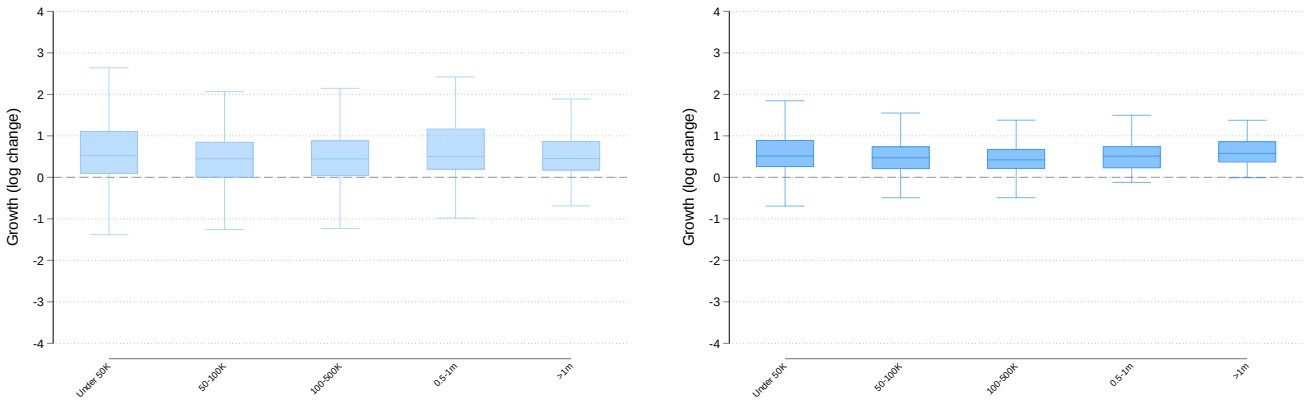
B Additional figures and tables

B1 Additional figures

Figure B1: Sectoral Growth Across the Urban Hierarchy: Cities

(a) Manufacturing

(b) Services

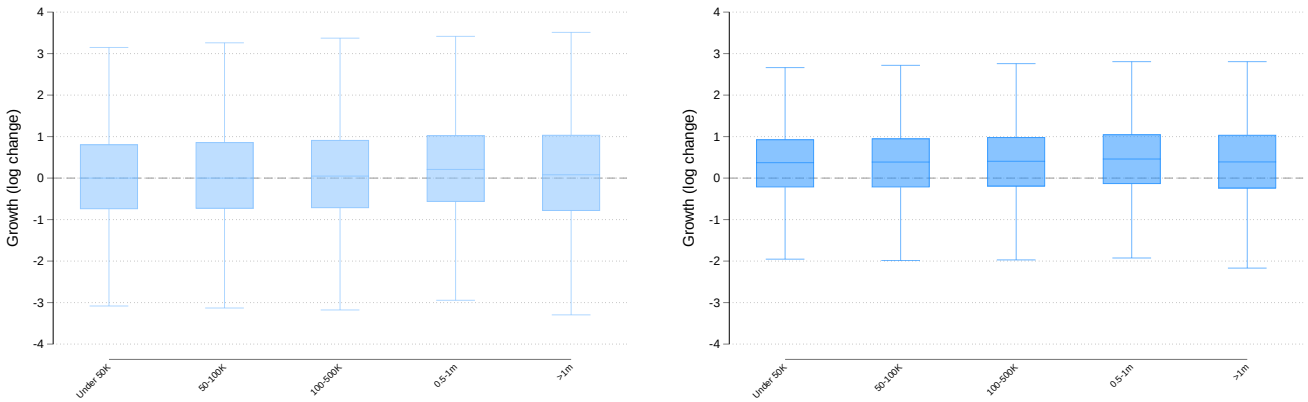


Notes: Non-farm employment growth 1998–2013 (long difference in log employment, using $\ln(x+1)$) for DDL cities, by baseline (2001) size class, split by sector: (a) manufacturing, (b) services. Both panels share a common vertical scale. Boxes span the interquartile range with the median; whiskers the adjacent values; outside values omitted; the dashed line marks zero. The figure covers the 1,443 cities that enter the urban-growth exposure (Table B1), of which 1,437 have sectoral employment in both years. Companion to the village version, Figure B2.

Figure B2: Sectoral Growth Across the Urban Hierarchy: Villages

(a) Manufacturing

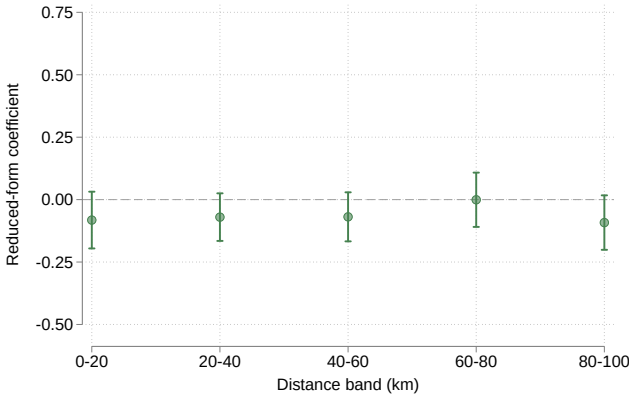
(b) Services



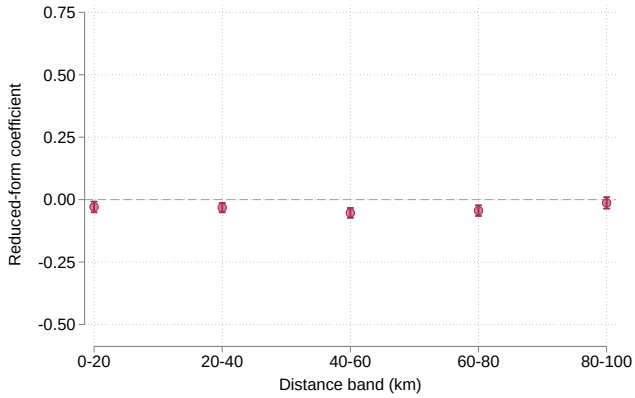
Notes: Non-farm employment growth 1998–2013 (long difference in log employment, using $\ln(x+1)$) for the ∼256K in-sample villages, by nearest-city baseline (2001) size class, split by sector: (a) manufacturing, (b) services. Both panels share a common vertical scale (matching Figure B1). Boxes span the interquartile range with the median; whiskers the adjacent values; outside values omitted; the dashed line marks zero.

Figure B3: Reduced-Form Effects by Distance Band: Secondary Outcomes

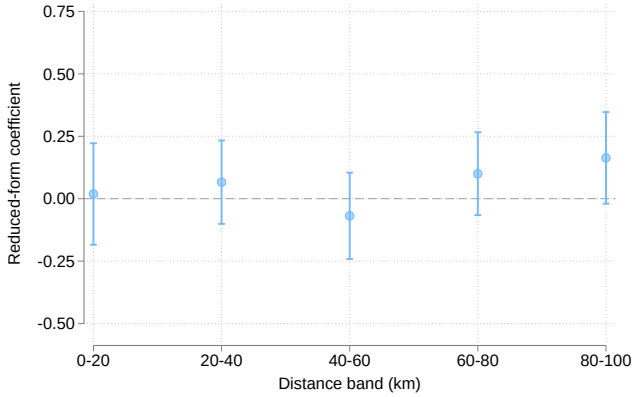
(a) Cultivator population



(b) Male population



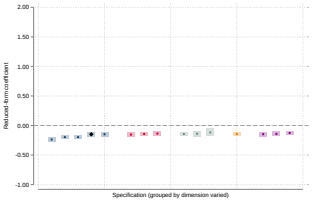
(c) Manufacturing employment



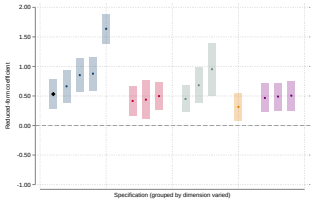
Notes: Companion to Figure 7. Each panel plots the reduced-form coefficient on the band-specific Bartik instrument from one joint 5-band regression of the corresponding outcome. Specification and controls as in Figure 7. Cultivators and male population are 2001–2011 changes; manufacturing employment is a 1998–2013 change. Vertical bars show 95% confidence intervals; standard errors clustered at the subdistrict level.

Figure B4: Specification-Curve Robustness: All Main Outcomes

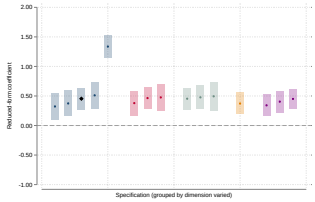
(a) Population growth



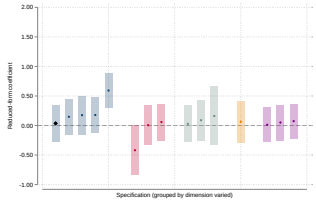
(b) Employment growth



(c) Services emp.



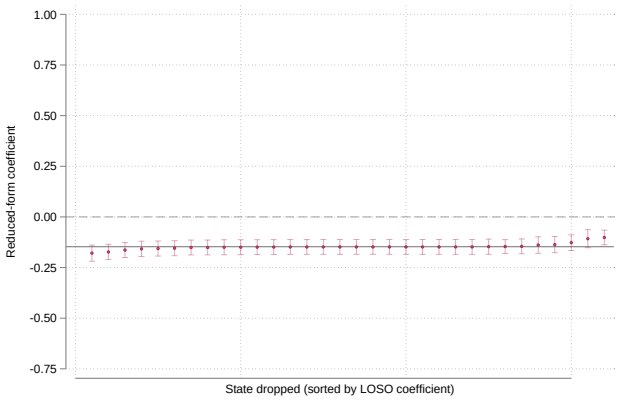
(d) Manuf. emp.



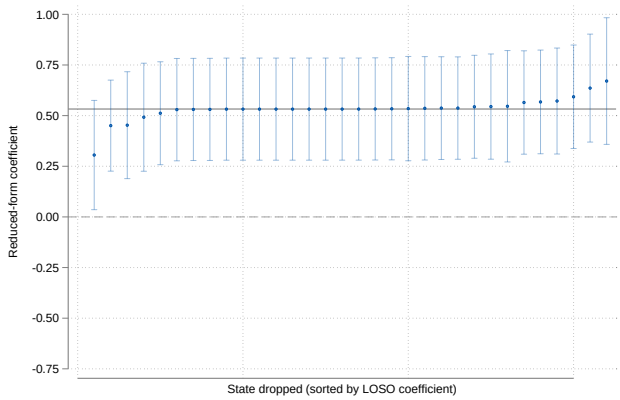
Notes: Companion to Figure 8, with the same fifteen specification variants per panel. Bars report reduced-form estimates, grouped by the dimension the variant changes, in the same order and colors as the main-text figure: controls (blue), the industries excluded from the instrument (red), the estimation sample (green), the weighting (orange), and the distance-decay parameter (purple). The marker is the point estimate and the shaded bar spans its subdistrict-clustered 95% confidence interval. The black diamond marks the main specification from Table 3. All panels share a common y-axis.

Figure B5: Leave-One-State-Out Reduced-Form Estimates

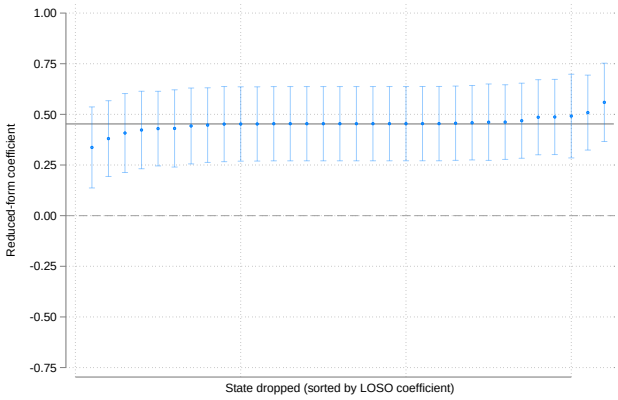
(a) Population growth



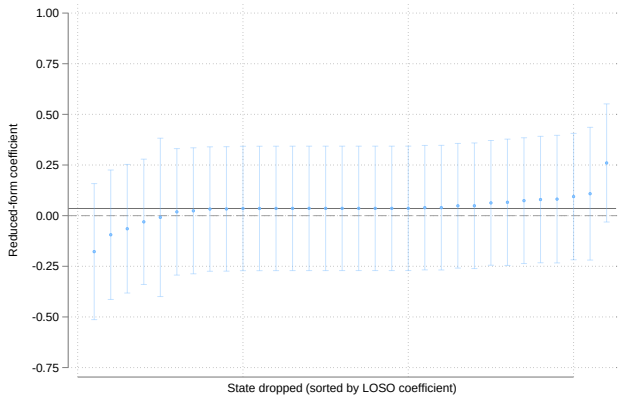
(b) Employment growth



(c) Services employment growth



(d) Manufacturing employment growth



Notes: Each point is a reduced-form coefficient estimated after dropping one Indian state or union territory from the sample; whiskers are 95% confidence intervals with subdistrict-clustered standard errors. All regressions use the preferred specification, including state fixed effects. The red solid line marks the within-outcome mean of the LOSO coefficients. The dashed line at zero is a shared reference.

B2 Additional tables

Table B1: City Summary Statistics

	N	Mean	Std Dev	Median
Panel A: Size and scale (baseline)				
Population, 2001	1,443	212,152	938,139	70,294
Population, 2011	1,443	259,522	1,150,780	83,851
Non-farm employment, 1998	1,443	23,854	153,613	6,436
Establishments, 1998	1,443	7,327	35,085	2,651
Average firm size (employees), 1998	1,443	2.8	3.7	2.4
Panel B: Sector and worker composition (baseline)				
Manufacturing employment share, 1998	1,443	0.240	0.144	0.203
Services employment share, 1998	1,443	0.749	0.150	0.788
Private-sector employment share, 1998	1,443	0.887	0.099	0.910
Cultivator share of main workers, 2001	1,443	0.109	0.087	0.083
Literacy rate, 2001	1,443	0.622	0.108	0.639
Panel C: Growth over the study period				
Population growth (log change, 2001–2011)	1,443	0.182	0.134	0.175
Non-farm employment growth (log change, 1998–2013)	1,437	0.606	0.995	0.451
Manufacturing employment growth (log change)	1,437	0.586	1.135	0.468
Services employment growth (log change)	1,437	0.625	1.048	0.477
Establishment-count growth (log change)	1,437	0.662	0.966	0.506
Panel D: Shift-share exposure inputs				
City-level Bartik shock	1,443	0.291	0.095	0.296
Sum of non-excluded industry shares	1,443	0.625	0.117	0.619

Notes: One row per city (DDL urban boundary) in the estimation universe, the 1,443 cities in the exposure measure with at least 10,000 residents in 2001 and baseline employment in the shift-share industries, out of the 1,507 cities defined in Section 2.1. Cities are delineated following the DDL urban boundaries; population is from the Population Census (2001, 2011) and employment, establishments, and firm size from the Economic Census (1998, 2013). Panel A reports size and scale at baseline; Panel B baseline sector and worker composition (employment shares are of total non-farm employment; the cultivator share and literacy rate are from the Census); Panel C growth over the study period, as log changes with the same definitions as the descriptive figures (population 2001–2011; employment, sectoral employment, and establishment counts 1998–2013, using $\ln(x+1)$); Panel D the city-level shift-share inputs. Six cities have no record in the 2013 Economic Census. Panel C therefore reports employment growth for 1,437 cities. These six cities enter the exposure measure with zero employment growth. City sizes are highly skewed, so the median is more representative of the typical city than the mean.

Table B2: First-Stage Regressions: Urban Employment Exposure on the Bartik Instrument

	(1) Preferred (subdist)	(2) Preferred (district)	(3) Minimal (subdist)
Bartik instrument	1.483***	1.483***	1.741***
	(0.118)	(0.285)	(0.104)
State fixed effects	✓	✓	
Geography (lat/lon)	✓	✓	✓
Log baseline population	✓	✓	
Incomplete-shares correction	✓	✓	✓
N	256,167	256,167	256,167
First-stage F	158.92	26.99	280.33
Mean dep. var.	0.323	0.323	0.323
BHJ shock-level F: 11.15 (retail exclusion, pop-growth sample; 22.7 effective independent shocks, at the industry level).			

Notes: Each column regresses village-level urban employment growth (the endogenous variable of the 2SLS in Table 3) on the retail-exclusion Bartik instrument. Standard errors in parentheses; clustering level specified in the column label. Coefficient stars use the column's own standard error: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. First-stage F is the Wald F on the instrument coefficient. Mean dep. var. is the sample mean of the endogenous variable. The final row reports the exposure-robust first-stage F at the shock level.

Table B3: Diagnostics for the Shares-Based Identification Route

	Aggregate (uncond.)	Aggregate (cond.)	Railway transport (cond.)	Education (cond.)	Other land transport (cond.)
Panel A. Rotemberg-weight concentration					
Rotemberg weight			0.208	0.141	0.127
Top-5 share of total positive weight	0.520				
Sum of negative weights (27 industries)	-0.317				
Panel B. Balance of the exposure shares (standardized coefficients)					
Log population 2001	-0.066***	-0.021***	-0.045***	-0.012	-0.025***
Log employment 1998	-0.007	0.074***	0.030***	-0.032***	0.001
Literacy rate 2001	0.185***	0.050***	-0.012	-0.015	-0.052***
Services emp share 1998	0.025***	0.013	-0.009	0.047***	0.005
Power supply 2001	0.169***	0.062***	-0.019***	-0.056***	-0.044***
Paved road access 2001	0.072***	0.028***	-0.012**	-0.031***	-0.008
Cultivator share 2001	0.049***	-0.064***	-0.055***	0.031**	0.023**
SC share 2001	-0.049***	-0.093***	-0.049***	0.038***	-0.019*
ST share 2001	0.049***	0.071***	0.046***	0.059***	0.018
Panel C. Share-based pre-trend (1990-vintage instrument predicts pre-period growth)					
	Coef.	Std. err.	p		
Employment growth, 1990–98	0.002	(0.010)	0.830		
Population growth, 1991–2001	0.000	(0.002)	0.874		

Notes: Diagnostics for the "shares" identification route of Goldsmith-Pinkham et al. (2020), on the retail-exclusion instrument. Panel A reports the Rotemberg weights of the three largest-weight industries, computed under the main specification's residualization, together with the concentration of the weights; the two remaining top-5 industries are other textiles manufacturing (0.116) and equipment manufacturing (0.093). Panel B regresses each standardized predetermined village characteristic on the standardized exposure-share objects named in the column headers: the aggregate included-industry share and the three largest-weight individual industry shares. The unconditional column is the bivariate association; the conditional columns control for the main-specification controls (state fixed effects, latitude, longitude, and log baseline population). The three single-industry columns also control for the sum of shares. The aggregate column cannot, because the aggregate share equals one minus that control. In the log-population row, log population is omitted from the controls. This conditional test matches the shares-exogeneity assumption as stated. The aggregate share remains significantly related to eight of the nine characteristics conditional on the controls. The assumption is therefore not credible in this setting, even though the share-based instrument does not predict pre-period growth (Panel C, computed with 1990-vintage shares under the same conditioning). The instrument itself inherits these correlations at the village level, which is why identification does not rest on unit-level comparisons. Standard errors are clustered at the subdistrict level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. SC and ST denote Scheduled Caste and Scheduled Tribe.

Table B4: Per-Industry Share Instruments and Overidentification

Instrument (single industry share)	Population	Employment	First-stage F
Railway transport (weight 0.208)	-0.010 (0.018)	0.202 (0.149)	132.8
Education (weight 0.141)	-0.052 (0.022)	0.446 (0.146)	53.9
Other land transport (weight 0.127)	-0.034 (0.019)	0.401 (0.126)	42.6
Other textiles mfg. (weight 0.116)	0.034 (0.024)	0.361 (0.164)	43.6
Equipment manufacturing (weight 0.093)	-0.292 (0.069)	-0.135 (0.297)	26.8
Joint, all five shares (2SLS)	-0.035 (0.009)	0.338 (0.063)	79.4
Hansen J statistic	48.36	3.59	
Hansen J p-value	<0.001	0.465	

Notes: Each of the first five rows uses a single industry's exposure share as the instrument for urban employment-growth exposure, under the main-specification controls, for the two outcomes named in the column headers. If the exposure shares were valid instruments for a common effect, these estimates would agree. The first-stage F column is from the population model; the employment models give nearly identical values. The joint row uses all five shares together, and the Sargan–Hansen test asks whether the five just-identified estimates are statistically compatible: for population they are not, with estimates ranging from 0.034 to −0.292, while for employment they are. A rejection implies that the shares are not valid instruments for a common effect, though it cannot separate invalid shares from heterogeneous effects across industries. Standard errors in parentheses are clustered at the subdistrict level.

Table B5: Inference Comparison: 2SLS Across Standard-Error Methods and Weak-IV Diagnostics

Outcome	Point est.	Subdist. SE	District SE	State SE	BHJ SE	AR p	AKM0 95% CI	RI p	Blocked RI p	N	Mean dep. var.
Population growth	-0.100	(0.016)	(0.031)	(0.060)	(0.048)	0.000	[-0.335, -0.009]	0.046	0.030	256,167	0.134
Employment growth	0.359	(0.090)	(0.136)	(0.112)	(0.115)	0.000	[0.074, 0.790]	0.022	0.022	256,167	0.202
Services employment growth	0.308	(0.067)	(0.116)	(0.113)	(0.112)	0.000	[0.021, 0.713]	0.036	0.022	252,670	0.359
Cultivators growth	0.074	(0.053)	(0.104)	(0.203)	(0.284)	0.169	[-1.423, 0.578]	0.824	0.828	250,473	-0.165
Male population growth	-0.101	(0.016)	(0.032)	(0.062)	(0.050)	0.000	[-0.351, -0.008]	0.054	0.029	256,167	0.132
Manufacturing employment growth	0.026	(0.112)	(0.195)	(0.228)	(0.423)	0.819	[-1.837, 1.134]	0.975	0.972	190,445	0.069

Notes: Each row gives the 2SLS point estimate under the main specification (state fixed effects, latitude and longitude, log baseline village population, the baseline level of the outcome, and the incomplete-shares correction), with four standard errors: clustered at the subdistrict, district, and state/UT levels, and BHJ exposure-robust at the industry-shock level. The Anderson–Rubin p-value is equivalent, with a single instrument, to the test that the reduced-form coefficient on the instrument is zero, computed under subdistrict clustering. The AKM0 column reports the null-imposed 95% confidence set from inverting the exposure-robust test at the shock level. The randomization-inference p-value uses 1,000 permutations of the national shocks; the blocked variant permutes shocks only within the seven broad sectors, so it does not rely on exchangeability across sectors. N and the mean of the dependent variable are from each row's estimation sample. Standard errors clustered at the subdistrict level are the baseline throughout the paper.

Table B6: Reduced-Form Robustness Across Specification Variants

	Pop. growth		Emp. growth		Services emp.		Manufacturing emp.	
Specification	Coef	SE	Coef	SE	Coef	SE	Coef	SE
No controls	-0.237***	(0.019)	0.852***	(0.144)	0.323***	(0.113)	0.148	(0.155)
Geo only	-0.197***	(0.017)	0.878***	(0.144)	0.374***	(0.111)	0.178	(0.152)
Baseline (preferred)	-0.148***	(0.019)	0.532***	(0.129)	0.454***	(0.094)	0.036	(0.157)
No state FE	-0.197***	(0.017)	1.637***	(0.127)	1.337***	(0.096)	0.593***	(0.150)
Nearest city <60km	-0.145***	(0.017)	0.450***	(0.115)	0.454***	(0.094)	0.029	(0.159)
Nearest city >20km	-0.118***	(0.031)	0.953***	(0.227)	0.494***	(0.125)	0.162	(0.251)
Nearest city >10km	-0.139***	(0.022)	0.682***	(0.153)	0.477***	(0.100)	0.090	(0.174)
No exclusion	-0.154***	(0.019)	0.416***	(0.127)	0.379***	(0.105)	0.005	(0.171)
Local services exclusion	-0.136***	(0.021)	0.438***	(0.165)	0.476***	(0.115)	-0.419**	(0.213)
Ag./wholesale/retail exclusion	-0.142***	(0.017)	0.498***	(0.118)	0.464***	(0.095)	0.059	(0.158)
Retail/rail/educ. exclusion	-0.171***	(0.021)	0.483***	(0.136)	0.390***	(0.104)	0.054	(0.170)
Distance decay $\theta = 0.5$	-0.147***	(0.019)	0.506***	(0.128)	0.405***	(0.094)	0.051	(0.155)
No distance decay ($\theta = 0$)	-0.142***	(0.018)	0.467***	(0.125)	0.341***	(0.093)	0.074	(0.150)
Distance decay $\theta = 2$	-0.126***	(0.017)	0.490***	(0.118)	0.451***	(0.085)	0.013	(0.147)
Population-weighted	-0.142***	(0.017)	0.316***	(0.117)	0.373***	(0.094)	0.065	(0.181)
OLS	-0.009***	(0.003)	0.130***	(0.019)	0.107***	(0.015)	0.119***	(0.021)
2SLS	-0.100***	(0.016)	0.359***	(0.090)	0.308***	(0.067)	0.026	(0.112)
No baseline Y	-0.148***	(0.019)	0.664***	(0.142)	0.511***	(0.116)	0.175	(0.165)
N (preferred spec)	256,167		256,167		252,670		190,445	
Mean dep. var. (preferred spec)	0.134		0.202		0.359		0.069	

Notes: Each cell is the reduced-form coefficient (with subdistrict-clustered standard error in parentheses) for the indicated outcome under one of 17 specification variants, relative to the "Baseline (preferred)" specification: state fixed effects, latitude and longitude, log baseline village population, the outcome-specific baseline, and the incomplete-shares correction. The "No state FE" row removes only the state fixed effects; the other variant rows modify the main specification in the indicated dimension. All rows use the retail-exclusion shift-share instrument except the four rows that explicitly vary the exclusion set; the retail/rail/education row additionally drops railway transport and education, the two industries with the largest Rotemberg weights, whose employment is largely administered by the state. The three distance-decay rows vary the decay parameter θ in the exposure weights (the main specification sets $\theta = 1$) while keeping the retail exclusion. The population-weighted row weights villages by baseline population. Outcomes and samples as in Table 3; the manufacturing column is estimated on villages with positive baseline manufacturing employment. The two "Nearest city" rows restrict the village sample by each village's distance to its own nearest city; the 100 km radius over which exposure is measured is the same in every row. N and Mean dep. var. in the footer are for the main specification's estimation sample; the sample-restriction rows use strictly smaller samples. The 2SLS row uses asymptotic z-based p-values; all reduced-form rows use t-based p-values. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table B7: Robustness Checks with Design-Based Inference

	Population	Total employment	Services	Resource extraction
	(1)	(2)	(3)	(4)
Main specification	-0.100	0.359	0.308	0.519
	(0.048)	(0.115)	(0.112)	(0.205)
AKM0 95% CI	[-0.335, -0.009]	[0.074, 0.790]	[0.021, 0.713]	[0.203, 1.757]
RI p-value	0.046	0.022	0.036	0.009
Control for own exposure to the national shocks	-0.099	0.337	0.313	0.507
	(0.047)	(0.122)	(0.116)	(0.204)
AKM0 95% CI	[-0.332, -0.011]	[-0.005, 0.746]	[0.018, 0.736]	[0.192, 1.747]
RI p-value				
Instrument without extraction and wholesale	-0.097	0.340	0.317	0.432
	(0.047)	(0.116)	(0.115)	(0.183)
AKM0 95% CI	[-0.366, 0.005]	[0.014, 0.849]	[-0.003, 0.811]	[0.134, 1.817]
RI p-value	0.050	0.031	0.031	0.020
Log 1991 instead of log 2001 population	–	0.310	0.268	0.491
		(0.112)	(0.092)	(0.200)
AKM0 95% CI		[-0.012, 0.673]	[-0.007, 0.554]	[0.183, 1.704]
RI p-value		0.044	0.041	0.010
Adjusted city definition (1991 towns)	-0.142	0.607	0.471	0.910
	(0.094)	(0.300)	(0.231)	(0.481)
AKM0 95% CI	[-1.043, 0.002]	[0.068, 3.041]	[0.063, 2.434]	[0.276, 6.183]
RI p-value	0.060	0.023	0.025	0.012

Notes: Each row reports the 2SLS estimate under the main specification of Table 3 with one change. The second row adds a control for the village's own exposure to the national industry shocks, built by applying the same national industry growth rates, over the same industries excluding retail, to the village's own 1998 employment composition, with the matching incomplete-shares control. Randomization inference is not reported for this row, because the control is built from the same shocks, so permuting them no longer isolates the effect of nearby urban growth. The third row builds the instrument without the extraction and wholesale industries. This leaves 78 of the 87 industry shocks. The fourth row replaces the log 2001 village population control with log population in 1991. This row reports no population estimate, because the population regression also controls for log 2001 population as the baseline level of the outcome. This row uses the 253,302 villages matched to the 1991 Population Census (249,885 for services). The fifth row uses the adjusted city definition described in Section 3.3, under which a shrid is part of a city only if it contained a town in the 1991 Census. Cities without a 1991 town are dropped. These include the 17 cities of Jammu and Kashmir, where the 1991 Census was not conducted. Cities whose 1991 towns together had fewer than 10,000 residents are also dropped, as are cities none of whose 1991 towns has a record in the 1998 Economic Census. Shrids outside the city boundaries that contained a 1991 town are excluded from the village sample. This row uses 265,720 villages (262,123 for services). Exposure-robust standard errors in parentheses. AKM0 95% CI and RI p-value are defined as in Table 4.

Table B8: Inference Robustness for the Additional Results

	2SLS est. (1)	Subdist. SE (2)	BHJ SE (3)	AKM0 95% CI (4)	RI p (5)	Mean dep. var. (6)
Panel A: By sector						
Trade	0.285***	(0.081)	(0.185)	[-0.125, 1.063]	0.152	0.248
Services	0.290***	(0.083)	(0.110)	[-0.028, 0.649]	0.043	0.464
Construction	0.145**	(0.064)	(0.183)	[-0.415, 0.704]	0.512	0.172
Resource extraction	0.519***	(0.104)	(0.205)	[0.203, 1.757]	0.009	0.140
Manufacturing	-0.079	(0.111)	(0.375)	[-1.505, 0.847]	0.879	-0.064
Transport	-0.069	(0.088)	(0.167)	[-0.747, 0.317]	0.694	0.377
Facilities	-0.008	(0.035)	(0.070)	[-0.204, 0.229]	0.942	0.149
Panel B: By organization of the establishment						
Private	0.334***	(0.104)	(0.163)	[-0.053, 0.980]	0.053	0.203
Government	0.208**	(0.099)	(0.178)	[-0.520, 0.614]	0.357	0.247
Hired	0.236***	(0.083)	(0.098)	[-0.038, 0.568]	0.057	0.443
Non-hired	0.324***	(0.115)	(0.198)	[-0.208, 1.016]	0.129	0.182
20+ establishments	0.361***	(0.090)	(0.115)	[0.075, 0.793]	0.016	0.224

Notes: Each row reports the 2SLS estimate of the effect of nearby urban employment growth on the log difference of Economic Census establishment employment in the indicated category between 1998 and 2013, under the same specification, sample (N = 256,167), and controls as Table 3, together with the conventional subdistrict-clustered standard error, the BHJ exposure-robust standard error, the null-imposed AKM0 95% confidence interval, and the randomization-inference p-value, defined as in Table 4. Categories are defined as in Table 5 and Table 6. Stars reflect the conventional standard errors. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table B9: Establishment Employment by Distance Band

	0–20 km	20–40 km	40–60 km	60–80 km	80–100 km
	(1)	(2)	(3)	(4)	(5)
Panel A: By sector					
Trade	0.252***	0.248***	0.015	-0.051	-0.075
	(0.074)	(0.061)	(0.062)	(0.063)	(0.074)
Services	0.287***	0.267***	0.110	-0.096	0.020
	(0.073)	(0.072)	(0.071)	(0.071)	(0.072)
Construction	0.265***	0.192***	0.025	0.063	0.103*
	(0.067)	(0.060)	(0.058)	(0.058)	(0.061)
Resource extraction	0.105*	0.264***	0.220***	0.371***	0.305***
	(0.063)	(0.063)	(0.064)	(0.079)	(0.091)
Manufacturing	0.040	0.057	-0.111	0.027	0.112
	(0.110)	(0.092)	(0.094)	(0.092)	(0.101)
Transport	0.196*	0.036	-0.034	0.069	-0.150
	(0.107)	(0.085)	(0.081)	(0.086)	(0.095)
Facilities	-0.035	-0.060*	0.050	-0.012	-0.041
	(0.047)	(0.033)	(0.032)	(0.036)	(0.033)
Panel B: By organization of the establishment					
Private	0.163*	0.167**	-0.007	0.091	0.143
	(0.084)	(0.074)	(0.074)	(0.083)	(0.092)
Government	0.206**	0.229***	0.246***	0.135	0.157*
	(0.092)	(0.085)	(0.089)	(0.086)	(0.089)
Hired	0.180**	0.124*	0.055	-0.089	0.094
	(0.079)	(0.074)	(0.077)	(0.074)	(0.075)
Non-hired	0.123	0.209***	0.056	0.145	0.138
	(0.089)	(0.079)	(0.081)	(0.092)	(0.101)
20+ establishments	0.150**	0.196***	0.121*	0.148**	0.177**
	(0.072)	(0.067)	(0.066)	(0.072)	(0.078)
Panel C: By gender					
Female employment	0.107	0.221***	0.068	0.119	0.173*
	(0.088)	(0.078)	(0.083)	(0.084)	(0.090)
Male employment	0.176**	0.187***	0.067	0.095	0.146*
	(0.072)	(0.064)	(0.065)	(0.071)	(0.077)
Female-owned establishments	-0.079	-0.095	-0.158*	-0.029	0.005
	(0.086)	(0.078)	(0.082)	(0.077)	(0.089)
N			256,151		

Notes: Reduced-form coefficients on the Bartik instrument interacted with 20-km distance bands, from the same joint five-band specification as Figure 7, for the establishment-employment categories of Table 5 in Panel A, of Table 6 in Panel B, and by gender in Panel C. Appendix Table B13 and Table B14 report design-based inference for these coefficients. Outcomes are log differences of Economic Census establishment employment between 1998 and 2013, with category definitions as in Table 5. Standard errors in parentheses are clustered at the subdistrict level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table B10: Sector Contributions to Employment Growth, by Distance Band

	Trade	Services	Construction	Resource extraction	Manufacturing	Transport	Facilities	Total
Aggregate	0.246***	0.169**	0.021	0.395***	-0.042	-0.098***	-0.008	0.682***
	(0.078)	(0.078)	(0.018)	(0.148)	(0.092)	(0.026)	(0.008)	(0.246)
0–20 km	0.094*	0.148***	0.016	-0.011	-0.105	-0.019	-0.012	0.111
	(0.050)	(0.055)	(0.013)	(0.025)	(0.070)	(0.025)	(0.008)	(0.137)
20–40 km	0.099**	0.107**	0.020	0.071*	-0.033	-0.007	-0.009*	0.247**
	(0.045)	(0.047)	(0.013)	(0.040)	(0.054)	(0.017)	(0.005)	(0.120)
40–60 km	-0.003	0.009	0.015	0.053	-0.073	-0.022	0.009*	-0.011
	(0.047)	(0.047)	(0.015)	(0.042)	(0.053)	(0.016)	(0.005)	(0.122)
60–80 km	0.003	-0.055	0.018	0.162**	0.064	-0.001	0.006	0.196
	(0.044)	(0.050)	(0.012)	(0.067)	(0.054)	(0.017)	(0.007)	(0.139)
80–100 km	-0.013	0.030	0.011	0.173**	0.122**	-0.048***	-0.007	0.267*
	(0.053)	(0.054)	(0.012)	(0.074)	(0.062)	(0.019)	(0.005)	(0.150)
N	251,062	251,062	251,062	251,062	251,062	251,062	251,062	251,062

Notes: Each cell reports the reduced-form coefficient on the aggregate Bartik instrument (first row) or on the band-specific instrument from the joint five-band specification of Figure 7 (remaining rows), where the outcome is the sector's contribution to village employment growth: the sector's 1998–2013 change in Economic Census establishment employment divided by the village's total establishment employment in 1998. Contributions are additive by construction, so the sector coefficients in each row sum to the Total column. The sample excludes villages in the top and bottom 1 percent of total employment growth, which keeps the identity exact. Appendix Table B15 reports design-based inference for these coefficients. Categories are defined as in Table 5. Standard errors in parentheses are clustered at the subdistrict level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table B11: Resident Population and Work by Distance Band

	0–20 km	20–40 km	40–60 km	60–80 km	80–100 km
	(1)	(2)	(3)	(4)	(5)
Panel A: Aggregate resident outcomes					
Population	-0.028***	-0.032***	-0.052***	-0.041***	-0.009
	(0.010)	(0.009)	(0.010)	(0.011)	(0.011)
Male population	-0.029***	-0.032***	-0.053***	-0.044***	-0.013
	(0.011)	(0.010)	(0.010)	(0.011)	(0.012)
Workforce	-0.071***	-0.053***	-0.039**	-0.013	-0.013
	(0.020)	(0.018)	(0.018)	(0.020)	(0.019)
Main workers	-0.029	-0.072**	-0.071**	-0.017	-0.099***
	(0.039)	(0.033)	(0.035)	(0.037)	(0.038)
Marginal workers	-0.127	0.011	0.174**	0.066	0.055
	(0.092)	(0.077)	(0.081)	(0.081)	(0.087)
Farm main workers	-0.094	-0.103*	-0.090	-0.022	-0.191***
	(0.068)	(0.055)	(0.059)	(0.061)	(0.061)
Non-farm main workers	0.111**	0.055	-0.030	-0.095**	-0.104**
	(0.056)	(0.046)	(0.046)	(0.046)	(0.050)
Panel B: Resident workers by gender					
Main workers, female	-0.208***	-0.277***	-0.248***	-0.126*	-0.324***
	(0.073)	(0.063)	(0.064)	(0.068)	(0.069)
Main workers, male	0.021	-0.022	-0.026	0.011	-0.044
	(0.035)	(0.029)	(0.031)	(0.033)	(0.035)
Marginal workers, female	-0.227**	-0.098	0.119	0.074	0.022
	(0.099)	(0.086)	(0.089)	(0.089)	(0.095)
Marginal workers, male	-0.054	0.072	0.203**	0.047	0.091
	(0.091)	(0.075)	(0.080)	(0.081)	(0.086)
N			256,151		

Notes: Reduced-form coefficients on the Bartik instrument interacted with 20-km distance bands, from the same joint five-band specification as Figure 7, for the resident-frame outcomes of Table 7 in Panel A and by gender in Panel B. Appendix Table B16 reports design-based inference for these coefficients. Outcomes are log differences of Population Census resident counts between 2001 and 2011, with variable definitions as in Table 7. Standard errors in parentheses are clustered at the subdistrict level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table B12: Design-Based Inference for the Distance-Band Estimates

	0–20 km	20–40 km	40–60 km	60–80 km	80–100 km
	(1)	(2)	(3)	(4)	(5)
Panel A: Population and employment (Figure 7)					
Population	-0.028	-0.032	-0.052	-0.041	-0.009
	(0.022)	(0.017)	(0.015)	(0.020)	(0.033)
AKM0 95% CI	[-0.088, 0.040]	[-0.080, 0.020]	[-0.079, 0.029]	[-0.089, 0.041]	[-0.074, 0.175]
RI p-value	0.327	0.175	0.060	0.159	0.843
Total employment	0.156	0.205	0.067	0.107	0.171
	(0.096)	(0.061)	(0.074)	(0.085)	(0.107)
AKM0 95% CI	[-0.156, 0.403]	[-0.046, 0.334]	[-0.102, 0.369]	[-0.053, 0.563]	[-0.085, 0.654]
RI p-value	0.284	0.051	0.436	0.214	0.256
Panel B: Agricultural laborers and extraction (Figure 8)					
Agricultural laborers	-0.206	-0.227	-0.311	-0.216	-0.209
	(0.145)	(0.139)	(0.173)	(0.135)	(0.100)
AKM0 95% CI	[-0.575, 0.283]	[-0.619, 0.212]	[-0.787, 0.308]	[-0.696, 0.178]	[-0.541, 0.126]
RI p-value	0.269	0.229	0.262	0.248	0.167
Extraction, high forest cover	0.147	0.187	0.232	0.480	0.490
	(0.132)	(0.132)	(0.089)	(0.106)	(0.225)
AKM0 95% CI	[-0.093, 0.749]	[-0.181, 0.594]	[-0.030, 0.486]	[0.306, 1.075]	[0.181, 2.291]
RI p-value	0.329	0.193	0.067	0.004	0.086
Extraction, low forest cover	0.076	0.267	0.118	0.146	0.019
	(0.100)	(0.088)	(0.101)	(0.088)	(0.141)
AKM0 95% CI	[-0.483, 0.240]	[-0.005, 0.535]	[-0.024, 0.906]	[-0.039, 0.659]	[-0.200, 1.048]
RI p-value	0.527	0.060	0.352	0.203	0.918
Panel C: Consumer-facing trade and services (Figure 9)					
Retail trade	0.223	0.208	-0.016	-0.057	-0.077
	(0.113)	(0.105)	(0.125)	(0.102)	(0.196)
AKM0 95% CI	[-0.147, 0.512]	[-0.138, 0.490]	[-0.278, 0.539]	[-0.318, 0.339]	[-1.125, 0.322]
RI p-value	0.187	0.187	0.930	0.624	0.751
Wholesale trade	0.176	0.278	0.128	0.044	0.012
	(0.073)	(0.061)	(0.059)	(0.069)	(0.147)
AKM0 95% CI	[0.023, 0.469]	[0.175, 0.591]	[0.003, 0.384]	[-0.231, 0.218]	[-0.553, 0.426]
RI p-value	0.033	0.009	0.097	0.584	0.959
Pure consumer services	0.083	0.090	-0.210	-0.258	-0.299
	(0.120)	(0.117)	(0.134)	(0.139)	(0.178)
AKM0 95% CI	[-0.202, 0.503]	[-0.250, 0.441]	[-0.622, 0.197]	[-0.807, 0.093]	[-1.179, 0.091]
RI p-value	0.588	0.540	0.264	0.124	0.153

Panel D: Contrasts

	Difference	Exposure-robust SE	Exposure-robust t	RI p (whole economy)	RI p (within sector)
Extraction, high minus low forest cover:					
0–20 km	0.071	(0.191)	0.37	0.762	0.795
20–40 km	-0.080	(0.153)	-0.53	0.598	0.653
40–60 km	0.115	(0.146)	0.78	0.553	0.579
60–80 km	0.334	(0.093)	3.58	0.031	0.013
80–100 km	0.472	(0.190)	2.48	0.094	0.059
60–100 km, average of the two rings	0.403	(0.128)	3.16	0.055	0.034
Pure consumer services, 40–60 minus 20–40 km	-0.300	(0.072)	-4.14	0.016	0.008
Retail trade, 40–100 minus 0–40 km	-0.263	(0.146)	-1.80	0.238	0.192
Wholesale trade, 60–100 minus 0–60 km	-0.168	(0.118)	-1.42	0.312	0.297
Pure consumer services, 40–100 minus 0–40 km	-0.339	(0.142)	-2.39	0.094	0.050

Notes: Reduced-form coefficients on the band-specific shift-share instruments from the joint five-band specifications of Figure 7, Figure 11, and Figure 10, with the same samples and controls. Exposure-robust standard errors, computed at the level of the industry shocks, in parentheses. AKM0 95% CI and RI p-value are defined as in Table 4. Because a permutation reassigns the shocks behind all five band instruments at once, each RI p-value tests the null of no effect in any band. Panel D reports differences between estimates, each with its exposure-robust standard error in parentheses. For extraction, each difference compares high-forest with low-forest villages, in one band or for the average of the two outermost bands. For the consumer-facing categories, each difference subtracts the estimate for nearer urban growth from the estimate for farther urban growth. The 40–60 minus 20–40 km difference for pure consumer services compares two adjacent bands of the five-band specification. The other consumer-facing differences come from specifications that restrict the band coefficients to be equal within 0–40 km and within 40–100 km (within 0–60 km and within 60–100 km for wholesale trade), with all other controls unchanged. The randomization-inference p-values in Panel D test the null that the two effects are equal.

Table B13: Design-Based Inference for the Distance-Band Estimates by Sector

	0–20 km	20–40 km	40–60 km	60–80 km	80–100 km
	(1)	(2)	(3)	(4)	(5)
Trade	0.252	0.248	0.015	-0.051	-0.075
	(0.107)	(0.111)	(0.130)	(0.095)	(0.206)
AKM0 95% CI	[-0.102, 0.523]	[-0.090, 0.570]	[-0.255, 0.600]	[-0.318, 0.290]	[-1.153, 0.351]
RI p-value	0.134	0.155	0.932	0.625	0.782
Services	0.287	0.267	0.110	-0.096	0.020
	(0.109)	(0.148)	(0.143)	(0.086)	(0.091)
AKM0 95% CI	[0.019, 0.655]	[-0.126, 0.758]	[-0.213, 0.698]	[-0.288, 0.287]	[-0.190, 0.451]
RI p-value	0.099	0.193	0.564	0.396	0.853
Construction	0.265	0.192	0.025	0.063	0.103
	(0.095)	(0.076)	(0.078)	(0.161)	(0.125)
AKM0 95% CI	[0.077, 0.665]	[-0.061, 0.391]	[-0.214, 0.259]	[-0.758, 0.381]	[-0.342, 0.485]
RI p-value	0.050	0.089	0.738	0.783	0.567
Resource extraction	0.105	0.264	0.220	0.371	0.305
	(0.069)	(0.079)	(0.067)	(0.090)	(0.163)
AKM0 95% CI	[-0.076, 0.326]	[0.051, 0.518]	[0.111, 0.610]	[0.208, 0.874]	[0.085, 1.609]
RI p-value	0.204	0.034	0.013	0.010	0.065
Manufacturing	0.040	0.057	-0.111	0.027	0.112
	(0.369)	(0.327)	(0.210)	(0.137)	(0.109)
AKM0 95% CI	[-1.388, 0.837]	[-1.456, 0.676]	[-1.102, 0.303]	[-0.514, 0.375]	[-0.348, 0.392]
RI p-value	0.934	0.922	0.689	0.863	0.410
Transport	0.196	0.036	-0.034	0.069	-0.150
	(0.185)	(0.130)	(0.122)	(0.169)	(0.194)
AKM0 95% CI	[-0.114, 1.116]	[-0.199, 0.671]	[-0.268, 0.560]	[-0.421, 0.649]	[-1.000, 0.332]
RI p-value	0.436	0.797	0.805	0.705	0.546
Facilities	-0.035	-0.060	0.050	-0.012	-0.041
	(0.060)	(0.051)	(0.030)	(0.031)	(0.023)
AKM0 95% CI	[-0.141, 0.245]	[-0.154, 0.185]	[-0.048, 0.133]	[-0.079, 0.131]	[-0.097, 0.060]
RI p-value	0.670	0.429	0.201	0.733	0.144

Notes: Coefficients of Panel A of Appendix Table B9. Exposure-robust standard errors, computed at the level of the industry shocks, in parentheses. AKM0 95% CI and RI p-value are defined as in Table 4. Because a permutation reassigns the shocks behind all five band instruments at once, each RI p-value tests the null of no effect in any band.

Table B14: Design-Based Inference for the Distance-Band Estimates by Establishment Type

	0–20 km	20–40 km	40–60 km	60–80 km	80–100 km
	(1)	(2)	(3)	(4)	(5)
Panel A: By organization of the establishment					
Private	0.163	0.167	-0.007	0.091	0.143
	(0.116)	(0.072)	(0.070)	(0.115)	(0.134)
AKM0 95% CI	[-0.234, 0.444]	[-0.179, 0.297]	[-0.172, 0.266]	[-0.140, 0.667]	[-0.196, 0.719]
RI p-value	0.346	0.126	0.934	0.479	0.416
Government	0.206	0.229	0.246	0.135	0.157
	(0.126)	(0.198)	(0.185)	(0.177)	(0.106)
AKM0 95% CI	[-0.137, 0.589]	[-0.307, 0.869]	[-0.297, 0.831]	[-0.689, 0.517]	[-0.196, 0.502]
RI p-value	0.187	0.354	0.353	0.598	0.265
Hired	0.180	0.124	0.055	-0.089	0.094
	(0.098)	(0.170)	(0.129)	(0.065)	(0.130)
AKM0 95% CI	[-0.133, 0.434]	[-0.269, 0.773]	[-0.286, 0.506]	[-0.279, 0.131]	[-0.379, 0.482]
RI p-value	0.151	0.582	0.755	0.225	0.587
Non-hired	0.123	0.209	0.056	0.145	0.138
	(0.147)	(0.108)	(0.066)	(0.118)	(0.133)
AKM0 95% CI	[-0.316, 0.533]	[-0.321, 0.401]	[-0.128, 0.276]	[-0.131, 0.645]	[-0.199, 0.705]
RI p-value	0.549	0.214	0.454	0.224	0.395
20+ establishments	0.150	0.196	0.121	0.148	0.177
	(0.095)	(0.065)	(0.080)	(0.084)	(0.107)
AKM0 95% CI	[-0.141, 0.409]	[-0.079, 0.330]	[-0.062, 0.444]	[-0.014, 0.582]	[-0.085, 0.648]
RI p-value	0.295	0.068	0.213	0.094	0.235
Panel B: By gender					
Female employment	0.107	0.221	0.068	0.119	0.173
	(0.232)	(0.191)	(0.081)	(0.090)	(0.126)
AKM0 95% CI	[-0.772, 0.616]	[-0.738, 0.553]	[-0.216, 0.286]	[-0.066, 0.554]	[-0.244, 0.582]
RI p-value	0.740	0.488	0.488	0.270	0.344
Male employment	0.176	0.187	0.067	0.095	0.146
	(0.087)	(0.091)	(0.103)	(0.097)	(0.118)
AKM0 95% CI	[-0.007, 0.521]	[-0.018, 0.543]	[-0.148, 0.527]	[-0.095, 0.589]	[-0.136, 0.685]
RI p-value	0.112	0.117	0.596	0.353	0.344
Female-owned establishments	-0.079	-0.095	-0.158	-0.029	0.005
	(0.352)	(0.385)	(0.164)	(0.093)	(0.114)
AKM0 95% CI	[-1.282, 0.780]	[-1.788, 0.675]	[-0.848, 0.205]	[-0.364, 0.232]	[-0.534, 0.266]
RI p-value	0.922	0.907	0.525	0.805	0.969

Notes: Coefficients of Panels B and C of Appendix Table B9. Exposure-robust standard errors, computed at the level of the industry shocks, in parentheses. AKM0 95% CI and RI p-value are defined as in Table 4. Because a permutation reassigns the shocks behind all five band instruments at once, each RI p-value tests the null of no effect in any band.

Table B15: Design-Based Inference for the Sector Contributions by Distance Band

	0–20 km	20–40 km	40–60 km	60–80 km	80–100 km
	(1)	(2)	(3)	(4)	(5)
Trade	0.094	0.099	-0.003	0.003	-0.013
	(0.084)	(0.054)	(0.055)	(0.060)	(0.112)
AKM0 95% CI	[-0.245, 0.269]	[-0.051, 0.267]	[-0.101, 0.285]	[-0.177, 0.202]	[-0.679, 0.193]
RI p-value	0.448	0.188	0.954	0.963	0.918
Services	0.148	0.107	0.009	-0.055	0.030
	(0.058)	(0.054)	(0.064)	(0.039)	(0.069)
AKM0 95% CI	[-0.162, 0.239]	[-0.205, 0.189]	[-0.143, 0.258]	[-0.192, 0.059]	[-0.159, 0.307]
RI p-value	0.100	0.193	0.909	0.219	0.735
Construction	0.016	0.020	0.015	0.018	0.011
	(0.015)	(0.021)	(0.017)	(0.023)	(0.016)
AKM0 95% CI	[-0.013, 0.082]	[-0.047, 0.077]	[-0.028, 0.076]	[-0.083, 0.072]	[-0.015, 0.119]
RI p-value	0.335	0.511	0.467	0.537	0.560
Resource extraction	-0.011	0.071	0.053	0.162	0.173
	(0.026)	(0.036)	(0.042)	(0.062)	(0.082)
AKM0 95% CI	[-0.085, 0.065]	[-0.055, 0.160]	[-0.019, 0.278]	[0.073, 0.587]	[0.061, 0.818]
RI p-value	0.677	0.150	0.207	0.018	0.045
Manufacturing	-0.105	-0.033	-0.073	0.064	0.122
	(0.227)	(0.178)	(0.139)	(0.062)	(0.058)
AKM0 95% CI	[-0.941, 0.412]	[-0.781, 0.341]	[-0.750, 0.195]	[-0.155, 0.242]	[-0.047, 0.340]
RI p-value	0.773	0.922	0.729	0.470	0.067
Transport	-0.019	-0.007	-0.022	-0.001	-0.048
	(0.033)	(0.030)	(0.027)	(0.035)	(0.022)
AKM0 95% CI	[-0.076, 0.140]	[-0.055, 0.152]	[-0.078, 0.095]	[-0.147, 0.085]	[-0.134, 0.016]
RI p-value	0.672	0.847	0.515	0.974	0.184
Facilities	-0.012	-0.009	0.009	0.006	-0.007
	(0.008)	(0.005)	(0.005)	(0.007)	(0.004)
AKM0 95% CI	[-0.024, 0.027]	[-0.018, 0.015]	[-0.002, 0.027]	[-0.007, 0.043]	[-0.018, 0.007]
RI p-value	0.279	0.206	0.098	0.487	0.183
Total	0.111	0.247	-0.011	0.196	0.267
	(0.284)	(0.181)	(0.131)	(0.147)	(0.169)
AKM0 95% CI	[-1.020, 0.708]	[-0.600, 0.587]	[-0.358, 0.452]	[-0.175, 0.778]	[-0.172, 0.980]
RI p-value	0.765	0.422	0.930	0.233	0.265

Notes: Coefficients of Appendix Table B10, estimated on the same trimmed sample. Exposure-robust standard errors, computed at the level of the industry shocks, in parentheses. AKM0 95% CI and RI p-value are defined as in Table 4. Because a permutation reassigns the shocks behind all five band instruments at once, each RI p-value tests the null of no effect in any band.

Table B16: Design-Based Inference for the Resident Outcomes by Distance Band

	0–20 km	20–40 km	40–60 km	60–80 km	80–100 km
	(1)	(2)	(3)	(4)	(5)
Panel A: Aggregate resident outcomes					
Population	-0.028	-0.032	-0.052	-0.041	-0.009
	(0.022)	(0.017)	(0.015)	(0.020)	(0.033)
AKM0 95% CI	[-0.088, 0.040]	[-0.080, 0.020]	[-0.079, 0.029]	[-0.089, 0.041]	[-0.074, 0.175]
RI p-value	0.327	0.175	0.060	0.159	0.843
Male population	-0.029	-0.032	-0.053	-0.044	-0.013
	(0.022)	(0.017)	(0.016)	(0.019)	(0.031)
AKM0 95% CI	[-0.088, 0.041]	[-0.077, 0.023]	[-0.081, 0.029]	[-0.092, 0.033]	[-0.077, 0.152]
RI p-value	0.315	0.178	0.062	0.132	0.744
Workforce	-0.071	-0.053	-0.039	-0.013	-0.013
	(0.045)	(0.039)	(0.028)	(0.037)	(0.032)
AKM0 95% CI	[-0.209, 0.049]	[-0.236, 0.019]	[-0.121, 0.050]	[-0.204, 0.059]	[-0.159, 0.062]
RI p-value	0.217	0.244	0.260	0.777	0.729
Main workers	-0.029	-0.072	-0.071	-0.017	-0.099
	(0.076)	(0.075)	(0.099)	(0.090)	(0.091)
AKM0 95% CI	[-0.302, 0.146]	[-0.275, 0.170]	[-0.295, 0.335]	[-0.292, 0.276]	[-0.543, 0.103]
RI p-value	0.744	0.437	0.595	0.866	0.427
Marginal workers	-0.127	0.011	0.174	0.066	0.055
	(0.137)	(0.152)	(0.149)	(0.181)	(0.215)
AKM0 95% CI	[-0.622, 0.191]	[-0.793, 0.262]	[-0.541, 0.464]	[-0.705, 0.490]	[-0.461, 1.026]
RI p-value	0.434	0.944	0.386	0.745	0.856
Farm main workers	-0.094	-0.103	-0.090	-0.022	-0.191
	(0.127)	(0.122)	(0.165)	(0.129)	(0.139)
AKM0 95% CI	[-0.403, 0.342]	[-0.342, 0.442]	[-0.405, 0.714]	[-0.425, 0.392]	[-0.845, 0.128]
RI p-value	0.504	0.494	0.664	0.878	0.311
Non-farm main workers	0.111	0.055	-0.030	-0.095	-0.104
	(0.171)	(0.107)	(0.045)	(0.064)	(0.090)
AKM0 95% CI	[-0.532, 0.489]	[-0.449, 0.255]	[-0.197, 0.084]	[-0.280, 0.124]	[-0.408, 0.187]
RI p-value	0.646	0.720	0.564	0.213	0.351
Panel B: Resident workers by gender					
Main workers, female	-0.208	-0.277	-0.248	-0.126	-0.324
	(0.152)	(0.102)	(0.155)	(0.114)	(0.121)
AKM0 95% CI	[-0.649, 0.231]	[-0.588, 0.017]	[-0.552, 0.490]	[-0.438, 0.290]	[-0.807, 0.002]
RI p-value	0.312	0.079	0.287	0.380	0.113
Main workers, male	0.021	-0.022	-0.026	0.011	-0.044
	(0.054)	(0.067)	(0.089)	(0.095)	(0.096)
AKM0 95% CI	[-0.187, 0.137]	[-0.206, 0.191]	[-0.254, 0.295]	[-0.302, 0.298]	[-0.529, 0.163]
RI p-value	0.725	0.786	0.794	0.933	0.743
Marginal workers, female	-0.227	-0.098	0.119	0.074	0.022
	(0.146)	(0.138)	(0.138)	(0.214)	(0.149)
AKM0 95% CI	[-0.592, 0.257]	[-0.791, 0.143]	[-0.484, 0.413]	[-0.975, 0.510]	[-0.457, 0.522]
RI p-value	0.262	0.572	0.476	0.730	0.904
Marginal workers, male	-0.054	0.072	0.203	0.047	0.091
	(0.163)	(0.193)	(0.173)	(0.171)	(0.259)
AKM0 95% CI	[-0.730, 0.272]	[-0.969, 0.386]	[-0.663, 0.525]	[-0.640, 0.474]	[-0.492, 1.337]
RI p-value	0.781	0.776	0.398	0.820	0.816

Notes: Coefficients of Appendix Table B11. Exposure-robust standard errors, computed at the level of the industry shocks, in parentheses. AKM0 95% CI and RI p-value are defined as in Table 4. Because a permutation reassigns the shocks behind all five band instruments at once, each RI p-value tests the null of no effect in any band.

Table B17: Consumer Services: Average Effects

	Local Consumer Svc	Pure Consumer Svc	Cons. Trade + Svc
	(1)	(2)	(3)
Panel A: OLS			
Urban emp growth	0.051***	0.014	0.055***
	(0.018)	(0.020)	(0.018)
Panel B: Reduced Form			
Bartik instrument	0.242**	-0.222*	0.310***
	(0.116)	(0.125)	(0.117)
Panel C: 2SLS			
Urban emp growth	0.164**	-0.150*	0.210**
	(0.081)	(0.084)	(0.083)
AKM0 95% CI	[-0.240, 0.868]	[-0.585, 0.719]	[-0.211, 0.992]
RI p-value	0.403	0.524	0.312
State fixed effects	✓	✓	✓
Geography (lat/lon)	✓	✓	✓
Log baseline population	✓	✓	✓
Baseline outcome	✓	✓	✓
Incomplete-shares correction	✓	✓	✓
N	256,167	256,167	256,167
First-stage F	158.41	158.71	158.56
Mean dep. var.	0.248	0.283	0.240

Notes: Outcomes are 1998–2013 log changes in employment for three consumer-facing industry aggregates: local consumer services (retail trade together with repair, restaurants, and personal and social services), pure consumer services (the same categories without retail trade), and consumer-facing trade and services (local consumer services plus wholesale trade). Specification, controls, and standard-error clustering are identical to Table 3. The AKM0 95% CI and RI p-value rows are defined as in Table 4. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table B18: Urban Growth and Village Population: Demographic Decomposition

	Pop. growth (1)	Male pop. (2)	Children pop. (3)	Children share (4)
Panel A: OLS				
Urban emp growth	-0.009***	-0.010***	-0.017***	-0.001
	(0.003)	(0.003)	(0.006)	(0.001)
Panel B: Reduced Form				
Bartik instrument	-0.148***	-0.150***	-0.216***	-0.007*
	(0.019)	(0.019)	(0.042)	(0.004)
Panel C: 2SLS				
Urban emp growth	-0.100***	-0.101***	-0.145***	-0.005*
	(0.016)	(0.016)	(0.031)	(0.003)
AKM0 95% CI	[-0.335, -0.009]	[-0.351, -0.008]	[-0.559, -0.029]	[-0.034, 0.005]
RI p-value	0.046	0.054	0.039	0.459
N	256,167	256,167	256,144	256,144
First-stage F	158.92	159.85	158.62	160.16
Mean dep. var.	0.134	0.132	-0.066	-0.030

Notes: Each column reports the same specification as Table 3's preferred column (subdistrict-clustered SE, state fixed effects, latitude, longitude, log total village population in 2001, the outcome-specific baseline level, and the incomplete-shares correction), with the outcome swapped. The child-share outcome is measured as the 2011-minus-2001 change in the share of the village population aged 0–6, so its coefficient is in percentage-point units rather than log-growth units. First-stage F is the Kleibergen–Paap statistic. "Mean dep. var." is computed on the estimation sample. The AKM0 95% CI and RI p-value rows are defined as in Table 4. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table B19: Employment Effects by Sector, by Village Population Density

	Trade	Services	Construction	Resource extraction	Manufacturing	Transport	Facilities
High density	0.491***	0.236*	-0.115	0.085	0.086	-0.025	-0.240***
	(0.141)	(0.140)	(0.148)	(0.112)	(0.200)	(0.175)	(0.080)
Low density	0.009	0.447***	0.223***	1.405***	-0.671***	-0.314*	0.089**
	(0.139)	(0.167)	(0.085)	(0.252)	(0.227)	(0.161)	(0.042)
High − low	0.482	-0.210	-0.338	-1.321	0.756	0.290	-0.329
	(0.260)	(0.262)	(0.351)	(0.361)	(0.439)	(0.425)	(0.157)
RI p-value, equal effects	0.162	0.468	0.557	0.044	0.216	0.539	0.176
N	256,161	256,161	256,161	256,161	256,161	256,161	256,161

Notes: Each column reports reduced-form coefficients on the Bartik instrument for the log difference of Economic Census employment in the indicated sector between 1998 and 2013, with the same specification and controls as Table 3. The first two rows estimate the effect separately in the villages above and below the median of baseline population density. The third row reports the difference between the two coefficients, which comes from a pooled specification that interacts the instrument and every control with the high-density indicator. The number in parentheses below the difference is its exposure-robust standard error. The next row reports the randomization-inference p-value for the null that the effect is the same in both groups. The exposure-robust first-stage F statistic is about 1 in high-density and 44 in low-density villages. Sectors are defined as in Table 5. Standard errors in parentheses are clustered at the subdistrict level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

C Model appendix

This appendix derives the results of Section 4. We first solve for the village wage at a given population and show that the village equilibrium exists and is unique. We then differentiate the equilibrium with respect to city size to obtain the responses of population, the wage, and welfare. We derive the conditions under which the population response changes sign at most once, and close with the response of employment in each sector.

An increase in urban productivity A_U raises the urban population N_U and the urban wage w_U, and it changes the urban services price p_N^U. We therefore express every village response per unit of N_U. We assume that visitors' spending in the village does not rise with the urban wage as the city grows. Growth in the urban wage then affects the village only through the migration pull.[24]

Urban growth affects the village through the services demand channel $\Phi(d)$ and the resource demand channel $\Psi(d)$ of equation (18), and through the migration pull $\tilde{M}$ of equation (20). The urban demand behind the two channels fades with distance, while the pull does not. We model this fading by assuming that the fraction of the city's services demand that reaches the village and the response of the resource price to city size both decay at constant rates, $\phi(d) = \phi_0\, e^{-\delta_\phi d}$ and $a(d) := \partial \ln p_R / \partial N_U = a_0\, e^{-\delta_a d}$, with $\delta_\phi > 0$ and $\delta_a > 0$. We impose no restriction on which of δ_ϕ and δ_a is larger. We also assume that, at the city's initial size N_U^0, the resource price is the same at every distance, $p_R(d, N_U) = \bar{p}_R\, e^{a(d)(N_U - N_U^0)}$. We evaluate derivatives with respect to distance at N_U^0. A resource price that differed across distance would be equivalent to a resource productivity A_R that differed across distance, and we hold A_R fixed across distance together with the other village parameters.

C1 Existence and uniqueness of the village equilibrium

Fix village population N_V, and let w be the wage that all three village sectors pay. The crop sector hires workers until the value of their marginal product at the national price equals the wage,

$$L_T^*(w) = \left[\frac{\alpha A_T\, \bar{T}^{1-\alpha}}{w}\right]^{1/(1-\alpha)}, \tag{25}$$

and the extraction sector does the same at the resource price,

$$L_R^*(w) = \left[\frac{\gamma\, p_R\, A_R\, \bar{F}^{1-\gamma}}{w}\right]^{1/(1-\gamma)}, \tag{26}$$

which is zero in villages without a resource endowment. Services employment depends on how much residents and visitors spend. Both groups spend the share s of their incomes on village services, and at the free-entry price $p_N^V = w/A_N$ the revenue of the services sector equals its wage bill. Services employment therefore satisfies $wL_N = s\,(wN_V + w_U \phi(d) N_U)$, which yields

$$L_N^*(w; N_V) = sN_V + \frac{s\, w_U\, \phi_0 N_U\, e^{-\delta_\phi d}}{w}. \tag{27}$$

The first term is the services employment that serves residents. It does not depend on the wage, because residents earn the same wage that services workers are paid and spend the fixed share s of it on services. The second term is the employment that serves the urban consumers who travel to the village. Like crop

[24] If visitors' spending rose with the urban wage, the services channel would be multiplied by one plus the elasticity of the urban wage with respect to city size. The signs of the wage, welfare, and crop responses derived below would not change. The stronger services channel would, however, raise the ratio R of equation (22) and move the crossover d^* farther from the city. Population could then rise near the city in villages where it otherwise falls. The stronger channel would also raise the threshold that the response of the resource price must exceed for extraction to expand.

and extraction employment, it falls as the wage rises. At a given population, the village wage $w_V(N_V)$ is the wage at which the three sectors together employ all residents,

$$H(w; N_V, N_U) := L_T^*(w) + L_R^*(w) + L_N^*(w; N_V) - N_V = 0. \tag{28}$$

Lemma 1 (Existence and uniqueness)**.** *For any strictly positive parameters, any resource endowment $\bar{F} \geq 0$, and any $s \in (0,1)$, equation (17) has a unique solution $N_V^* \in (0, \bar{N})$.*

Proof. The wage is well defined. At a given N_V, crop employment, extraction employment, and the services employment that serves visitors all fall as the wage rises,

$$\frac{\partial H}{\partial w} = -\frac{1}{1-\alpha}\frac{L_T^*(w)}{w} - \frac{1}{1-\gamma}\frac{L_R^*(w)}{w} - \frac{L_N^*(w; N_V) - sN_V}{w} = -\frac{\Lambda}{w} < 0,$$

where Λ is the expression in equation (19), evaluated at the wage w. This expression is positive, because crop employment is positive at any wage and the other two terms are never negative. As w goes to zero, crop employment grows without bound and H tends to $+\infty$. As w grows without bound, every term that depends on the wage vanishes and H tends to $sN_V - N_V = -(1-s)N_V < 0$. Since H is continuous and strictly decreasing in w, it has exactly one root $w_V(N_V)$. Because $\partial H/\partial w \neq 0$ at that root, the implicit function theorem implies that $w_V(N_V)$ is continuously differentiable.

The wage falls as population rises. Differentiating $H(w_V(N_V); N_V, N_U) = 0$ with respect to N_V,

$$\frac{dw_V}{dN_V} = -\frac{\partial H/\partial N_V}{\partial H/\partial w} = -\frac{(1-s)\, w_V}{\Lambda} < 0, \tag{29}$$

where $\partial H/\partial N_V = s - 1$. An additional resident supplies one unit of labor, and their spending creates demand for s units of services labor. The remaining $1 - s$ units must be absorbed by the sectors whose demand falls with the wage, which hire them only at a lower wage.

Village utility falls as population rises. Since $\mathcal{V}(N_V) = (1-s)\ln w_V(N_V) + s \ln A_N$ in equation (17), equation (29) implies $d\mathcal{V}/dN_V = -(1-s)^2/\Lambda < 0$.

One crossing. The right-hand side of equation (17) increases with $\mathcal{V}$ and therefore falls strictly with N_V. As N_V goes to zero, the wage at which the three sectors employ all residents grows without bound. The right-hand side then tends to $\bar{N}$ and exceeds N_V. At $N_V = \bar{N}$ the wage is finite, and the right-hand side is below $\bar{N}$. The difference between the right-hand side and N_V is thus continuous and strictly decreasing. Because it is positive near zero and negative at $\bar{N}$, it vanishes at exactly one $N_V^* \in (0, \bar{N})$. □

C2 The population response and the three forces

Write the population equation (17) as $N_V = F(N_V; N_U)$ and differentiate,

$$\frac{dN_V^*}{dN_U} = \frac{\partial F/\partial N_U}{1 - \partial F/\partial N_V}. \tag{30}$$

The denominator exceeds one, since $\partial F/\partial N_V < 0$ by the proof of Lemma 1. The response therefore has the sign of $\partial F/\partial N_U$. Since F rises with village utility and falls with city utility, $\partial F/\partial N_U$ has the sign of the gain in village utility from urban growth minus the gain in city utility.

Urban growth affects village utility only through the wage. At a given population, it raises the wage through two demand channels. The first is the spending of urban consumers who travel to the village, $s\, w_U$ each. The second is the resource price, which a larger city raises at the rate $a(d)$. Differentiating equation (28) with respect to N_U at a given population,

$$\frac{\partial H}{\partial N_U} = \frac{s\, w_U\, \phi(d)}{w_V} + \frac{L_R^*\, a(d)}{1-\gamma}.$$

The first term is the additional services labor that visitors' spending requires. The second is the additional extraction labor that the higher resource price makes profitable, where $1/(1-\gamma)$ is the elasticity of extraction labor demand with respect to the resource price. Dividing by $-\partial H/\partial w = \Lambda/w_V$, we obtain the wage response at a given population,

$$\left.\frac{\partial w_V}{\partial N_U}\right|_{N_V} = \Phi(d) + \Psi(d), \qquad \Phi(d) = \frac{s\, w_U\, \phi_0\, e^{-\delta_\phi d}}{\Lambda}, \qquad \Psi(d) = \frac{a_0\, e^{-\delta_a d}\, w_V\, L_R^*}{(1-\gamma)\,\Lambda},$$

which is equation (18). Since village utility is $(1-s)\ln w_V$ up to a constant, urban growth raises it by $(1-s)[\Phi(d)+\Psi(d)]/w_V$. This gain is positive at every distance, because $\phi(d)$ is positive at every distance.

On the other side of the comparison, urban growth raises city utility by the migration pull $\tilde{M}$ of equation (20). Because urban growth does not change the migration cost, the pull is the same at every distance. Collecting the two sides,

$$\frac{dN_V^*}{dN_U} = \frac{\tilde{A}\,[(1-s)[\Phi(d)+\Psi(d)]/w_V - \tilde{M}]}{1+\tilde{A}\,B}, \qquad \tilde{A} := \frac{N_V(1-N_V/\bar{N})}{\sigma}, \quad B := \frac{(1-s)^2}{\Lambda}, \tag{31}$$

where $\tilde{A}$ measures how many residents move to the city when city utility rises by one unit relative to village utility, and $B = -d\mathcal{V}/dN_V$ measures how much village utility falls with one more resident. The sign of the response is the sign of the bracket, as stated in equation (21). Its size is the bracket multiplied by $\tilde{A}/(1+\tilde{A}B)$. This factor rises with $\tilde{A}$ and is zero when $\tilde{A}$ is zero.

Far from the city, the population response approaches zero. Since services employ at least sN_V workers, crop employment never exceeds $(1-s)\bar{N}$, which implies that the village wage cannot fall below $\alpha A_T \bar{T}^{1-\alpha}[(1-s)\bar{N}]^{-(1-\alpha)}$. Village utility is therefore bounded below, while the migration cost rises with distance without bound. The share of workers who stay then approaches one in remote villages, which drives $\tilde{A} = N_V(1-N_V/\bar{N})/\sigma$ toward zero. If the migration cost leveled off instead, the share of workers who stay would remain bounded away from one and the response would not vanish.

Closer to the city, the share of workers who stay need not rise with distance. Differentiating equations (17) and (28) with respect to distance,

$$\frac{dN_V^*}{dd} = \frac{\tilde{A}\,[\mu'(d) - (1-s)\,\delta_\phi\, N_U\, \Phi(d)/w_V]}{1+\tilde{A}\,B}.$$

The share of workers who stay rises with distance wherever the migration cost rises faster than village utility falls as urban services demand fades, that is, wherever $\mu'(d) > (1-s)\,\delta_\phi\, N_U\, \Phi(d)/w_V$. Since $d\tilde{A}/dN_V = (1-2N_V/\bar{N})/\sigma$, the term $\tilde{A}$ then falls with distance wherever more than half of the village's workers stay. Both conditions hold at every distance in the parameterization of Figure 9.

When $R \leq 1$, population falls with urban growth at every distance beyond the city's edge (see Proposition 1 below). The size of the decline is the product of the factor $\tilde{A}/(1+\tilde{A}B)$ and the gap between the migration pull and the demand effect, which widens with distance under the condition of that proposition. The decline is largest at an intermediate distance if two conditions hold. Near the city, the gap must widen proportionally faster than the factor falls. This is the case when R is close to one, because the gap then starts close to zero. Far from the city, the migration cost must grow without bound. The factor then approaches zero, and the decline fades with it. When the migration pull is much stronger than the demand effect at the city's edge, the decline can instead be largest there. When the migration cost levels off, the decline can grow with distance.

C3 Wage and welfare responses

The total wage response adds the effect of the population response on the wage to the direct effect of the two demand channels,

$$\frac{dw_V^*}{dN_U} = \frac{\partial w_V}{\partial N_V}\frac{dN_V^*}{dN_U} + \left.\frac{\partial w_V}{\partial N_U}\right|_{N_V},$$

where the first term is positive in villages that lose population and negative in villages that gain it. Substituting dN_V^*/dN_U from equation (31) and $\partial w_V/\partial N_V = -(1-s)w_V/\Lambda$ from equation (29), we obtain

$$\frac{dw_V^*}{dN_U} = \frac{\Phi(d) + \Psi(d) + \tilde{A}\,(1-s)\,w_V\,\tilde{M}/\Lambda}{1 + \tilde{A}\,(1-s)^2/\Lambda} > 0. \tag{32}$$

The two demand channels are non-negative, and the term in $\tilde{M}$ is strictly positive since $\tilde{M} > 0$. The village wage therefore rises at every distance. Where the village gains population, the additional workers damp the direct effect of the demand channels, but never by enough to reverse it. Where the village loses population, migration adds to that effect, because a village that loses workers clears its labor market at a higher wage.

We measure the welfare of the villagers who stay by the log of their real wage, which equals their indirect utility $V^V = (1-s)\ln w_V + s\ln A_N$ up to a constant. Differentiating and using equation (32),

$$\frac{dV^V}{dN_U} = \frac{1-s}{w_V}\frac{dw_V^*}{dN_U} > 0, \tag{33}$$

and welfare rises at every distance.

C4 The regime dichotomy

By equation (31), the population response has the sign of the gain in village utility, $(1-s)\,[\Phi(d)+\Psi(d)]/w_V$, minus the migration pull $\tilde{M}$. To follow this difference across distance in full equilibrium, we express it in terms of each sector's share of Λ. Let $\chi(d) := (L_N^* - sN_V)/\Lambda$ be the share that comes from services employment serving visitors, and $\omega_R(d) := L_R^*/[(1-\gamma)\Lambda]$ and $\omega_T(d) := L_T^*/[(1-\alpha)\Lambda]$ the shares that come from extraction and crops. The three shares add up to one, $\chi+\omega_T+\omega_R = 1$. Finally, let $\varepsilon_a(d) := N_U\,a(d)$ be the elasticity of the resource price with respect to city size, and $\varepsilon_M := N_U\,\tilde{M}$ the migration pull expressed as an elasticity with respect to city size. Since $\Phi\,N_U/w_V = \chi$ and $\Psi\,N_U/w_V = \varepsilon_a\,\omega_R$, the difference multiplied by N_U is

$$\Sigma_N(d) \;=\; (1-s)\,\big[\chi(d) + \varepsilon_a(d)\,\omega_R(d)\big] \;-\; \varepsilon_M, \tag{34}$$

and the population response is positive exactly where $\Sigma_N(d)$ is positive. Every term of Σ_N except ε_M varies with distance, directly through ϕ and a and indirectly through the equilibrium wage and sector shares. The number of times Σ_N crosses zero therefore depends on the two decay rates and on how the wage and the sector shares change with distance.

First, the village wage falls with distance,

$$-\frac{d\ln w_V}{dd} \;=\; W(d) \;=\; \frac{\chi\,\delta_\phi + m\,\mu'(d)}{1 + m\,(1-s)} \;\geq\; 0, \qquad m := \frac{\tilde{A}\,(1-s)}{\Lambda}, \tag{35}$$

because a more distant village receives less urban services demand and, facing a higher migration cost, retains a larger share of its workers at any given wage. The expression holds for any differentiable migration cost, and W is non-negative wherever that cost rises with distance. Second, χ falls at every distance,

$$\frac{d\ln\chi}{dd} \;=\; -\,\delta_\phi\,(1-\chi) \;-\; W\left[\frac{\omega_T\,\alpha}{1-\alpha} + \frac{\omega_R\,\gamma}{1-\gamma}\right] \;<\; 0. \tag{36}$$

It falls because the urban demand behind it decays at the rate δ_ϕ, and because the falling wage raises crop and extraction employment proportionally more than employment serving visitors. Crop and extraction labor demands are more elastic with respect to the wage. Third, the extraction share need not fall. It changes with distance at the rate

$$\frac{d \ln \omega_R}{dd} \;=\; \chi\, \delta_\phi \;+\; W \Big[\frac{\chi\, \gamma}{1-\gamma} + \frac{\omega_T\, (\gamma - \alpha)}{(1-\alpha)(1-\gamma)} \Big], \tag{37}$$

which can be positive. The first term arises because the decay of visitors' demand shrinks Λ and raises the other shares in proportion. The falling wage adds to this rate when $\gamma \geq \alpha$, because extraction labor demand is then at least as elastic with respect to the wage as crop labor demand. Extraction labor demand is also always more elastic than the demand for workers serving visitors. The falling wage lowers the rate only when γ lies far enough below α that $\omega_T(\alpha - \gamma)/(1-\alpha) > \chi\gamma$.

A rising ω_R can interrupt the decline of Σ_N. If resource demand decays more slowly than ω_R rises with distance, the resource term $\varepsilon_a\, \omega_R$ rises with distance and can raise the demand effect again after the services channel has faded. Proposition 1 therefore assumes that resource demand decays at least as fast as ω_R rises. This assumption compares δ_a with the rate at which an equilibrium share changes with distance. When $\gamma \leq \alpha$, the bracket in equation (37) is at most $\chi\gamma/(1-\gamma)$, which implies that the rate at which ω_R rises is at most $\chi\,[\delta_\phi + W\gamma/(1-\gamma)]$. That rate is then small wherever χ is small, which is the case far enough from the city.[25]

Proposition 1 (Regime dichotomy)**.** ***The population response takes one of two regimes.*** *Suppose resource demand decays at least as fast as the extraction share drifts,*

$$\delta_a \;\geq\; \frac{d \ln \omega_R(d)}{dd} \qquad \textit{at every distance,} \tag{38}$$

and let R be the regime index of equation (22). If $R > 1$, the population response is positive at distances closer than a single crossover d^*, defined by $\Sigma_N(d^*) = 0$, and negative beyond it. If $R \leq 1$, the population response is negative at every distance beyond the city's edge. In villages without a resource endowment, ω_R is zero at every distance, condition (38) is vacuous, and the dichotomy needs no condition at all.

Proof. Under condition (38), the resource term of Σ_N does not rise with distance, since $d \ln(\varepsilon_a\, \omega_R)/dd = -\,\delta_a + d \ln \omega_R/dd \leq 0$. The services term falls strictly by equation (36), and Σ_N therefore falls strictly with distance. At the city's edge, substituting $\chi = \Phi\, N_U/w_V$ and $\varepsilon_a\, \omega_R = \Psi\, N_U/w_V$ into equation (34) and comparing with equation (22) yields $\Sigma_N(0) = \varepsilon_M\, (R-1)$, where the wage in R is the wage at the city's edge. Far from the city, χ vanishes with ϕ, and $\varepsilon_a\, \omega_R$ vanishes with a because ω_R stays below one. The difference Σ_N therefore approaches $-\varepsilon_M < 0$. If $R > 1$, the difference Σ_N is positive at the city's edge and negative far from it. Because it falls strictly, it crosses zero at exactly one distance d^*. If $R \leq 1$, Σ_N

[25] When the condition of Proposition 1 fails, $\varepsilon_a\, \omega_R$ can rise over a range of distances, and the population response can change sign more than once. The population response after the second change of sign need not be small. The condition is sufficient for the population response to change sign at most once, but it is not necessary. To see how often the response changes sign at most once when the condition fails, we solve the full equilibrium of the model at 240 distances between 0.25 and 200 kilometers for independent uniform draws of the parameters. In every draw we fix $s = 0.25$, $\alpha = 2/3$, and $w_U = 2.5$, and we set crop productivity, the level of urban utility, and ϕ_0 so that a village 16 kilometers from the city, the median distance in our sample, retains 90 percent of its workers and employs two percent of them in services sold to urban visitors. We discard draws in which this calibration would require more than the city's entire demand for services to reach a village at the city's edge. Among 200 draws with γ between 0.2 and 0.8 and wide ranges for the other parameters, the condition fails in 30, and the population response changes sign at most once in every one of them. Among 448 draws with γ between 0.2 and 0.97, the condition fails in 112, and the response changes sign at most once in all of them except three, which all have $\gamma \geq 0.91$. In 288 additional draws concentrated on high values of γ, the response changes sign more than once in ten, all with $\gamma \geq 0.89$. At such values the fixed endowment earns roughly a tenth of extraction revenue or less, compared with the one-third share of land in crop revenue.

is non-positive at the city's edge and falls thereafter, which implies that the response is negative at every distance beyond the edge. In villages without a resource endowment, $\Sigma_N = (1-s)\,\chi - \varepsilon_M$ falls strictly by equation (36), and the same argument applies with no condition. No closed form is available for d^*, because the equilibrium shares are not exact exponentials in distance. □

C5 Sectoral responses

Crop employment depends only on the village wage, because the crop good sells at a fixed national price. Equation (25) then implies $d \ln L_T^* = -d \ln w_V/(1-\alpha)$ along any equilibrium path.

Proposition 2 (Unconditional crop contraction)**. *Crop employment falls everywhere.*** *Under $\tilde{M} > 0$, village crop employment contracts at every distance,*

$$\frac{dL_T^*}{dN_U} = -\frac{1}{1-\alpha}\,\frac{L_T^*}{w_V}\,\frac{dw_V^*}{dN_U} < 0.$$

Proof. Differentiating equation (25) along the equilibrium path yields the displayed expression, which is negative because $dw_V^*/dN_U > 0$ by equation (32). □

Near the city, the wage rise that contracts crop employment can come from urban demand for services and resource output, from the migration pull, or from both. All three sources fade with distance, and far from the city the wage rise comes mainly from the one that fades most slowly. Where a village has a resource endowment and urban demand for its resource output fades more slowly than both urban services demand and the number of workers close to the margin of migrating, the wage rise far from the city comes mainly from resource demand. Extraction then expands and draws workers out of crops (see Proposition 3 below). In villages without a resource endowment, the crop contraction fades with distance, because urban services demand and the number of workers close to the margin of migrating both fade.

Proposition 3 (Conditional extraction response)**. *Extraction employment rises only above a threshold.*** *Extraction employment rises with urban growth if and only if*

$$\varepsilon_a(d) > T_R(d) := \frac{\chi(d) + m\,\varepsilon_M}{1 - \omega_R(d) + m\,(1-s)},$$

where χ, ω_R, m, ε_a, and ε_M are as defined in Section C4. Far from the city both χ and m vanish while $1-\omega_R$ stays positive. $T_R(d)$ therefore approaches zero, although $\varepsilon_a(d)$ approaches zero as well.

Proof. Log-differentiating equation (26) along the equilibrium path,

$$\frac{d \ln L_R^*}{dN_U} = \frac{1}{1-\gamma}\left[a(d) - \frac{d \ln w_V^*}{dN_U}\right].$$

The response is therefore positive if and only if $\varepsilon_a(d)$ exceeds the elasticity of the village wage with respect to city size, which by equation (32) is $N_U\, d \ln w_V^*/dN_U = [\chi + \varepsilon_a\,\omega_R + m\,\varepsilon_M]/[1 + m(1-s)]$. Moving $\varepsilon_a\,\omega_R$ to the left-hand side yields the stated threshold. Its denominator is positive because $\omega_R < 1$. The share ω_R is below one because the land endowment ensures that the village always employs some workers in crops. □

Far from the city, the sign of the extraction response depends on whether $\varepsilon_a(d)$ or $T_R(d)$ approaches zero faster. Extraction expands at long distances when resource demand fades more slowly than both urban services demand and the number of workers close to the margin of migrating, and contracts when it fades faster than at least one of them. At long distances χ fades at the rate δ_ϕ and m at the rate $\mu'(d)/\sigma$. Extraction therefore expands far from the city when $\delta_a < \delta_\phi$ and $\mu(d)/\sigma - \delta_a d$ grows without bound. The exponential migration cost of Figure 9 satisfies the second condition. Extraction contracts far from the city when $\delta_a > \delta_\phi$.

Services employment can rise or fall. Differentiating the services market condition $w_V L_N^* = s(w_V N_V + w_U \phi_0 N_U e^{-\delta_\phi d})$ along the equilibrium path,

$$\frac{dL_N^*}{dN_U} = \frac{s\, w_V\, dN_V^*/dN_U + s\, w_U\, \phi_0 e^{-\delta_\phi d} - (L_N^* - sN_V)\, dw_V^*/dN_U}{w_V}, \tag{39}$$

whose first term is the change in residents' spending and has the sign of the population response. The second term is the additional spending of visitors, and the third is the higher cost of hiring the workers who serve them.

Proposition 4 (Conditional services response)**.** ***Services employment can rise where population falls.*** *Services employment rises with urban growth, $dL_N^*/dN_U > 0$, if and only if*

$$s\, w_U\, \phi_0 e^{-\delta_\phi d} > (L_N^* - sN_V)\, \frac{dw_V^*}{dN_U} - s\, w_V\, \frac{dN_V^*}{dN_U}.$$

The left-hand side is largest near the city. The condition can therefore hold there even where $dN_V^*/dN_U < 0$.

Proof. The condition states that the right-hand side of equation (39) is positive. To see that it can hold where population falls, let $\varepsilon_V := N_U\, d\ln w_V^*/dN_U$ be the elasticity of the village wage with respect to city size. Since $s\, w_U\, \phi_0 e^{-\delta_\phi d} = w_V(L_N^* - sN_V)/N_U$, equation (39) becomes

$$\frac{dL_N^*}{dN_U} = s\, \frac{dN_V^*}{dN_U} + \frac{L_N^* - sN_V}{N_U}\,(1 - \varepsilon_V).$$

By the expression for ε_V in the proof of Proposition 3, $1-\varepsilon_V = [\omega_T+\omega_R(1-\varepsilon_a)+m(1-s-\varepsilon_M)]/[1+m(1-s)]$, which is positive whenever $\varepsilon_a \le 1$ and $\varepsilon_M \le 1-s$. Services employment then rises wherever population falls by less than $(L_N^* - sN_V)(1-\varepsilon_V)/(sN_U)$. Under these conditions the bound is positive at every distance, because employment serving visitors is. □

In the limit in which urban demand for services vanishes, $\phi(d) = 0$, the last two terms of equation (39) are zero, and services employment changes by the share s of the change in population. In two sectors, employment can rise where population falls. In services, this happens through the demand of urban visitors. In extraction, it happens wherever the response of the resource price is large enough to outweigh the upward pressure that urban services demand and migration put on the village wage (see Proposition 3). Crop employment, by contrast, falls at every distance (Proposition 2).

Proposition 5 (Recomposition)**.** ***Where only resource demand persists, employment recomposes.*** *Consider the limiting configuration $\phi(d) = 0$ and $\tilde{A} = 0$, with $a(d) > 0$ and a positive resource endowment. Then*

$$\frac{dN_V^*}{dN_U} = 0, \qquad \frac{dL_N^*}{dN_U} = 0, \qquad \frac{d\ln L_R^*}{dN_U} = \frac{a(d)\,[1-\omega_R(d)]}{1-\gamma} > 0, \qquad \frac{d\ln L_T^*}{dN_U} = -\frac{a(d)\,\omega_R(d)}{1-\alpha} < 0,$$

and the two level changes offset exactly, $dL_T^* + dL_R^* = 0$. In a village without a resource endowment every response in this configuration is zero.

Proof. When $\phi(d) = 0$, services market clearing implies $L_N^* = sN_V$, and therefore $dL_N^* = s\, dN_V^*$. When $\tilde{A} = 0$, equation (31) implies $dN_V^*/dN_U = 0$. It follows that $dL_N^*/dN_U = 0$, and the wage response reduces to $dw_V^*/dN_U = \Psi(d)$. Log-differentiating extraction labor demand along the equilibrium path, we obtain $d\ln L_R^*/dN_U = [a(d) - \Psi/w_V]/(1-\gamma) = a(d)[1-\omega_R]/(1-\gamma)$, and equation (25) implies $d\ln L_T^*/dN_U = -\Psi/[w_V(1-\alpha)] = -a(d)\,\omega_R/(1-\alpha)$. In this configuration $L_N^* - sN_V = 0$, which implies $\Lambda = L_T^*/(1-\alpha) + L_R^*/(1-\gamma)$, $L_R^*/(1-\gamma) = \omega_R\Lambda$, and $L_T^*/(1-\alpha) = (1-\omega_R)\Lambda$. The two level changes per unit of N_U are then $dL_R^* = a\,\omega_R(1-\omega_R)\Lambda$ and $dL_T^* = -a\,\omega_R(1-\omega_R)\Lambda$, which sum to zero. In a village without a resource endowment, Ψ is zero, and the wage and every sectoral response vanish. □

In this configuration population does not respond, and services employment remains at the share s of it. Total employment in crops and extraction is therefore fixed, and extraction can expand only by drawing workers out of crops.